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Quantum perceptron algorithms and machine learning techniques for quantum separability

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Affidavit

I, undersigned, Xiaoyu Sun, hereby declare that the work presented in this manuscript is my own work, carried out under the scientific supervision of Hachem Kadri and Giuseppe Di Molfetta, in accordance with the principles of honesty, integrity and responsibility inherent to the research mission. The research work and the writing of this manuscript have been carried out in compliance with both the french national charter for Research Integrity and AMU charter on the fight against plagiarism.

This work has not been submitted previously either in this country or in another country in the same or in a similar version to any other examination body.

Marseille, the 18th of March 2026

Liste de publications et participation aux conférences

Liste des publications et/ou brevets réalisées dans le cadre du projet de thèse :

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1. Ecole d'été d'informatique 2023
   https://ecoles-cea-edf-inria.fr/history/ecole-informatique-2023/
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3. Journées Informatique Quantique (JIQ 2026)
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Résumé et mots clés

Avec le développement rapide de la puissance de calcul et des techniques algorithmiques, l'apprentissage automatique a profondément influencé la recherche scientifique et les applications quotidiennes. Parallèlement, les progrès de l'informatique quantique ont introduit des paradigmes algorithmiques fondamentalement nouveaux et la possibilité d'accélérer considérablement les calculs. Ces avancées motivent naturellement l'émergence du domaine de l'apprentissage automatique quantique. Dans cette thèse de doctorat, nous étudions deux problèmes liés à différents aspects de ce nouveau champ de recherche.

La première partie consiste à utiliser des algorithmes et des techniques de calcul quantiques pour résoudre des problèmes relevant de l'apprentissage automatique traditionnel. Plus précisément, nous nous concentrons sur la tâche de classification binaire linéaire, qui peut être résolue classiquement par l'algorithme du perceptron. Dans ce travail, nous examinons d'abord plusieurs algorithmes de perceptron quantiques existants, puis nous en proposons de nouveaux, basés sur la programmation linéaire, qui améliorent la complexité des requêtes et les opérations arithmétiques au sein de modèles de calcul quantiques idéalisés. Notre objectif est d'apporter un éclairage constructif sur l'efficacité statistique des différents modèles de perceptron et d'offrir de nouvelles perspectives sur l'apprentissage par perceptron quantique.

La seconde partie consiste à utiliser des techniques d'apprentissage automatique pour contribuer à la résolution du problème de l'information quantique. Plus précisément, nous étudions le problème de la séparabilité quantique, pour lequel déterminer si un état quantique est intriqué ou séparable est un problème NP-difficile. Dans ce travail, nous examinons d'abord la convergence d'une approximation de l'algorithme de Frank-Wolfe, plus rapide que l'algorithme standard pour résoudre le problème de séparabilité. Nous démontrons ensuite l'efficacité du test de témoin d'intrication proposé, puis l'utilisons pour construire des ensembles de données d'intrication destinés à l'entraînement de modèles de classification modernes en apprentissage automatique. Les méthodes que nous examinons peuvent servir de prototypes préliminaires pour la détection rapide de l'intrication, avec des applications potentielles dans la certification de l'intrication dans les expériences quantiques.

Mots clés : apprentissage automatique quantique, classification binaire, algorithmes de perceptron quantique, intrication et séparabilité quantiques, algorithme de Frank-Wolfe

Abstract and keywords

With the rapid development of computational power and algorithmic techniques, machine learning has had a substantial impact on both scientific research and everyday applications. At the same time, progress in quantum computing has introduced fundamentally new algorithmic paradigms and the possibility of significant computational speedups. These developments naturally motivate the emerging field of quantum machine learning. In this PhD thesis, we study two problems related to different aspects of this new field of research.

The first part is to use quantum algorithms and quantum computing to solve problems in the traditional machine learning area. Specifically, we focus on the linear binary classification task, which can be solved classically by the perceptron algorithm. In this work, we first examine several existing quantum perceptron algorithms, and then we propose new ones based on linear programming that achieve improvements in query complexity and arithmetic operations within idealised quantum computational models. We aim to provide constructive insights into the statistical efficiency of variant perceptron models and offer new perspectives on quantum perceptron learning.

The second part is to use machine learning techniques to assist in solving the quantum information problem. Specifically, we study the quantum separability problem, where determining whether a quantum state is entangled or separable is known to be NP-hard in general. In this work, we first examine the convergence behaviours of an approximated Frank-Wolfe algorithm, which is used in solving the separability problem and is more time-efficient than the standard one. Then we demonstrate the effectiveness of the proposed entanglement witness test, and use it to constitute bounded entanglement datasets to train modern machine learning classification models. The methods we examine can serve as early-stage prototypes for fast entanglement detection, with potential applications in entanglement certification in quantum experiments.

Keywords: quantum machine learning, binary classification, quantum perceptron algorithms, quantum entanglement and separability, Frank-Wolfe algorithm

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Table des matières

Table des figures

Liste des tableaux

1 Introduction

Sommaire

1.1 Overview

1.1.1 Machine learning

Broadly speaking, machine learning (ML) is a field of study that uses computational methods to learn from experience and identify generalisable predictive patterns. The experience is usually referred to as data that has been collected and made available in a digital form. For example, the MNIST dataset, which contains 28×28 greyscale images of handwritten digits (LeCun et al. 2010). In this case, the predictive patterns can be the labels of the digits (0-9).

Depending on the data types and goals, machine learning can be classified into three basic paradigms : supervised learning, unsupervised learning, and reinforcement learning. For supervised learning, a learning model is given input with its desired specific output (or labels), which is a supervisory signal. The supervised learning models or algorithms are asked to predict outputs on new input instances. The most common types of supervised learning tasks are classification and regression. For unsupervised learning, the goal is to discover patterns using data that are not labelled, classified, or categorised, among which a typical application is clustering with the well-known k-means clustering algorithm. In reinforcement learning (RL), algorithms are designed to take actions in a dynamic environment to maximise rewards, where data is interactions and rewards.

Over the past decades, the increasing demand to learn from high-dimensional data has driven significant interest in machine learning. And more recently, with the rapid development of computational power and algorithmic techniques, machine learning has made a substantial impact on both scientific research and everyday applications. For example, classification algorithms have enabled major advances in image and audio recognition, as well as in malicious website and spam detection. Bandit algorithms from reinforcement learning are widely used in recommendation systems. Nowadays, increasingly sophisticated ML methods have played a transformative role across multiple domains. In biology, tools such as AlphaFold have achieved breakthroughs in protein structure prediction (Jumper et al. 2021), while generative models are applied for drug discovery. In physics and astronomy, machine learning techniques are applied to particle detection for CERN experiments (Baldi et al. 2014 ; Adam-Bourdarios et al. 2015). In healthcare, they support clinical diagnosis and medical image processing. In finance, machine learning is widely used for fraud detection and customer profiling. And as has been recognised, large language models (LLMs) such as generative pre-trained transformers (GPTs) (Achiam et al. 2023), combined with natural language processing (NLP) for text generation, and other forms of creative AI (i.e. Sora), have been widely used by the population.

1.1.2 Quantum computing

In the early 1920s, the modern quantum theory of non-relativistic quantum mechanics was established to explain physical phenomena observed at atomic scales. During the same period in the twentieth century, computer science emerged on the other side of the academic community. In the 1936 paper of Turing et al. 1936, the Church–Turing thesis states that the concept of computability can be characterised by *Turing computable*, where Turing defined the Turing machine (programmable computers) and the concept of universal Turing machine (UTM), which can be used to simulate any other Turing machines. The broad acceptance of it established the notion of modern computer science. In 1980, Paul Benioff introduced a quantum mechanical model of Turing machines (Benioff 1980) (a quantum Turing machine) based on the work of reversible Turing machines (Bennett 1973). About the same time, Feynman conjectured that a quantum computer is necessary for simulating quantum mechanics and proposed a basic model for a quantum computer (Feynman 1982), which is conventionally recognised as the start of quantum computing.

After the conceptualising of quantum computing, the first universal quantum computer was proposed by Deutsch 1985. Subsequently, various quantum algorithms appear to solve or calculate problems, such as the Deutsch–Jozsa algorithm (Deutsch et Jozsa 1992), to determine if a function is constant or balanced ; the Bernstein–Vazirani algorithm (Bernstein et al. 1993), to learn a string encoded in a function ; the Simon's algorithm (Simon 1997), to determine a function is one-to-one or two-to-one ; and the well-known Shor's algorithms for finding prime factors of an integer (Shor 1994). The first quantum error correction scheme was also proposed by Shor 1995. In 1996, Grover's search algorithm was proposed to achieve quadratic speedup on any unstructured search problem (Lov K. Grover 1996), and the same year, the conjecture of Feynman was shown to be correct by Lloyd 1996.

Alongside theoretical developments, experimental progress has been made across multiple physical platforms, including nuclear magnetic resonance (NMR) (Chuang et al. 1998), superconducting circuits (Jones et al. 1998), optical (O'Brien et al. 2003), and ion-trap (Gulde et al. 2003) devices.

Nowadays, quantum computing has integrated knowledge from quantum algorithms, quantum information processing and experimental quantum systems. In 2018, the term noisy intermediate-scale quantum (NISQ) was introduced by PRESKILL 2018, referring to quantum processors with tens to thousands of qubits operating without full error correction. Although this is still the era we are currently in, the complexity and scale of quantum computers have been increasing significantly in recent years. In 2019, the 53-qubit Sycamore processor demonstrated so-called "quantum supremacy" (ARUTE et al. 2019) on quantum random circuit sampling. In 2020, the Jiuzhang photonic quantum computer performed Gaussian boson sampling with up to 76 photons (H.-S. ZHONG et al. 2020). More recently, IBM has developed increasingly large-scale quantum processors, including the 127-qubit Eagle (2021), the 433-qubit Osprey (2022), and the 1121-qubit Condor (2023) (IBM 2025), surpassing the 1,000-qubit threshold.

1.1.3 Quantum machine learning

One of the central features of quantum computation is the superposition principle, which allows quantum states to represent high-dimensional data compactly. Additionally, quantum computing exhibits inherent parallelism, enabling the use of quantum linear algebra subroutines such as the quantum Fourier transform (Michael A NIELSEN et al. 2010a) and HHL algorithm (Aram W HARROW et al. 2009) that operate in polylog(D) time under certain conditions.

Meanwhile, the rapid development of both machine learning and quantum computing has naturally led to the emergence of quantum machine learning (QML). Depending on the nature of the data and computational resources, QML can be broadly classified into three categories as have been suggested (BIAMONTE et al. 2017) : quantum algorithms for classical data (QaCd), classical algorithms for quantum data (CaQd), and quantum algorithms for quantum data (QaQd). Sometimes, the definition of data or input data is hard to separate ; more specifically, we mean classical/quantum algorithms for classical/quantum tasks. Below, we briefly review each category.

QaCd Progress in quantum computing has introduced fundamentally new algorithmic paradigms and the possibility of significant computational speedups. This has greatly motivated the use of quantum computing for machine learning tasks (LLOYD, MOHSENI et al. 2013 ; SCHULD, SINAYSKIY et al. 2015 ; SCHULD et KILLORAN 2019), also known as quantum-enhanced ML, in which quantum computation is widely viewed as a promising framework for addressing challenges associated with high-dimensional data, one of the central difficulties in modern machine learning. Several major directions have emerged in this context, each leveraging different quantum algorithmic primitives. For example, linear algebra-based quantum algorithms, such as the HHL algorithm with fast matrix inversion (Aram W HARROW et al. 2009), have been applied in least-squares linear regression (WIEBE, BRAUN et al. 2012) and least-squares support vector machine (REBENTROST et al. 2014). Amplitude amplification-based algorithms, as well as Grover's search, have been applied for the k-nearest neighbours (KNN) algorithm (WIEBE, KAPOOR et al. 2014) and the training of the perceptron algorithm (KAPOOR et al. 2016).

Beyond these primitives, more flexible frameworks have been developed, such as variational quantum algorithms (VQAs) (CEREZO et al. 2021), which combine parametrised quantum circuits with classical optimisation loops, have been widely applied to classification, regression, and generative modelling tasks. Closely related to this paradigm, quantum neural networks (QNNs) provide trainable quantum circuit architectures analogous to classical neural networks, with applications in supervised learning and generative modelling (Schuld, Bocharov et al. 2020; Tian et al. 2023). These models have been viewed as strong candidates for demonstrating quantum advantages in the NISQ era.

In addition, general-purpose quantum optimisation algorithms, such as the quantum approximate optimisation algorithm (QAOA) (Farhi et al. 2014), have been explored for solving combinatorial optimisation problems and may be incorporated into machine learning pipelines, although they are not specifically designed for learning tasks.

CaQd Meanwhile, many frontier problems in modern physics revolve around the quantum realm. Enhanced computational methods and algorithmic tools are, therefore, essential for advancing our understanding of quantum systems. CaQd focuses on applying classical machine learning techniques to analyse quantum data, which includes tasks such as quantum state tomography and reconstruction (Torlai et al. 2018; Carrasquilla, Torlai et al. 2019), phase classification in condensed matter physics (Carrasquilla et Melko 2017), and quantum error mitigation(QEM) (Strikis et al. 2021). Advanced ML models also provide powerful tools for predicting the properties of quantum many-body systems (Carleo et al. 2017) and simulating open quantum systems dynamics (Reh et al. 2021).

QaQd Originating from Feynman's proposal to use quantum systems to simulate quantum dynamics and avoid the exponential overhead of classical methods (Feynman 1982), quantum machine learning provides a natural framework for processing quantum data. In this setting, algorithms operate directly on quantum states generated by physical systems, which could enable more efficient extraction of quantum features and correlations than using classical algorithms (Biamonte et al. 2017). For example, quantum principal component analysis (PCA) can extract properties of quantum states with significantly reduced operations compared to classical approaches (Lloyd, Mohseni et al. 2014) when given multiple copies.

Given that the data of quantum chemistry and many-body physics are intrinsically quantum, QaQd is widely regarded as a promising application to solve problems in these fields. In classically intractable regimes, quantum generative models have been designed to reconstruct states efficiently (Kieferova et al. 2016). In addition, general-purpose quantum algorithms, such as quantum simulators for Hamiltonian inference (Wiebe, Kapoor et al. 2014) and the variational quantum eigensolver (VQE) (Peruzzo et al. 2014), have been widely explored for quantum chemistry and many-body physics problems, although they are not specifically designed for machine learning tasks.

Challenges and open problems Despite its theoretical appeal, QML faces several significant real-world challenges.

First, *data encoding* remains a major bottleneck. Many quantum algorithms assume that classical data can be efficiently loaded into quantum states (e.g., via amplitude encoding), which is not known to be achievable in general without incurring exponential overhead (Aaronson 2015).

Second, *noise and decoherence* severely limit current quantum devices. Quantum algorithms, especially those relying on deep circuits or precise interference patterns, are highly sensitive to errors. Maintaining coherence over long computation times remains a major experimental challenge.

Third, *hardware scalability* is still an open problem. Building large-scale quantum computers requires precise control over many qubits, with stringent requirements on error rates, connectivity, and coherence times. The overhead introduced by quantum error correction may significantly increase the number of physical qubits required for practical applications.

Fourth, *algorithmic limitations* must be carefully considered. Many proposed quantum speedups rely on idealised assumptions, such as efficient oracle access or specific input structures, which may not hold in real-world applications. Moreover, several quantum algorithms have been "dequantized" (Tang 2019 ; Arrazola et al. 2020), meaning that classical algorithms with comparable performance have been developed.

Finally, *practical advantage* remains uncertain. While quantum algorithms provide provable speedups for certain problems, identifying tasks where these advantages translate into real-world applications is still an open challenge. In particular, large-scale machine learning tasks often involve substantial data input/output overhead, which can negate potential quantum advantages.

Nevertheless, the exploration of quantum advantages remains theoretically valuable, as it may shed light on the boundaries of classical complexity. Moreover, from a practical standpoint, early experimental results and quantum circuit simulations have already demonstrated promising quantum advantages in some machine-learning tasks (Schuld, Bocharov et al. 2020 ; Saggio et al. 2021). Overall, QML represents a promising but still developing field at the intersection of quantum computing and machine learning. While significant theoretical progress has been made, achieving practical quantum advantage will likely require advances across both hardware and algorithm design, as well as a deeper understanding of the interplay between quantum resources and learning theory.

1.2 Contributions

In this PhD thesis, we study two problems related to distinct aspects of quantum machine learning. In Chapter 3, we present the first part of the thesis, where we explore the possibility of leveraging quantum superposition, interference and parallelism (introduced in Section 2.1.4) in quantum algorithms and quantum computing to assist the problem of the QaCd category.

Specifically, we focus on the linear binary classification task, which can be solved classically by the perceptron algorithm. In this work, we first examine several existing quantum perceptron algorithms and then propose new ones based on linear programming that improve query complexity and arithmetic operations in idealised quantum computational models. We aim to provide constructive insights into the statistical efficiency of variant perceptron models and offer new perspectives on quantum perceptron learning.

In Chapter 4, we present the second part of the thesis, where we study how to use traditional ML techniques and algorithmic tools to assist in solving a quantum information problem, which falls into the category of CaQd.

Specifically, we study the quantum separability problem. While quantum entanglement has been widely identified as a quantum advantage resource (introduced in Section 2.1.4), determining whether a quantum state is entangled or separable is known to be NP-hard in general. In this work, we first examine the convergence behaviours of an approximated Frank-Wolfe algorithm, which is used in solving the separability problem and is more time-efficient than the standard one. Then we demonstrate the effectiveness of the proposed entanglement witness test and use it to constitute bounded entanglement datasets to train modern machine learning classification models. The methods we examine can serve as early-stage prototypes for fast entanglement detection, with potential applications in entanglement certification for quantum experiments.

1.3 Organization

In Chapter 2, we introduce some background and preliminaries of both quantum computing and machine learning that are going to be used in or related to this thesis. In Chapter 3, we present our first contribution of the thesis regarding quantum perceptron algorithms, which is based on the arXiv preprint (Sun et al. 2025) (to appear soon on Quantum Machine Intelligence). In this work, previous variants of quantum perceptron algorithms are examined, and new ones are proposed that achieve improved query complexity while preserving favourable properties for small-margin regimes. In Chapter 4, we present the second contribution of the thesis on solving the quantum separability problem via an approximated Frank-Wolfe and machine learning techniques. In this work, we provide theoretical guarantees and experimental validations for the effectiveness of the previously proposed methods (Casalé, Di Molfetta, Anthoine et al. 2023). Finally, in Chapter 5, we summarise several partially explored sub-problems that warrant further investigation.

2 Backgrounds and preliminaries

Sommaire

2.1 Quantum computing

2.1.1 Hilbert space and quantum states

The notion of a quantum state is central to quantum mechanics and, in particular, to quantum information theory (Michael A Nielsen et al. 2010a). Informally, the state of a system encodes all information necessary to predict the outcomes of measurements performed on that system. A quantum state lives in the *Hilbert spaces*. Below, we introduce the space first.

Hilbert space In quantum information theory, states are naturally described within the framework of Hilbert spaces. In the finite-dimensional setting considered here, a Hilbert space $\mathcal{H}$ is a complex vector space equipped with an inner product $\langle\cdot|\cdot\rangle : \mathcal{H} \times \mathcal{H} \to \mathbb{C}$. It is also known as an inner product space. This structure allows one to define notions such as length and orthogonality. The norm of a vector $|\psi\rangle \in \mathcal{H}$ is given by

$$\||\psi\rangle\| = \sqrt{\langle\psi|\psi\rangle}. \tag{1}$$

Two vectors $|\psi\rangle$ and $|\phi\rangle$ are orthogonal if $\langle\psi|\phi\rangle = 0$, and a vector is normalised when $\langle\psi|\psi\rangle = 1$. An orthonormal basis $\{|i\rangle\}$ satisfies

$$\langle i|j\rangle = \delta_{ij}, \quad \sum_i |i\rangle\langle i| = \mathbb{I}. \tag{2}$$

Composite quantum systems are described using *tensor products* of Hilbert spaces. Depending on the number of constituent subsystems, a quantum system can be *bipartite* or *multipartite*.

A bipartite quantum system consists of two subsystems, typically denoted A and B. The joint system is represented by the tensor product space $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$. A general pure state of a bipartite system can be written as

$$|\Psi\rangle = \sum_{i,j} c_{ij}\, |i\rangle \otimes |j\rangle, \tag{3}$$

where $c_{ij} \in \mathbb{C}$ and $\sum_{i,j} |c_{ij}|^2 = 1$. If $\{|i\rangle\}$ and $\{|j\rangle\}$ are orthonormal bases for $\mathcal{H}_A$ and $\mathcal{H}_B$, then $\{|i\rangle \otimes |j\rangle\}$ forms an orthonormal basis for $\mathcal{H}_A \otimes \mathcal{H}_B$. Linear operators on composite systems are constructed via tensor products. For operators A on $\mathcal{H}_A$ and B on $\mathcal{H}_B$, their tensor product $A \otimes B$ acts as

$$(A \otimes B)(|\psi\rangle \otimes |\phi\rangle) = A|\psi\rangle \otimes B|\phi\rangle.$$

More generally, a multipartite quantum system consists of $N > 2$ subsystems. Its Hilbert space is given by the tensor product $\mathcal{H} = \bigotimes_{k=1}^{N} \mathcal{H}_k$. A general pure state of a multipartite system takes the form

$$|\Psi\rangle = \sum_{i_1,\ldots,i_N} c_{i_1\ldots i_N}\, |i_1\rangle \otimes \cdots \otimes |i_N\rangle, \tag{4}$$

where $\{|i_k\rangle\}$ is an orthonormal basis of $\mathcal{H}_k$ for each subsystem k.

Quantum states A *pure* qubit can be described as a unit vector in a two-dimensional complex Hilbert space $\mathbb{C}^2$. In this picture, the state is written in Dirac "bra-ket" notation as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \tag{5}$$

where $|0\rangle$ and $|1\rangle$ form an orthonormal basis, and the complex coefficients satisfy the normalisation condition

$$|\alpha|^2 + |\beta|^2 = 1.$$

The pure qubit expression provides an intuitive way of understanding quantum superposition. Such that a classical bit is either in state 0 or state 1, a qubit can exist in a continuum between them.

Other than the vectorised formula, another way to describe a quantum system more generally is via density matrices, where a density matrix ρ must satisfy the following conditions :

— ρ is Hermitian, i.e., $\rho = \rho^\dagger$,
— ρ is positive semidefinite,
— $\mathrm{Tr}(\rho) = 1$.

This framework includes pure states as a special case. In this case, a *pure state* quantum system can be characterised fully by $\rho = |\psi\rangle\langle\psi|$, whose state is known exactly.

However, sometimes a quantum system cannot be represented by a single vector. Instead, it can be a *statistical mixture* (or an ensemble) of pure quantum states,

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \tag{6}$$

where $\{p_i\}$ is a probability distribution. In this case, the quantum system ρ is said to be in a *mixed state*.

2.1.2 Quantum systems' evolution

The evolution of a quantum system describes how its state changes over time. For an *isolated (closed)* quantum system, this evolution is governed by *unitary* transformations. Specifically, if the system is in state $|\psi(t_1)\rangle$ at time t_1, then its state at a later time t_2 is given by

$$|\psi(t_2)\rangle = U(t_1, t_2)\,|\psi(t_1)\rangle,$$

where U is a unitary operator satisfying $U^{-1} = U^\dagger$. In the Schrödinger picture, time evolution is given by

$$i\hbar \frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle,$$

where $\hbar$ is the Planck constant, and H is a Hermitian operator known as the Hamiltonian of the system. The Hamiltonian encodes the physical properties of the system, such as its energy and interactions, and fully determines its dynamics. With a time-independent Hamiltonian, the system can be solved by $|\psi(t)\rangle = \exp\left(-iH(t-t_0)\right) = U(t, t_0)\,|\psi(t_0)\rangle$, where the unitary evolution operator

$$U(t_1, t_2) = \exp\left(-iH(t_2 - t_1)\right),$$

establishing the equivalence between the continuous-time description and the unitary evolution picture. For density operators, the evolution of a closed system is given by

$$\rho(t_2) = U(t_1, t_2)\,\rho(t_1)\,U^\dagger(t_1, t_2).$$

In practice, many quantum systems are not perfectly isolated and interact with their environment. Nevertheless, unitary evolution often provides a good approximation or an effective description, especially when considering controlled operations such as quantum gates.

2.1.3 Quantum measurement

Though quantum states can be in superposition, their properties can be experimentally verified via measurement outcomes, and the density operator formalism provides a consistent operational description. Within this formulation, the probability of obtaining measurement outcome m is given by

$$p(m) = \mathrm{Tr}(\rho O_m), \tag{7}$$

where $\{O_m\} = \{M_m^\dagger M_m\}$ denotes a set of measurement operators, following the completeness relation $\sum_m M_m^\dagger M_m = I$, ensuring that probabilities sum to one. For a pure state $|\psi\rangle$, the probability of obtaining measurement outcome m is

$$p(m) = \langle\psi|O_m|\psi\rangle,$$

and the post-measurement state $|\psi\rangle'$ becomes

$$|\psi\rangle' = \frac{M_m|\psi\rangle}{\sqrt{\langle\psi|M_m^\dagger M_m|\psi\rangle}} = \frac{M_m|\psi\rangle}{\sqrt{p(m)}}.$$

For a general mixed state described by a density operator ρ, the post-measurement state conditioned on obtaining outcome m is given by

$$\rho_m = \frac{M_m \rho M_m^\dagger}{\mathrm{Tr}(M_m^\dagger M_m \rho)} = \frac{M_m \rho M_m^\dagger}{p(m)}. \tag{8}$$

Computational basis measurements A simple example is the *measurement of a qubit in the computational basis* $\{|0\rangle, |1\rangle\}$, with measurement operators $M_0 = |0\rangle\langle 0|$ and $M_1 = |1\rangle\langle 1|$. For a pure state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, the outcomes occur with probabilities $|\alpha|^2$ and $|\beta|^2$, respectively, and the state collapses to the corresponding basis state.

Projective measurements An important special case of the general measurement is *projective measurements* (also known as von Neumann measurements). For a pure state

$|\psi\rangle$, the probability of obtaining outcome of eigenvalue m is

$$p(m) = \langle\psi|P_m|\psi\rangle,$$

where P_m are orthogonal projectors onto the eigenspaces corresponding to eigenvalues m. And the post-measurement state is

$$\frac{P_m|\psi\rangle}{\sqrt{p(m)}}.$$

Projective measurements satisfy $P_m^2 = P_m$, $P_m^\dagger = P_m$, and $P_m P_{m'} = \delta_{mm'} P_m$, together with the completeness relation $\sum_m P_m = I$. A projective measurement is associated with an observable M, which admits a spectral decomposition

$$M = \sum_m m P_m.$$

In particular, expectation values of observables take the form

$$\mathbb{E}(M) = \sum_m m\, p(m) = \langle\psi|M|\psi\rangle = \mathrm{Tr}(\rho M).$$

POVM measurements More general measurements are described by *positive operator-valued measures* (POVMs), where the POVM element is defined as

$$E_m = M_m^\dagger M_m,$$

with a POVM is defined by the complete set $\{E_m\}$, which are positive semidefinite operators satisfying the completeness relation $\sum_m E_m = I$. For a pure state $|\psi\rangle$, the probability of outcome m is then

$$p(m) = \langle\psi|E_m|\psi\rangle.$$

Unlike projective measurements, POVM elements are not required to be orthogonal projectors, and therefore allow a more general description of measurement statistics. In practice, POVMs provide a convenient framework when only measurement probabilities are of interest, while projective measurements offer a simpler and more intuitive description when the post-measurement state and repeatability are relevant. In particular, POVMs are essential when dealing with noisy systems, partial measurements, or situations where the measurement process is not repeatable.

2.1.4 Quantum resources for computational advantage

Quantum computational advantage is generally understood to arise from a combination of uniquely quantum features, most notably superposition, interference, and entanglement. While each of these can be described independently, it is their interplay that enables algorithms with no efficient classical counterpart.

Quantum superposition A pure qubit can exist in a superposition of computational basis states,

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1,$$

where α and β are complex probability amplitudes. Unlike classical probabilities, these amplitudes can interfere, leading to distinctly quantum effects. Upon measurement in the computational basis, the state yields outcomes $|0\rangle$ and $|1\rangle$ with probabilities $|\alpha|^2$ and $|\beta|^2$, respectively. However, before measurement, the state of the system is not classically definite, but it evolves deterministically under unitary dynamics.

Geometrically, a pure qubit corresponds to a point on the surface of the Bloch sphere, while a mixed state lies inside the Bloch sphere (see (Michael A. Nielsen et al. 2010b, p. 15)). This highlights that a qubit is described by continuous parameters; however, it does not imply that a qubit can store an arbitrary amount of classical information, as any measurement yields only a single classical bit. Instead, the advantage of superposition lies in enabling interference effects that can be exploited during computation.

Quantum interference The linearity of quantum evolution implies that operations act simultaneously on all components of a superposition, a feature often heuristically referred to as *quantum parallelism*. However, the computational advantage does not come solely from parallel evaluation, since measurement collapses the state.

The essential resource is interference. Because amplitudes are complex, different computational paths can combine, leading to constructive or destructive interference. The probability of an outcome depends on the squared magnitude of the sum of amplitudes, allowing algorithms to suppress incorrect results while amplifying correct ones. In this sense, quantum algorithms can be viewed as procedures for engineering interference patterns that encode global properties of a problem (Cleve et al. 1998).

Quantum entanglement Entanglement is a form of correlation with no classical analogue, arising in composite systems. A bipartite state is said to be entangled if it cannot be written as a convex combination of products of states of its subsystems. Such states exhibit correlations that cannot be explained by shared classical randomness alone.

Entanglement is widely regarded as a key resource for quantum advantage, particularly in multi-qubit systems (R. Horodecki et al. 2009). It enables non-classical correlations across subsystems and plays a central role in quantum communication, error correction, and many quantum algorithms (Jozsa et al. 2003). However, the precise role of entanglement remains subtle, and it is generally understood to act in conjunction with other resources such as interference and coherence.

2.1.5 Quantum computing models

2.1.5.1 Quantum circuits

The quantum circuit model provides a standard framework for describing quantum computation. In this model, computation is represented as a sequence of quantum gates acting on a collection of qubits. A quantum circuit is typically depicted as a set of horizontal wires, each corresponding to a qubit, with time progressing from left to right. Gates act on one or more qubits and are represented by unitary operators.

Mathematically, an n-qubit state is represented as a vector in a 2^n-dimensional Hilbert space, and quantum gates correspond to unitary transformations on this space. Some common quantum gates and their matrix representations are shown below in Table 2.1.

As an illustrative example of a quantum circuit, Figure 2.1 shows a two-qubit quantum circuit generating a *Bell state*.

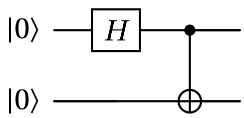


FIGURE 2.1 : A two-qubit quantum circuit generating a Bell state.

Starting from the initial state $|00\rangle$, the Hadamard gate H acting on the first qubit produces

$$|00\rangle \xrightarrow{H\otimes I} \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle).$$

Subsequently, applying a controlled-NOT (CNOT) gate yields

$$\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle),$$

which is a maximally entangled Bell state.

A key feature of quantum circuits is reversibility : all quantum gates must be unitary, and therefore invertible. As a consequence, operations such as copying or deleting quantum information are not allowed, reflecting fundamental constraints such as the no-cloning theorem. Measurement is treated as a separate operation, converting quantum information into classical outcomes according to the *Born rule*.

The quantum circuit model is known to be *universal* (YAO 1993), in the sense that a *universal quantum computer* (or quantum Turing machine) can simulate all the others with no more than polynomial overhead. Established within the circuit model, a *universal gate set* is a set of quantum gates such that any unitary operation can be approximated to arbitrary accuracy (in operator norm) by a finite sequence of gates drawn from the set, and the Solovay–Kitaev theorem guarantees that this approximation can be done efficiently (KITAEV 1997). In practice, universality is achieved by combining arbitrary single-qubit operations with at least one entangling two-qubit gate, such as the controlled-NOT (CNOT). This reflects the fundamental structure of quantum computation : local operations together with entanglement generation are sufficient to implement any quantum process. Below, we introduce several universal gate sets that are commonly used in the literature.

Clifford + T gate set One of the most widely used universal gate sets is $\{H, S, T, \text{CNOT}\}$. The subset $\{H, S, \text{CNOT}\}$, known as the Clifford group, is not universal by itself and

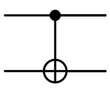
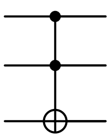
Operator	**Gate**	**Matrix**
Pauli-X	$\boxed{X}$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
Pauli-Y	$\boxed{Y}$	$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$
Pauli-Z	$\boxed{Z}$	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Hadamard H	$\boxed{H}$	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$
Phase $S = \sqrt{Z}$	$\boxed{S}$	$\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$
$T = \sqrt{S}$ gate	$\boxed{T}$	$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$
$R_x(\theta)$	$\boxed{R_x(\theta)}$	$\begin{pmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$
$R_y(\theta)$	$\boxed{R_y(\theta)}$	$\begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$
$R_z(\theta)$	$\boxed{R_z(\theta)}$	$\begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}$
CNOT		$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$
Toffoli		$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$

TABLE 2.1 : Common quantum gates.

can be efficiently simulated classically (Gottesman–Knill theorem). The addition of the non-Clifford T gate promotes the set to universality.

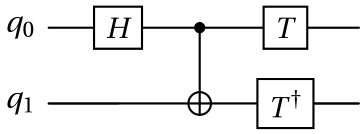


FIGURE 2.2 : A representative circuit from Clifford + T universal gate set.

Single-qubit rotations + CNOT gate set Universality can also be achieved using continuous single-qubit rotations together with an entangling gate. A common choice consists of rotation operators $R_x(\theta)$, $R_y(\theta)$, $R_z(\theta)$ or equivalently phase shift gates $P(\varphi)$ combined with CNOT. This formulation is particularly natural in physical implementations where Hamiltonian dynamics generate continuous rotations.

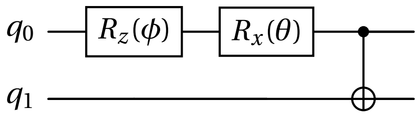


FIGURE 2.3 : A representative circuit from single-qubit rotations + CNOT universal gate set.

H + Toffoli gate set Another universal construction is given by the Toffoli (CCNOT) gate together with the Hadamard gate. The Toffoli gate alone is universal for reversible classical computation, and the addition of the Hadamard gate extends this to universal quantum computation.

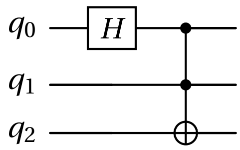


FIGURE 2.4 : A representative circuit from the Hadamard +Toffoli universal gate set.

Remarks Although different universal gate sets exist, they are computationally equivalent : any quantum circuit expressed in one set can be efficiently translated into another, only with a polynomial increase in resources. In practice, the choice of gate set depends on the physical platform and error-correction considerations. Discrete gate sets such as Clifford+T are particularly important for fault-tolerant quantum computation, while continuous rotation-based sets are more natural in analogue or NISQ devices.

2.1.5.2 Alternative models of quantum computation

While the circuit model is the most widely used, several alternative models of quantum computation have been developed. These models are computationally equivalent in the sense that they can efficiently simulate one another, but they offer different perspectives on the origin of quantum advantage.

Measurement-based quantum computation In measurement-based quantum computation (MBQC), computation is driven by measurements on a highly entangled resource state, such as a cluster state. Instead of applying a sequence of unitary gates, the computation proceeds via adaptive measurements, with classical feedforward determining future measurement settings. Entanglement is prepared at the outset and then consumed during the computation (Raussendorf et al. 2003).

Adiabatic quantum computation Adiabatic quantum computation encodes the solution to a problem in the ground state of a Hamiltonian. The system is initialised in the ground state of a simple Hamiltonian and then slowly evolved into a final Hamiltonian whose ground state encodes the desired solution. Under suitable conditions, the adiabatic theorem guarantees that the system remains close to the ground state throughout the evolution (Aharonov, Van Dam et al. 2008).

Quantum walk-based computation Quantum walks provide another model of quantum computation, generalising classical random walks by allowing coherent superpositions over paths. There are two main types : continuous-time (Childs 2009) and discrete-time (Lovett et al. 2010) quantum walks. In both cases, it has been shown that quantum walk models are computationally universal, meaning that they can efficiently simulate any quantum circuit.

The (continuous/discrete-time) quantum walk evolution is governed by unitary dynamics, leading to interference effects that can significantly alter the spreading behaviour compared to classical walks. Quantum walks have been used to design algorithms with provable speedups, particularly for search and graph-related problems.

Equivalence of models Despite their differing formulations, these models are computationally equivalent : any computation performed in one model can be mapped to another with no more than polynomial overhead. This equivalence reinforces the idea that quantum computational advantage arises from underlying physical principles—such as superposition, interference, and entanglement—rather than from a particular computational framework.

2.1.6 Quantum algorithms

Oracle models Quantum algorithms exploit superposition, interference, and entanglement to achieve computational advantages over classical approaches. Many of the earliest examples are naturally formulated in the *oracle model*, where access to a function is provided through a *black-box* unitary operator, where one can only query it with inputs and get outputs.

Formally, for a Boolean function $f : \{0,1\}^n \to \{0,1\}$, then an oracle is defined as a unitary operator U_f acting on $n+1$ qubits as

$$U_f \,|x\rangle\,|y\rangle = |x\rangle\,|y \oplus f(x)\rangle\,, \tag{9}$$

where $\oplus$ denotes addition modulo 2. An equivalent phase-oracle representation can be obtained by preparing the second register in the state $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, yielding

$$U_f \,|x\rangle\,|-\rangle = (-1)^{f(x)}\,|x\rangle\,|-\rangle\,, \tag{10}$$

which induces the effective transformation $|x\rangle \mapsto (-1)^{f(x)}\,|x\rangle$

In the query (black-box) model, the computational cost of an algorithm is measured by the number of oracle queries it makes. It is very useful for isolating the role of quantum resources, as both classical and quantum algorithms are given the same oracle access, performance differences should therefore be attributed to quantum features such as superposition and interference.

Deutsch-Jozsa algorithm The Deutsch-Jozsa algorithm (Deutsch et Jozsa 1992) is one of the earliest examples of a quantum algorithm in the oracle model. The problem is defined as follows : given a Boolean function $f : \{0,1\}^n \to \{0,1\}$, promised to be either constant (all 0 or all 1 for all inputs) or balanced (0 on exactly half of the inputs, and on the other half 1), determine which is the case. Classically, one needs to make $O(2^n)$ queries to the function f to determine it with no error, but the Deutsch-Jozsa algorithm shows that it is possible to only query a quantum oracle U_f once, which is exponentially faster than any possible deterministic classical algorithm. In the following, we briefly introduce this algorithm.

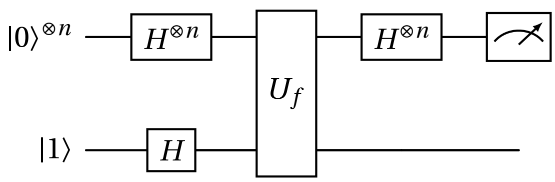


Figure 2.5 : Quantum circuit for the Deutsch-Jozsa algorithm.

The algorithm uses $n+1$ qubits initialised in the state

$$|\psi_0\rangle = |0\rangle^{\otimes n}\,|1\rangle\,.$$

Applying Hadamard gates to all qubits yields

$$|\psi_0\rangle \xrightarrow{H^{\otimes n}\otimes H} |\psi_1\rangle = \frac{1}{\sqrt{2^{n+1}}}\sum_{x=0}^{2^n-1} |x\rangle\,(|0\rangle - |1\rangle).$$

Applying the oracle gives

$$|\psi_1\rangle \xrightarrow{U_f} |\psi_2\rangle = \frac{1}{\sqrt{2^{n+1}}}\sum_{x=0}^{2^n-1} (-1)^{f(x)}\,|x\rangle\,(|0\rangle - |1\rangle).$$

Finally, applying Hadamard gates to the first n qubits yields

$$|\psi_2\rangle \xrightarrow{H^{\otimes n}\otimes I} |\psi_3\rangle = \frac{1}{2^n}\sum_{y=0}^{2^n-1}\left[\sum_{x=0}^{2^n-1}(-1)^{f(x)}(-1)^{x\cdot y}\right]|y\rangle\,.$$

In particular, the probability of measuring $|0\rangle^{\otimes n}$ is

$$\left|\frac{1}{2^n}\sum_{x=0}^{2^n-1}(-1)^{f(x)}\right|^2,$$

which equals 1 if f is constant and 0 if f is balanced. Thus, a single oracle query suffices to determine the nature of f with certainty. In contrast, any deterministic classical algorithm requires $2^{n-1}+1$ queries in the worst case. However, this separation does not extend to the bounded-error classical setting, where a probabilistic algorithm can solve the problem with high probability using a constant number of queries. Nevertheless, the Deutsch-Jozsa algorithm illustrates how quantum interference can be used to extract global properties of a function.

Grover's search algorithm Another important example of a quantum algorithm in the oracle model is *Grover's search* algorithm (Lov K. Grover 1996). The problem is defined as follows : given a function $f:\{0,1\}^n \to \{0,1\}$ marking M solutions among $N=2^n$ possible inputs, find an x such that $f(x)=1$.

As in the Deutsch-Jozsa algorithm, the oracle can be written in phase form as

$$O\,|x\rangle = (-1)^{f(x)}\,|x\rangle\,, \tag{11}$$

such that solutions are marked by a phase flip. The algorithm starts by preparing the uniform superposition

$$|\psi\rangle = \frac{1}{\sqrt{N}}\sum_{x=0}^{N-1}|x\rangle\,,$$

and then repeatedly applies the Grover operator

$$G = (2\,|\psi\rangle\langle\psi| - I)\,O, \tag{12}$$

which consists of the oracle followed by a reflection $R_\psi = 2|\psi\rangle\langle\psi| - I$ about $|\psi\rangle$. The reflection operator is also sometimes called Grover's diffusion operator D with entries $D_{ij} = 2/N - 1$ for $i = j$, and $D_{ij} = 2/N$ for $i \neq j$.

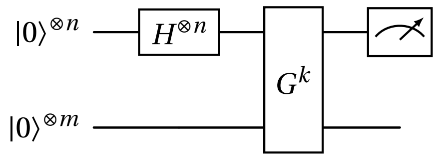


FIGURE 2.6 : Quantum circuit with k Grover iterations.

To describe the evolution, define the orthonormal states

$$|\alpha\rangle = \frac{1}{\sqrt{M}} \sum_{x:\, f(x)=1} |x\rangle, \quad |\beta\rangle = \frac{1}{\sqrt{N-M}} \sum_{x:\, f(x)=0} |x\rangle .$$

The initial state can be written as

$$|\psi\rangle = \sqrt{\frac{M}{N}}\, |\alpha\rangle + \sqrt{\frac{N-M}{N}}\, |\beta\rangle = \sin\theta\, |\alpha\rangle + \cos\theta\, |\beta\rangle ,$$

where $\sin(\theta) = \sqrt{M/N}$.

Each application of G corresponds to a rotation by an angle θ in the two-dimensional subspace spanned by $\{|\alpha\rangle, |\beta\rangle\}$. After k iterations, the state becomes

$$G^k |\psi\rangle = \sin((2k+1)\theta)\, |\alpha\rangle + \cos((2k+1)\theta)\, |\beta\rangle .$$

After $R = O\left(\sqrt{\frac{N}{M}}\right)$ iterations and measuring the final state, it yields a solution with high probability, giving a quadratic speedup over classical search.

Quantum complexity perspective Quantum complexity theory studies the computational power of quantum computers in terms of resources such as time and query complexity. A central class is *bounded-error quantum polynomial time* (BQP), consisting of decision problems solvable by a quantum computer in polynomial time with bounded error probability.

Oracle problems provide a useful framework for comparing classical and quantum models. While the Deutsch-Jozsa algorithm demonstrates a separation in deterministic query complexity, stronger results are obtained for problems such as Simon's problem, which exhibits an exponential separation between classical and quantum algorithms even in the bounded-error setting, yielding an oracle separation between BQP and the bounded-error probabilistic polynomial time (BPP) complexity classes. Meanwhile, the Grover's algorithm, although it only provides a quadratic speedup, is versatile to a wide range of problems related to unstructured search.

These results indicate that quantum computational advantage arises from the ability to coherently query and interfere over many inputs simultaneously. However, the precise relationship between classical and quantum complexity classes remains an open question in general. In fact, it is known that BPP ⊆ BQP, since quantum computers can efficiently simulate classical probabilistic computation. However, whether this inclusion is strict remains an open problem. On the other hand, the relationship between BQP and NP remains unknown (Arora et al. 2009).

2.2 Machine learning

2.2.1 PAC learning framework

In computational learning theory, Probably Approximately Correct (PAC) learning provides a general framework for reasoning about when learning from data is possible. Introduced by Valiant 1984, it captures the basic requirement that a learner, given a finite sample, should select a hypothesis from a prescribed class that generalises well beyond the observed data.

We begin by introducing the basic notation used in the PAC learning model, and we leave the formal definition of PAC learning to a later Section 2.2.4.1. Let $\mathscr{X}$ denote the input space and $\mathscr{Y} = \{0,1\}$ the label space, corresponding to binary classification. A *concept* is a mapping $c : \mathscr{X} \to \mathscr{Y}$, and a *concept class* $\mathscr{C}$ is a set of such functions.

We assume that examples are drawn independently and identically distributed (i.i.d.) according to an unknown distribution $\mathscr{D}$ over $\mathscr{X}$. The learner is given a sample $S = (x_1,\dots,x_N) \sim \mathscr{D}^N$ together with labels $(c(x_1),\dots,c(x_N))$, and selects a hypothesis h_S from a hypothesis class $\mathscr{H}$.

Definition 2.2.1 (Generalization error)**.** *(Mohri et al. 2018, p. 10) Given a hypothesis $h \in \mathscr{H}$, a target concept $c \in \mathscr{C}$, and a distribution $\mathscr{D}$, the generalization error (or risk) is defined as*

$$R(h) = \mathbb{P}_{x\sim\mathscr{D}}\big[h(x) \neq c(x)\big]. \tag{13}$$

Definition 2.2.2 (Empirical error)**.** *(ibid., p. 10) Given a sample $S = (x_1,\dots,x_m)$, the empirical error of $h \in \mathscr{H}$ is*

$$\widehat{R}_S(h) = \frac{1}{m}\sum_{i=1}^{m} \mathbf{1}_{h(x_i)\neq c(x_i)}. \tag{14}$$

2.2.2 Classification

Training sample As was introduced in Section 1.1.1, a main task in supervised ML is classification, and depending on the classes of labels, it can be binary or multi-class. We begin by introducing the binary classification setting, which is also the main target for this thesis.

Let $\mathscr{X}$ denote the input space and $\mathscr{Y} = \{-1,+1\}$ the label space. A dataset consists of pairs $(\boldsymbol{x}, y) \in \mathscr{X} \times \mathscr{Y}$. We assume that data are generated according to an unknown distribution $\mathscr{D}$ over $\mathscr{X} \times \mathscr{Y}$. A training sample $\mathscr{S} = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N) \sim \mathscr{D}^N$ is a sequence consisting of N i.i.d examples. When the order of the sequence does not matter, we use the notation $\mathscr{S} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^N \sim \mathscr{D}^N$ to refer to a training set.

In this thesis, we focus on a deterministic setup, rather than a stochastic setup, where the labels $y_i = f(\boldsymbol{x}_i)$ are uniquely determined by some measurable function f with probability one. In the following Section 2.2.2 and Section 2.2.3, we greatly adapt the notations and definitions from (Mohri et al. 2018, Chapter 5).

Linear classification A natural choice of hypothesis class is the set of linear classifiers

$$\mathscr{H} = \{\boldsymbol{x} \mapsto \operatorname{sgn}(\boldsymbol{w}^T \boldsymbol{x} + b) : \boldsymbol{w} \in \mathbb{R}^{D'},\ b \in \mathbb{R}\}. \tag{15}$$

The decision boundary is the hyperplane $\boldsymbol{w}^T \boldsymbol{x} + b = 0$, where $\boldsymbol{w}$ is the normal vector and b is the intercept. As a standard simplification, the intercept term can be absorbed into the parameter vector by augmenting the input space, yielding classifiers of the form

$$h(\boldsymbol{x}) = \operatorname{sgn}(\boldsymbol{w}^T \boldsymbol{x}), \quad \boldsymbol{w} \in \mathbb{R}^D, \tag{16}$$

where $D = D' + 1$. In the linearly separable case, there exists $\boldsymbol{w}$ [1] such that $y_i \boldsymbol{w}^T \boldsymbol{x}_i > 0$ [2], for all $i \in [N]$. However, such a separating hyperplane is not unique, and infinitely many solutions may exist. To distinguish among them, it is necessary to introduce a measure of separation. An important quantity for this purpose is the *geometric margin*.

Definition 2.2.3. *The geometric margin of a linear classifier h at a point x is defined as the Euclidean distance from x to the decision boundary :*

$$\rho_h(\boldsymbol{x}) = \frac{|\boldsymbol{w}^T \boldsymbol{x}|}{\|\boldsymbol{w}\|}. \tag{17}$$

For a labelled sample $S = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$, where $y_i \in \{-1, +1\}$, the geometric margin of the classifier *over the sample* is defined as the minimum margin over all training points :

$$\rho_h = \min_{i \in [N]} \frac{y_i(\boldsymbol{w}^T \boldsymbol{x}_i)}{\|\boldsymbol{w}\|}. \tag{18}$$

This quantity corresponds to the distance between the decision boundary and the closest point in the training set. It therefore captures the robustness of the classifier : a larger margin implies that all training points lie farther from the decision boundary, making the classifier less sensitive to perturbations or noise in the input.

Maximum-margin principle In the linearly separable case, there exist infinitely many hyperplanes that perfectly separate the data. One may consider selecting the hyperplane

1. Throughout this thesis, unless otherwise stated, we refer to the hyperplane $\boldsymbol{w}^T \boldsymbol{x} = 0$ by its normal vector $\boldsymbol{w}$.
2. We adapt the convention that $\operatorname{sgn}(x) = 1$, if $x > 0$; and $\operatorname{sgn}(x) = -1$, if $x \le 0$.

that maximises the geometric margin :

$$\max_{\boldsymbol{w}} \min_{i \in [N]} \frac{y_i(\boldsymbol{w}^T \boldsymbol{x}_i)}{\|\boldsymbol{w}\|}. \tag{19}$$

Equivalently, this optimisation problem can be reformulated as

$$\min_{\boldsymbol{w}} \frac{1}{2}\|\boldsymbol{w}\|^2 \quad \text{subject to} \quad y_i(\boldsymbol{w}^T \boldsymbol{x}_i) \geq 1, \ \forall i \in [N]. \tag{20}$$

The solution $\boldsymbol{u}$ to this problem is known as the *maximum-margin hyperplane*, also the *hard-margin SVM* solution (to be discussed in Section 2.2.3).

The geometric margin provides a direct geometric interpretation of classification confidence : it measures how far a point is from the decision boundary in the hyperplane's normal direction. Maximising this margin leads to classifiers that are more robust and often exhibit better generalisation performance.

2.2.3 Binary classification models

2.2.3.1 Online perceptron algorithm

Now, given N normalized linearly separable training examples $\{\boldsymbol{x}_1, .., \boldsymbol{x}_N\} \in \mathbb{R}^D$, $||\boldsymbol{x}_i|| = 1$,[3] with corresponding labels $\{y_1, .., y_N\}, y_i \in \{+1, -1\}$. The perceptron itself is a weight vector (hyperplane) $\boldsymbol{w}$ that follows a prediction rule $\hat{y}_i = \text{sgn}(\boldsymbol{w}^T \boldsymbol{x}_i)$ for every unlabelled instance $\boldsymbol{x}_i$, as in Eqn. (16) .

Perceptron learning aims to find a hyperplane $\boldsymbol{w}^T \boldsymbol{x}_i y_i > 0$, $\forall i \in [N]$, such that all training examples are classified correctly (zero empirical error $\widehat{R}_S$, see Eqn.(13)). The online perceptron algorithm (Rosenblatt 1958) is a foundational online learning method introduced in the early 1960s. At each round, the model maintains a weight vector $\boldsymbol{w}_t$, makes a prediction based on the rule, and updates $\boldsymbol{w}_t$ upon misclassification. Such that $\boldsymbol{w}_{t+1} \leftarrow \boldsymbol{w}_t + y'_i \boldsymbol{x}'_i$, where $(\boldsymbol{x}'_i, y'_i)$ is a training example pair misclassified by $\boldsymbol{w}_t$.

It is well known that if the training sample is linearly separable by an SVM solution $\boldsymbol{u}$ with a maximum-margin γ, such that $\boldsymbol{u}^T \boldsymbol{x}_i y_i \geq \gamma$ for all $i \in [N]$. The online perceptron algorithm converges after at most $O(1/\gamma^2)$ updates, yielding a linear classifier that correctly classifies all N training points (Novikoff 1962). Geometrically, the solution lies in the *version space*, defined as $\mathcal{VS} := \{\boldsymbol{w} \mid \boldsymbol{w}^T \boldsymbol{x}_i y_i > 0, \forall i \in [N]\}$. As only directions of $\boldsymbol{w}$ matter for classification, we can limit the length of $||\boldsymbol{w}|| \leq 1$, the $\mathcal{VS}$ is then a bounded polyhedron (a polytope) satisfying N half-space constraints, also known as a convex *feasible set* in the context of linear programming. Notably, the solution to the hard-margin SVM problem corresponds to the centre of the largest inscribed sphere within $\mathcal{VS}$ as pointed out by Tong et al. 2001.

Notice that the solution of a classical online perceptron algorithm converges to an arbitrary point $\boldsymbol{w}_r \in \mathcal{VS}$, so the problem of perceptron learning can be reinterpreted as a feasibility problem through the lens of linear programming (Grötschel et al. 2012;

3. We use $||\cdot||$ to represent the ℓ_2 norm throughout the thesis.

Nemirovski 1994). Given this, we point out that general feasibility algorithms can also serve as primitive approaches to address the perceptron learning problem with high-dimensional data, such as the ellipsoid method (Khachiyan 1979 ; L. Yang et al. 2009), and various cutting-plane methods (P. Vaidya 1989 ; Atkinson et al. 1995 ; Bertsimas et al. 2004 ; Louche et al. 2015 ; Lee, Sidford et Wong 2015). A detailed summary can be found in a recent paper of Jiang et al. 2020, Table 1.

2.2.3.2 Hard-margin SVMs

The hard-margin SVMs are a class of supervised learning algorithms that construct a linear classifier by maximising the geometric margin between two classes. By appropriate scaling, this problem can be equivalently written as a convex optimisation problem of Eqn. (20), which is known as the *primal optimisation problem* of SVM. The objective function is strictly convex, and the constraints are affine, ensuring the existence of a unique global optimum. And notably, the primal SVM problem is indeed a convex quadratic programming (QP) instance, which could be solved by the block coordinate descent method.

Support vectors From the Karush-Kuhn-Tucker (KKT) conditions, the solution admits the representation

$$\boldsymbol{w} = \sum_{i=1}^{N} \alpha_i y_i \boldsymbol{x}_i,$$

where $\alpha_i \geq 0$ are Lagrange multipliers. Training points with $\alpha_i > 0$ are called *support vectors* and lie on the margin boundaries,

$$y_i(\boldsymbol{w}^T \boldsymbol{x}_i) = 1.$$

These points uniquely determine the decision boundary, while all other points do not influence the solution.

Dual formulation By introducing Lagrange multipliers and eliminating the primal variables, the SVM problem can be expressed in its dual form :

$$\max_{\alpha} \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{N} \alpha_i \alpha_j y_i y_j (x_i \cdot x_j),$$

subject to

$$\alpha_i \geq 0, \quad \sum_{i=1}^{N} \alpha_i y_i = 0.$$

Notice that the dual problem depends only on inner products between data points, and this forms the basis of kernel methods.

2.2.3.3 Soft-margin SVM

In practical settings, data are rarely linearly separable. In such cases, the hard-margin constraints $y_i(\boldsymbol{w}^T\boldsymbol{x}_i) \geq 1$, for all $i \in [N]$ cannot be satisfied simultaneously. To address this, the constraints are relaxed by introducing non-negative slack variables $\xi_i \geq 0$,

$$y_i(\boldsymbol{w}^T\boldsymbol{x}_i + b) \geq 1 - \xi_i, \quad \forall i \in [N].$$

Each slack variable ξ_i quantifies the extent to which the corresponding sample violates the margin constraint. In particular, points with $\xi_i > 0$ either lie within the margin or are misclassified. Thus, the slack variables provide a measure of deviation from ideal separability.

This relaxation introduces a trade-off between two competing objectives : on the one hand, maximising the margin (which corresponds to minimising $\|\boldsymbol{w}\|^2$), and on the other hand, penalising violations of the margin constraints. This leads to the *soft-margin SVM* formulation

$$\min_{\boldsymbol{w},b,\boldsymbol{\xi}} \quad \frac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_{i=1}^{N}\xi_i \quad \text{subject to} \quad y_i(\boldsymbol{w}^T\boldsymbol{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \tag{21}$$

where $C > 0$ is a regularisation parameter controlling the balance between margin maximisation and constraint violation.

As in the separable case, this is a convex optimisation problem : the objective function is strictly convex, and the constraints are affine. Consequently, the problem admits a unique global minimiser. The optimisation problem can be further analysed via the KKT conditions, leading to an equivalent dual formulation in which the solution depends only on inner products between training points.

2.2.3.4 Kernel methods

In many applications, the data are not linearly separable in the original input space. One approach is to map the data to a higher-dimensional feature space where linear separation becomes possible. Let $\Phi : \mathscr{X} \to \mathscr{H}$ denote such a mapping into a (possibly high-dimensional) Hilbert space.

A key observation is that, in the dual formulation of SVMs, the classifier depends only on inner products of training samples $\boldsymbol{x}_i \cdot \boldsymbol{x}_j$. This allows one to replace these inner products with a kernel function

$$K(\boldsymbol{x}, \boldsymbol{x}') = \langle \Phi(\boldsymbol{x}), \Phi(\boldsymbol{x}') \rangle, \tag{22}$$

without explicitly computing the mapping Φ. This technique is known as the *kernel trick*, and enables SVMs to learn *non-linear* decision boundaries in the original input space. Common choices of kernels include :

Polynomial kernel

$$K(\boldsymbol{x},\boldsymbol{x}') = (\boldsymbol{x}\cdot\boldsymbol{x}' + c)^d, \tag{23}$$

which corresponds to a mapping into a space of polynomial features up to degree d.

Gaussian (RBF) kernel

$$K(\boldsymbol{x},\boldsymbol{x}') = \exp\left(-\frac{\|\boldsymbol{x}-\boldsymbol{x}'\|^2}{2\sigma^2}\right), \tag{24}$$

which induces an infinite-dimensional feature space and yields highly flexible non-linear decision boundaries. Recently, quantum kernels have also been proposed (Schuld et Killoran 2019), where classical data are encoded into quantum states via a feature map (quantum circuit), and kernel values are obtained by estimating inner products between these quantum states. This allows one to implicitly work in a high-dimensional Hilbert space that may be difficult to access classically.

By replacing inner products with kernel functions, SVMs can be extended to non-linear classification while retaining the same optimisation framework.

2.2.4 Generalization performance

In statistical learning theory, the framework of *probably approximately correct* (PAC) learning formalises the notion of learnability under uncertainty. In the following, we introduce the PAC learning framework, and a few well-known generalisation bounds have been proven. Our definitions and notations follow the book of Mohri et al. 2018, and readers may look for detailed proofs of the theorems below.

2.2.4.1 PAC learning and generalization bounds

Definition 2.2.4 (PAC learning)**.** *(ibid., p. 11) A concept class $\mathscr{C}$ is said to be PAC-learnable if there exists an algorithm $\mathscr{A}$ and a polynomial function* poly$(\cdot)$ *such that for any $\epsilon,\delta > 0$, for any distribution $\mathscr{D}$ over $\mathscr{X}$, and for any target concept $c \in \mathscr{C}$, the following holds. If $\mathscr{A}$ is given a sample $S \sim \mathscr{D}^N$ of size*

$$N \geq \mathrm{poly}\big(1/\epsilon, 1/\delta, n, \mathrm{size}(c)\big),$$

then with probability at least $1-\delta$,

$$\Pr_{S\sim\mathscr{D}^N}\left[R(h_S) \leq \epsilon\right] \geq 1-\delta,$$

where $h_S = \mathscr{A}(S)$ and $R(h)$ denotes the generalization error. If, in addition, $\mathscr{A}$ runs in polynomial time, then $\mathscr{C}$ is said to be efficiently PAC-learnable.

Theorem 1 (Finite $\mathscr{H}$, consistent case)**.** *(ibid., p. 15) Let $\mathscr{H}$ be a finite hypothesis class and assume that the learning algorithm returns a hypothesis h_S such that $\widehat{R}_S(h_S) = 0$. Then, for any $\epsilon,\delta > 0$,*

$$\Pr_{S\sim\mathscr{D}^N}\left[R(h_S) \leq \epsilon\right] \geq 1-\delta$$

provided that

$$N \geq \frac{1}{\epsilon}\left(\log|\mathscr{H}| + \log\frac{1}{\delta}\right).$$

Equivalently, with probability at least $1-\delta$,

$$R(h_S) \leq \frac{1}{N}\left(\log|\mathscr{H}| + \log\frac{1}{\delta}\right). \tag{25}$$

Theorem 2 (Finite $\mathscr{H}$, inconsistent case)**.** *(Mohri et al. 2018, p. 20) Let $\mathscr{H}$ be a finite hypothesis class. Then, for any $\delta > 0$, with probability at least $1-\delta$, the following holds :*

$$\forall h \in \mathscr{H}, \quad R(h) \leq \widehat{R}_S(h) + \sqrt{\frac{\log|\mathscr{H}| + \log\frac{2}{\delta}}{2N}}. \tag{26}$$

This bound shows that the true risk is controlled by the empirical risk plus a complexity term that decreases as $O(1/\sqrt{m})$. Unlike the consistent case, it applies to arbitrary hypotheses and does not require $\widehat{R}_S(h) = 0$.

2.2.4.2 VC-dimension and generalization bounds

When the hypothesis class $\mathscr{H}$ is infinite, its cardinality is no longer a meaningful measure of complexity. Instead, a key quantity is the Vapnik-Chervonenkis (VC) dimension, which captures the expressive power of $\mathscr{H}$ in a combinatorial way.

Definition 2.2.5 (Shattering)**.** *(ibid., p. 36) A set of points $S = \{x_1, \ldots, x_N\} \subset \mathscr{X}$ is said to be shattered by a hypothesis class $\mathscr{H}$ if for all possible labellings $(y_1, \ldots, y_N) \in \{0,1\}^N$, there exists a hypothesis $h \in \mathscr{H}$ such that $h(x_i) = y_i$ for all i.*

Definition 2.2.6 (VC-dimension)**.** *(ibid., p. 36) The VC-dimension of a hypothesis class $\mathscr{H}$ is defined as*

$$\mathrm{VCdim}(\mathscr{H}) = \max\{m : \mathscr{H} \textit{ shatters a set of size } m\}. \tag{27}$$

The VC-dimension provides a measure of model complexity that applies to both finite and infinite hypothesis classes. For example, the class of hyperplanes in $\mathbb{R}^D$ has VC-dimension $D+1$.

Theorem 3 (VC-dimension generalization bound)**.** *(ibid., p. 43) Let $\mathscr{H}$ be a hypothesis class with VC-dimension D. Then, for any $\delta > 0$, with probability at least $1-\delta$, the following holds for all $h \in \mathscr{H}$:*

$$R(h) \leq \widehat{R}_S(h) + \sqrt{\frac{2D\log(\frac{eN}{D})}{N}} + \sqrt{\frac{\log(1/\delta)}{2N}}. \tag{28}$$

This bound shows that the generalisation error depends on the ratio D/N, highlighting that good generalisation requires the sample size N to be large compared to the complexity D of the hypothesis class.

Theorem 4 (VC-dimension bound, realizable case)**.** *(Mohri et al. 2018, p. 57) Assume the realisable setting, i.e. there exists $h^\star \in \mathcal{H}$ such that $R(h^\star) = 0$, and the algorithm returns a consistent hypothesis h_S with $\widehat{R}_S(h_S) = 0$. Then, with high probability,*

$$R(h_S) = O\left(\frac{D\log(N/D)}{N}\right). \tag{29}$$

2.2.4.3 Leave-one-out analysis

The leave-one-out (LOO) error provides a useful tool to analyse the generalisation performance of learning algorithms. It is defined by removing each training point in turn, retraining the model, and evaluating the error on the omitted point.

Definition 2.2.7 (Leave-one-out error)**.** *(ibid., p. 85) Let h_S denote the hypothesis returned by a learning algorithm $\mathscr{A}$ trained on a sample S of size N. The leave-one-out (LOO) error is defined as*

$$\widehat{R}_{\mathrm{LOO}}(\mathscr{A}) = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}_{h_{S\setminus\{x_i\}}(x_i)\neq y_i}. \tag{30}$$

Lemma 5 (Unbiasedness of LOO error)**.** *The expected leave-one-out error is an unbiased estimate of the expected generalisation error on a sample of size $N-1$:*

$$\mathbb{E}_{S\sim\mathscr{D}^N}[\widehat{R}_{\mathrm{LOO}}(\mathscr{A})] = \mathbb{E}_{S'\sim\mathscr{D}^{N-1}}[R(h_{S'})].$$

This connection allows one to derive generalisation bounds based on algorithm-specific properties.

Theorem 6 (SVM leave-one-out bound)**.** *(ibid., p. 86) For support vector machines, the LOO error can be controlled by the number of support vectors. In particular, let h_S be the hypothesis returned by an SVM trained on a sample S. Then,*

$$\mathbb{E}_{S\sim\mathscr{D}^N}[R(h_S)] \leq \mathbb{E}_{S\sim\mathscr{D}^{N+1}}\left[\frac{N_{\mathrm{SV}}(S)}{N+1}\right], \tag{31}$$

where $N_{\mathrm{SV}}(S)$ denotes the number of support vectors.

This bound highlights that for SVMs, the generalisation error is controlled by the fraction of training points that define the decision boundary.

Theorem 7 (Perceptron leave-one-out bound)**.** *(ibid., p. 193) Assume the realisable setting, i.e. the data is linearly separable. Let h_S be the hypothesis returned by the perceptron algorithm. Then,*

$$\mathbb{E}_{S\sim\mathscr{D}^N}[R(h_S)] \leq \mathbb{E}_{S\sim\mathscr{D}^{N+1}}\left[\min\left(\frac{M(S), r_S^2/\gamma_S^2}{N+1}\right)\right], \tag{32}$$

where $M(S)$ is the number of updates, r_S the radius of the data, and γ_S is the geometric margin.

In the linear separable case, the number of updates is bounded by $M(S) \leq r_S^2/\gamma_S^2$, which yields a margin-based generalisation bound.

Summary LOO analysis provides a simple and interpretable way to relate generalisation performance to structural properties of the learned model, such as the number of support vectors for SVMs and geometric margin or update complexity for the perceptron.

3 On quantum perceptron learning via quantum search

Sommaire

3.1 Introduction

With the growing interest in quantum machine learning, the perceptron, a fundamental building block in traditional machine learning, has emerged as a valuable model for exploring quantum advantages. In this work, we make two principal contributions.

First, we revisit the *quantum version space perceptron* (QVSP) algorithm proposed by Kapoor et al. 2016, identifying and correcting a flawed complexity assumption. We prove that the complexity of the algorithm is dimension-dependent, which has significant implications for the feasibility of the sampling-based quantum perceptron algorithm in high-dimensional regimes.

Second, we propose and analyse two *quantum-enhanced* cutting-plane algorithms for perceptron learning. Specifically, we leverage established quantum subroutines such as *Grover's search* (see Figure 2.6) and *quantum walk search*, and provide detailed algorithmic constructions together with query and arithmetic complexity analyses. Our results establish improved complexity bounds under an idealised implementation framework and noise-free quantum computational models, offering insights into the trade-offs between margin dependence, dimensional dependence, and quantum resources. These findings provide a refined understanding of quantum perceptron models and their theoretical computational complexity properties.

3.1.1 Motivation and contributions

In supervised machine learning, classification requires algorithms to give the correct label to unseen (test) data based on knowledge recognised from labelled training data. The simplest task in classification is likely binary classification on linearly separable data, which can be solved by the *online perceptron algorithm* (Rosenblatt 1958), with a tight update bound that only depends on the geometric margin of data, proved by Novikoff 1962; Block 1962. Although other classification methods, such as *support vector machine* (SVM), which relies on advanced optimisation techniques to find optimal separating hyperplanes, the online perceptron algorithm uses a simple, iterative update rule and serves as the basic building unit of *multilayer perceptron* (MLP), i.e. the single-layer perceptron, which is also a minimal *feedforward neural network* (FNN). Due to its elementariness and simplicity, it is considered an ideal theoretical lens to study statistical and computational trade-offs of quantum machine-learning algorithms, as pointed out by Kapoor et al. 2016. However, prior quantum perceptron models, particularly the QVSP algorithm introduced by ibid., rest on an assumption about margin-based sampling (in their proof of Theorem 2), which is theoretically flawed in higher feature space dimensions $D > 2$. This raises the critical question of exploring high-dimensional available quantum perceptron algorithms.

In this Chapter, we make two principal contributions. First, we revisit the QVSP algorithm proposed by ibid. and prove that the probability of sampling a D-dimensional hyperplane from a standard normal distribution that perfectly classifies the data scales as $\Omega(\gamma^D)$ in the worst case, rather than $\Theta(\gamma)$, where γ is the geometric margin. This correction is based on a critical assumption of ibid. and extends to subsequent works in the quantum perceptron literature (Roget et al. 2022; Liao et al. 2024). This finding prompts a question about whether quantum computing can address margin dependency and what quantum perceptron algorithms should be applied in high-dimensional feature spaces.

Our second contribution answers this question by proposing two quantum-enhanced cutting-plane algorithms for perceptron learning. Specifically, a *hybrid* quantum-classical *cutting-plane random walk* (HCP-RW) and a *fully-quantum cutting-plane quantum walk* (QCP-QW). Compared to the classical online perceptron algorithm (Rosenblatt 1958), these algorithms possess an intrinsic advantage in margin dependency brought by linear programs (LPs), which enjoys $O^*(D\log(1/\gamma))$ [1] number of updates. Meanwhile, within idealised oracle-based quantum models, they also yield a sublinear dependence of $O(\sqrt{N})$ in the number of training examples, as with the *online quantum perceptron* (OQP) algorithm (Kapoor et al. 2016). Further, the QCP-QW algorithm provides improved complexity scaling by $O^*(D^{1.5})$ speedup in both the number of queries [2] and arithmetic operations compared with the HCP-RW. Together, the two contributions provide a refined understanding of the behaviour of earlier quantum perceptron models and introduce new quantum-enhanced perceptron learning algorithms grounded in convex optimisation.

While both the HCP-RW and QCP-QW algorithms focus on theoretical computational complexity analysis, we acknowledge that, the reported speedups should be viewed as characterising the asymptotic potential of quantum perceptron models under idealised setup [3] rather than guarantees of real-time performance. The implementation on near-term noisy intermediate-scale quantum (NISQ) or early fault-tolerant devices requires error correction or mitigation schemes are beyond the scope of the present work. Therefore, we provide detailed pseudocode, complexity analyses and algorithmic flow for QCP-QW, and defer simulations, small-scale tests and fault-tolerant schemes to future work.

The rest of the Chapter is organised as follows. In the following Section 3.2, we first re-examine some known quantum perceptron models (Kapoor et al. 2016; Roget et al. 2022; Liao et al. 2024) and provide the formal definitions of the *classical/quantum queries* related to this work. Section 3.3 corrects an error in the QVSP algorithm and provides a simulation validating this correction. In Section 3.4, we deliberate on quantum-enhanced perceptron learning algorithms by first explaining the *quantum weak online learning* framework and then explicating a hybrid ellipsoid (HE) algorithm and two cutting-plane-based algorithms, the quantum-classical hybrid algorithm HCP-RW and a fully-quantum algorithm QCP-QW that improved on the HCP-RW algorithm, elaborating on how to use quantum walk search (Magniez et al. 2007) to speed up sampling. In Section 3.6, we conclude the main contribution of this work and point out future directions. Finally, in Section 3.7, we reevaluate the generalisation performances of various quantum perceptron models and introduce another framework for active learning.

Here we present the query complexities of the classical and quantum-enhanced algorithms in advance in Table 3.1 below. The asymptotic results are obtained under $\gamma \to 0$, $D \geq 2$ is a constant, and N is the dataset size [4]. While the same table appears later in

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1. We use the asterisk notation $O^*(\cdot)$ to suppress the error parameter ϵ and factors like $\log(D)$, $\log\log(1/\gamma)$ and $\log\log(N)$ throughout the thesis.

2. Throughout the Chapter, the terms classical and quantum query, as well as query complexity, follow the definitions given in Section 3.2.

3. Note that throughout the thesis, we phrase the terms of "query complexity", "computational complexity" and "arithmetic complexity" etc., as the complexities under idealised, noise-free quantum computational model, which should not be interpreted as real-device performances.

4. Except for the version space perceptron, which is not applicable, other algorithms' complexities are analysed under the classical/quantum weak online learning setting as in Section 3.4.1.

Section 3.4.5 as well, here we add several details to the original works, and we distinguish between classical and quantum algorithms using the notation (C/Q).

Table 3.1: A summary of algorithms and query complexities.

Algorithm	Complexity
Version Space Perceptron (C)	$O^*(\frac{N}{\gamma^D})$
QVSP (Q)	$O^*(\sqrt{\frac{N}{\gamma^D}})$
Online Perceptron (C)	$O^*(\frac{1}{\gamma^2}\log(1/\gamma^2)N)$
OQP (Q)	$O^*(\frac{1}{\gamma^2}\log(1/\gamma^2)\sqrt{N})$
Ellipsoid Algorithm (C)	$O^*(D^2\log\left(\frac{1}{\gamma}\right)(N+D^4\log\left(\frac{1}{\gamma}\right)))$
HE (Q) (3)	$O^*(D^2\log\left(\frac{1}{\gamma}\right)(\sqrt{N}+D^4\log\left(\frac{1}{\gamma}\right)))$
CP-RW (Hit-and-Run) (C)	$O^*(D\log\left(\frac{1}{\gamma}\right)(N+D^5\log\left(\frac{1}{\gamma}\right)))$ [a]
HCP-RW (Q) (4)	$O^*(D\log\left(\frac{1}{\gamma}\right)(\sqrt{N}+D^{4.5}\sqrt{\log\left(\frac{1}{\gamma}\right)}))$ [a]
QCP-QW (Q) (5)	$O^*(D\log\left(\frac{1}{\gamma}\right)(\sqrt{N}+D^3\sqrt{\log\left(\frac{1}{\gamma}\right)}))$ [b]

[a] For Hit-and-Run CP-RW and HCP-RW introduced in Section 3.4.3, we simplify the result using $O(\log^2 N) \le O(D)$. Otherwise, tighter bounds can be achieved by replacing an $O^*(D)$ factor in $O^*(D^5)$ and $O^*(D^{4.5})$ with $\log^2 N$.

[b] This complexity is derived under the implementation assumptions discussed in the feasibility section 3.4.4.3.

In the first column, the QVSP was proposed by Kapoor et al. 2016, where Grover's search is used to accelerate the search for a perfect hyperplane. Under the correction we proved later in Section 3.3, the query complexity is bounded by $O^*(N/\sqrt{\gamma^D})$; taking into account the improvement by Liao et al. 2024, it becomes $O^*(\sqrt{N/\gamma^D})$. The version space perceptron algorithm is the classical counterpart of QVSP, in which one samples from the parameter space according to the multivariate normal distribution $\mathcal{N}(\mathbf{0},\mathbb{1})$ and checks the feasibility of each sampled hyperplane.

In the second column, the online perceptron refers to the well-known algorithm proposed by Rosenblatt 1958, and its convergence result follows from Novikoff 1962. The OQP was proposed by Kapoor et al. 2016, where Grover's search is applied to the dataset to accelerate the identification of misclassified examples.

In the third column, the ellipsoid algorithm, proposed by Khachiyan 1979, was the first polynomial-time algorithm for solving linear programs. The HE proposed in this work simply incorporates Grover's search to accelerate the identification of a separating hyperplane.

In the last column, the CP-RW with Hit-and-Run was proposed by Bertsimas et al. 2004, while the HCP-RW and QCP-QW are proposed in this work. These algorithms exploit not only Grover's search but also quantum counting and quantum walk search (with quantisation of Hit-and-Run Markov chain) algorithms to achieve additional speedups.

3.2 Quantum perceptron models

3.2.1 OQP and QVSP algorithms

Motivated by the classical perceptron algorithm, several quantum variants have been proposed to accelerate perceptron training, improve statistical efficiency, and enhance generalisation performance (Kapoor et al. 2016; Roget et al. 2022; Liao et al. 2024).

In all the perceptron models, a *classical query* on a training data pair $(\boldsymbol{x}'_i, y'_i)$ refers to an evaluation of a Boolean function $f_{\boldsymbol{w}}$, where

$$f_{\boldsymbol{w}}(\boldsymbol{x}'_i y'_i) = 1 \iff \boldsymbol{w}^T \boldsymbol{x}'_i y'_i \leq 0 \text{ (misclassified)}, \tag{33}$$

acting on one data pair $(\boldsymbol{x}'_i, y'_i)$ each time.

A *quantum query* on a training data $|\boldsymbol{x}'_i y'_i\rangle$ refers to an evaluation of a quantum oracle (as in Eqn.(9), Eqn.(10)) $F_{\boldsymbol{w}}$ which acts coherently on a superposition of state vectors,

$$F_{\boldsymbol{w}} |\boldsymbol{x}'_i y'_i\rangle = (-1)^{f_{\boldsymbol{w}}(\boldsymbol{x}'_i y'_i)} |\boldsymbol{x}'_i y'_i\rangle , \tag{34}$$

or equivalently,

$$F |\boldsymbol{w}\rangle |\boldsymbol{x}'_i y'_i\rangle |0\rangle = |\boldsymbol{w}\rangle |\boldsymbol{x}'_i y'_i\rangle |0 \oplus f_{\boldsymbol{w}}(\boldsymbol{x}'_i y'_i)\rangle .$$

The critical assumption for performing such quantum queries is the availability of an efficient quantum data access scheme, as discussed in detail by Kapoor et al. 2016. The scheme is commonly modelled as a quantum oracle, enabling coherent loading of a superposition over training examples.

In the OQP algorithm, assume there is an efficient quantum data access scheme, and each training pair can be encoded as a quantum state of the form $|x_i, y_i\rangle$. Compared to the classical online perceptron algorithm, the OQP algorithm reduces the number of classical queries from $O^*(N/\gamma^2)$ in the classical setting to $O^*(\sqrt{N}\log(1/\gamma^2)/\gamma^2)$ quantum queries by using Grover's search (as introduced in Section 2.1.6) to locate misclassified examples efficiently. However, the number of perceptron updates remains $O^*(1/\gamma^2)$, the same as in the classical case. As a result, this algorithm can still be limited by poor convergence behaviour when the dataset has a very small margin γ.

To address this issue, ibid. proposed the QVSP algorithm (see Algorithm 1 in Section 3.2.2), where Grover's search is used as a subroutine to amplify the probability of finding a hyperplane in version space to at least 1/4, and an outer loop of repeated sampling ensures the success probability is close to one. Argued by ibid., this method improves the overall query complexity from $O^*(N/\gamma)$ to $O^*(N/\sqrt{\gamma})$, offering faster convergence for small-margin data.[5] Being a fully-quantum algorithm, in QVSP, a unitary operator $G_{\mathscr{S}}$ is defined to act on an arbitrary hyperplane $|\boldsymbol{w}\rangle$ of the parameter space,

$$G_{\mathscr{S}} |\boldsymbol{w}\rangle = (-1)^{1+(f_w(\boldsymbol{x}_1,y_1)\vee f_w(\boldsymbol{x}_2,y_2)\vee\cdots\vee f_w(\boldsymbol{x}_N,y_N))} |\boldsymbol{w}\rangle , \tag{35}$$

which can be achieved by using $O^*(N)$ quantum queries (Eqn.(34)) that a negative phase

5. However, we show in the next section that this argument does not hold.

is added when $\boldsymbol{w}$ classifies all examples correctly.

3.2.2 Subsequent improvement

In the subsequent works, Roget et al. 2022 revisited these quantum perceptron models by explicitly analysing the trade-off between computational complexity and statistical performance, and proposed a hybrid quantum perceptron (HQP) algorithm. Here, we present the comparison between the QVSP in Algorithm 1 and HQP in Algorithm 2.

It can be seen that, in HQP, instead of updating a single hypothesis as in the OQP, the algorithm first samples K candidate hyperplanes from a multivariate normal distribution, similar to QVSP. Then, for each candidate hyperplane, a Grover search is performed over the data sample to detect a misclassified example. If no such example is found, the hyperplane is accepted as a valid separator. This combines the quadratic speedup in N from quantum search with the improved dependence on the margin γ arising from the version space formulation, leading to $O^*(\sqrt{N}\log\big(1/\gamma\big)/\gamma)$ quantum queries.

Moreover, the authors derive a generalisation guarantee by leveraging the leave-one-out (LOO) analysis (see Section 2.2.4.3) to bound the expected risk of the output hypothesis

$$\mathbb{E}_{S\sim\mathscr{D}^N}\big[R(h_S)\big] \leq \sqrt{\frac{\pi}{2}}\frac{\log(1/\epsilon)}{N+1}\mathbb{E}_{S\sim\mathscr{D}^{N+1}}\left[\frac{1}{\gamma_S}\right], \tag{36}$$

which improves over the classical online perceptron bound (see Eqn.(7)) and demonstrates that quantum enhancements can yield both computational and statistical advantages.

Shortly after, a further computational complexity improvement of QVSP was proposed by Liao et al. 2024, based on a quantum counting subroutine to realize the unitary operator of Eqn.(35) with $O^*(\sqrt{N})$ quantum queries, reducing the total query complexity from $O^*(N/\sqrt{\gamma})$ to $O^*(\sqrt{N/\gamma})$.

However, these two papers are both based on the statistical argument of Kapoor et al. 2016 that the probability of sampling a vector $\sim\mathcal{N}(\mathbf{0},\mathbb{1})$ in $\mathbb{R}^D$ that lies in the version space is $\Theta(\gamma)$. In the next section, we show that this argument is not exact.

3.3 Corrected analysis of version space sampling

3.3.1 Review of prior analysis

In the QVSP algorithm proposed by ibid., it is stated that the probability of sampling according to the normal distribution in $\mathbb{R}^D$ for a perfect hyperplane scales as $\Theta(\gamma)$. Then, Grover's search is used as a substitution for the Monte-Carlo sampling and provides a square-root speedup over the margin. However, in this section, we show that this result is not exact and we give the correction in Theorem 10.

For the maximum margin separator $\boldsymbol{u}$ and a point $\boldsymbol{x}_k$ on the margin, one has $y_k\boldsymbol{u}^T\boldsymbol{x}_k = \gamma$. When a perturbed vector $\boldsymbol{w}$ close to $\boldsymbol{u}$ that also leads to perfect classification, one has $0 = \cos(\pi/2) < y_k\boldsymbol{w}^T\boldsymbol{x}_k = l_k < \cos(\pi/2 - 2\arcsin\big(\gamma\big)) \leq 2\gamma$, $0 < \gamma < 1$. As $\gamma \to 0$, it can be approximated by $0 < l_k < 2\gamma$. In the proof of ibid., the condition $\{\boldsymbol{w} \mid 0 < l_k < 2\gamma\}$

Algorithm 1 QVSP

Input: training set $\{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$
Output: $\boldsymbol{w}$ separating hyperplane
Sample $\{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_K\} \sim \mathcal{N}(\boldsymbol{0}, \mathbb{1})$
for $j = 1$ to $\lceil \log(1/\epsilon) \rceil$ **do**
    $m \leftarrow Qsearch\big(\{\boldsymbol{w}_j\}_{j=1}^K\big)$ ▷ search over hyperplanes
    if $y_i \boldsymbol{w}_m^\top \boldsymbol{x}_i > 0, \ \forall i$ **then**
        return $\boldsymbol{w}_m$
    end if
end for
return $\boldsymbol{w}_1$

Algorithm 2 HQP

Input: training set $\{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$
Output: $\boldsymbol{w}$ separating hyperplane
Sample $\{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_K\} \sim \mathcal{N}(\boldsymbol{0}, \mathbb{1})$
for $k = 1$ to K **do**
    $C \leftarrow 1$
    for $j = 1$ to $\lceil \log(1/\epsilon) \rceil$ **do**
        $(\boldsymbol{x}_m, y_m) \leftarrow Qsearch\big(\boldsymbol{w}_k, \{(\boldsymbol{x}_i, y_i)\}_{i=1}^N\big)$
        if $y_m \boldsymbol{w}_k^\top \boldsymbol{x}_m \leq 0$ **then**
            $C \leftarrow 0$ ▷ misclassified point found
        end if
    end for
    if $C = 1$ **then**
        return $\boldsymbol{w}_k$
    end if
end for
return $\boldsymbol{w}_1$

is perceived as an equivalent for $\{\boldsymbol{w} \mid \boldsymbol{w} \in \mathcal{VS}\}$. However, we point out that this is incorrect as the former is merely a necessary condition for the latter. By definition, $\{\boldsymbol{w} \mid \boldsymbol{w} \in \mathcal{VS}\} = \{\boldsymbol{w} \mid y_i \boldsymbol{w} \cdot \boldsymbol{x}_i > 0, \forall i \in [N]\}$, but $\{\boldsymbol{w} \mid y_i \boldsymbol{w} \cdot \boldsymbol{x}_i > 0, \forall i \in [N]\} \subsetneq \{\boldsymbol{w} \mid 0 < l_k < 2\gamma\}$. As a consequence, the statement of Kapoor et al. 2016, Theorem 2 that $\Pr[\boldsymbol{w} \in \mathcal{VS}] = \Pr\big[0 < l_k < 2\gamma\big] = \Theta(\gamma)$ is not true.

3.3.2 Corrected theorem and proof

Lemma 8. *The sufficient condition for $\boldsymbol{w} \in \mathcal{VS}$ in $\mathbb{R}^D$ is that the angular distance between the hyperplane $\boldsymbol{w}$ and the maximal margin separator $\boldsymbol{u}$ satisfies $\measuredangle(\boldsymbol{w}, \boldsymbol{u}) < \arcsin \gamma$.*

Proof. By the triangular inequality on angular distance, it holds that

$$\begin{aligned}\measuredangle(\boldsymbol{w}, y_i \boldsymbol{x_i}) &\leq \measuredangle(\boldsymbol{w}, \boldsymbol{u}) + \measuredangle(\boldsymbol{u}, y_i \boldsymbol{x_i}) \leq \measuredangle(\boldsymbol{w}, \boldsymbol{u} + \frac{\pi}{2} - \arcsin\gamma) \\ &< \arcsin\gamma + \frac{\pi}{2} - \arcsin\gamma = \frac{\pi}{2}.\end{aligned}$$

Therefore $\boldsymbol{w}^T \boldsymbol{x_i} y_i = ||\boldsymbol{w}|| \cos \measuredangle(\boldsymbol{w}, y_i \boldsymbol{x_i}) > 0$ for all $i \in [N]$, indicating $\boldsymbol{w} \in \mathcal{VS}$. □

Lemma 9. *The probability that sampling from $\mathcal{N}(\boldsymbol{0}, \mathbb{1})$ for a D-dimensional vector $\boldsymbol{w}$ that satisfies $\measuredangle(\boldsymbol{w}, \boldsymbol{u}) < \arcsin \gamma$ is $\frac{1}{2} \mathrm{I}_{\gamma^2}(\frac{D-1}{2}, \frac{1}{2})$, where $\mathrm{I}_x(a, b)$ is the regularized incomplete beta function.*

Proof. The set $\{\boldsymbol{w} \mid \measuredangle(\boldsymbol{w},\boldsymbol{u}) < \arcsin\gamma\}$ uniquely determine a hyperspherical sector in a D-dimensional unit ball B_1,[6] which has colatitude angle $\arcsin\gamma$ and its volume is given by $V^{sector}(\arcsin\gamma) = \frac{1}{2}V(B_1)\mathrm{I}_{\gamma^2}(\frac{D-1}{2},\frac{1}{2})$ (S. Li 2010). Since the multivariate normal distribution is isotropic, the probability of sampling a vector $\boldsymbol{w}$ from $\mathcal{N}(\mathbf{0},\mathbb{1})$ or $\mathcal{U}_{B_1}$[7] that fall into the hyperspherical sector are the same. Consequently,

$$\Pr\left[\{\boldsymbol{w}|\measuredangle(\boldsymbol{w},\boldsymbol{u}) < \arcsin\gamma\}\right] = \frac{V^{sector}(\arcsin\gamma)}{V(B_1)} = \frac{1}{2}\mathrm{I}_{\gamma^2}(\frac{D-1}{2},\frac{1}{2}).$$

□

Theorem 10. *Given a training dataset $\{(\boldsymbol{x}_i,y_i)\}_{i=1}^N$ separated by a maximal geometric margin of γ in $\mathbb{R}^D$, the probability of sampling a D-dimensional vector $\boldsymbol{w}\sim\mathcal{N}(\mathbf{0},\mathbb{1})$ which perfectly separates the data can be geometrically defined as the quotient of the volumes of version space and the D-dimensional unit ball B_1, such that*

$$\Pr[\boldsymbol{w}\in\mathcal{VS}] = \frac{V(\mathcal{VS})}{V(B_1)} = \begin{cases}\Omega(\gamma^D), & \text{when } \gamma\to 0,\ D\geq 2 \text{ is a constant;}\\ \Omega(\frac{\gamma^D}{\sqrt{D}}), & \text{when } \gamma\to 0 \text{ and } D\to\infty.\end{cases}$$

Proof. We note that only directions of $\boldsymbol{w}$ determine classification results, and since the normal distribution $\mathcal{N}(\mathbf{0},\mathbb{1})$ is isotropic, the probabilities of sampling a vector $\boldsymbol{w}$ from $\mathcal{N}(\mathbf{0},\mathbb{1})$ or from $\mathcal{U}_{B_1}$ that it falls in the version space are the same. As a result, one has $\Pr[\boldsymbol{w}\in\mathcal{VS}] = \frac{V(\mathcal{VS})}{V(B_1)}$. From Lemma 8, we have $\{\boldsymbol{w} \mid \measuredangle(\boldsymbol{w},\boldsymbol{u}) < \arcsin\gamma\} \subseteq \{\boldsymbol{w} \mid \boldsymbol{w}\in\mathcal{VS}\}$, so the relationship of their probabilities satisfies $\Pr[\boldsymbol{w}\in\mathcal{VS}] \geq \Pr\left[\{\boldsymbol{w} \mid \measuredangle(\boldsymbol{w},\boldsymbol{u}) < \arcsin\gamma\}\right]$. By asymptotic approximation of the incomplete beta function of Lemma 9 (detailed analysis is provided in Appendix A.1), when $D\geq 2$ is a constant,

$$\Pr\left[\{\boldsymbol{w}|\measuredangle(\boldsymbol{w},\boldsymbol{u}) < \arcsin\gamma\}\right] \sim \frac{\gamma^{D-1}}{(D-1)}/\mathrm{B}(\frac{D-1}{2},\frac{1}{2}) \geq \frac{\gamma^{D-1}}{\pi(D-1)}. \tag{37}$$

And in the limit of $D\to\infty$, it yields

$$\Pr\left[\{\boldsymbol{w}|\measuredangle(\boldsymbol{w},\boldsymbol{u}) < \arcsin\gamma\}\right] \sim \frac{\gamma^{D-1}}{(D-1)}/\sqrt{\frac{2\pi}{D-1}} \sim \frac{\gamma^{D-1}}{\sqrt{2\pi(D-1)}}. \tag{38}$$

Hence, the result of Theorem 10 follows. □

Classically, a straightforward Monte-Carlo sampling method can be applied, such that samples K hyperplanes according to $\mathcal{N}(\mathbf{0},\mathbb{1})$ with a failure possibility at most some δ, would require

$$(1-p)^K \leq \exp\left(-pK\right) \leq \delta.$$

Therefore, the number of samples $K = \lceil\frac{1}{p}\ln(1/\delta)\rceil = O(\frac{1}{\gamma^D}\log(1/\delta))$ grows exponentially with the data dimension $D\geq 2$ in the limit of small margin. This exponentially large

6. We use B_R to refer to a $D-$dimensional ball with radius R and its centre at origin.

7. We use $\mathcal{U}_{B_1}$ to denote a uniform distribution inside of B_1 with probability density function defined by $f(\boldsymbol{x}) = 1/V(B_1)$, $\forall\boldsymbol{x}\in B_1$.

number of samples can be explained by a well-known phenomenon in sampling, dubbed the "*curse of dimensionality*", describing an exponential increase in volume when adding an extra dimension. The same problem regarding exponentially small version space has also been addressed by Fine et al. 2002; Gonen et al. 2013. Substituting the Monte-Carlo sampling with Grover's search, the query complexity of the QVSP algorithm should scale as $O^*(N\sqrt{K}) = O^*(N/\sqrt{\gamma^D})$ in the worst case, instead of $O^*(N/\sqrt{\gamma})$ as stated in the previous result. Notwithstanding with a quantum counting subroutine (Liao et al. 2024), it scales as $O^*(\sqrt{N/\gamma^D})$. This is a negative result that suggests that the QVSP algorithm can be ineffective when applied to high-dimensional data with small margins.

3.3.3 Simulation of margin decay

We present in Figure 3.1 a simulation result on an artificially designed hard dataset (Mohri et al. 2018; Roget et al. 2022) that has an exponentially small margin with data dimension.[8] It works as a concrete counterexample to show that the prior claims of $\Theta(\gamma)$ fail.

As we can see from it, the probability of sampling a $\boldsymbol{w} \sim \mathcal{N}(0, \mathbb{1})$ that lies in the version space (black solid line) is lower bounded by the regularised incomplete beta function given in Lemma 9 (sky blue solid line). When $D \geq 2$ is a constant, $\Pr[\boldsymbol{w} \in \mathcal{VS}]$ is lower bounded by $\frac{\gamma^{D-1}}{\pi(D-1)}$ (vermilion dashed line) as shown in Eqn.(37), and when $D \to \infty$, it is asymptotically lower bounded by $\frac{\gamma^{D-1}}{\sqrt{2\pi(D-1)}}$ (orange dashed line), as shown in Eqn.(38). It confirms our result of Theorem 10, that the probability of sampling a perfect classifier from a standard normal distribution is lower bounded by $\Omega(\gamma^D)$ in the worst case.

Discussion of the version space sampling method It is worth noting that the lower bound of $\Omega(\gamma^D)$ corresponds to a worst-case scenario, which could be too loose for certain datasets, where a small margin does not necessarily imply a small version space. Especially, the datasets with low-rank data matrices, which typically arise with small sample size, sparse data or small effective/intrinsic dimension.

Particularly, one can represent a dataset by an $X \in \mathbb{R}^{N\times D}$ data matrix, where the i^{th} row represents the $\boldsymbol{x}_i y_i$ vector, and the rank r of the matrix characterises the dimension of the linear subspace spanned by the data points. When the rank is significantly smaller than the ambient feature dimension $r \ll D$, a substantially larger version space can manifest, such that the success probability $\Pr[\boldsymbol{w} \in \mathcal{VS}]$ may be significantly larger than the worst-case lower bound. More precisely, we explicate this observation in the following Remark 11.

Remark 11. *Let* $X := [\boldsymbol{x}_1 y_1, \cdots, \boldsymbol{x}_N y_N] \in \mathbb{R}^{N\times D}$ *be the data matrix, and let* $r = \mathrm{rank}(X)$. *Define the subspace*

$$\mathcal{S} := \mathrm{span}\{\boldsymbol{x}_i y_i\}_{i=1}^N \subseteq \mathbb{R}^D.$$

8. This artificially designed dataset is from Mohri et al. 2018, Chapter 8, $\boldsymbol{x}_i' = (\underbrace{(-1)^i, \ldots, (-1)^i, (-1)^{i+1}}_{i \text{ first components}}, 0, \ldots, 0)$, $y_i = (-1)^{i+1}$.

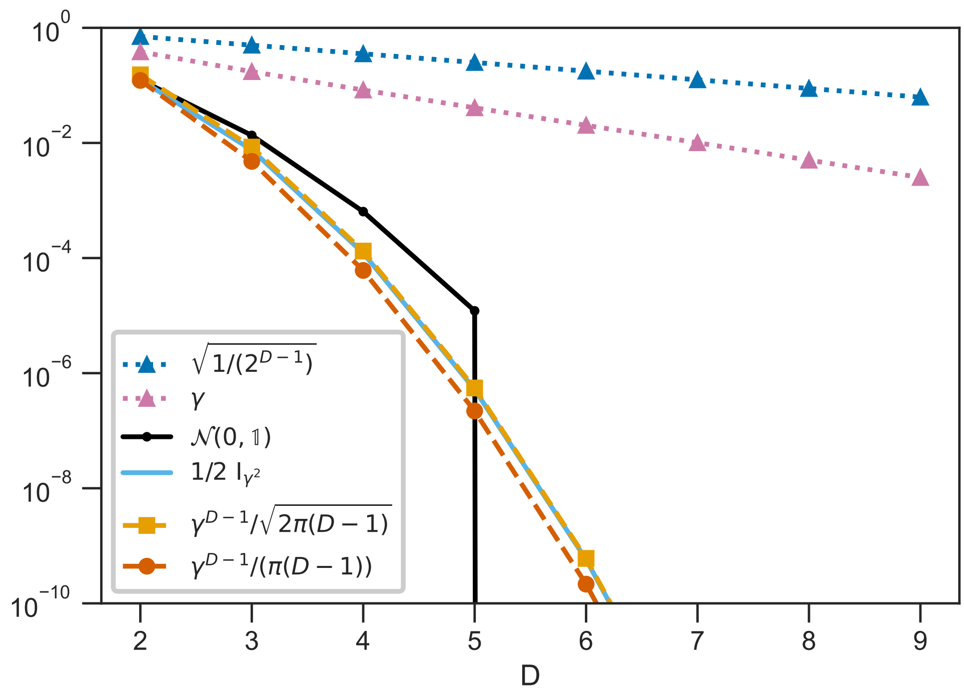

Figure 3.1: The perfect sampling probability relationships (log scale) on a hard dataset from Mohri et al. 2018 (normalised norms). The black solid line denotes the perfect sampling probability; the sky blue solid line denotes the bound in Lemma 9; the vermilion dashed line and the orange dashed line represent the bounds from Eqn.(37) and Eqn.(38) respectively. The margin γ (approximated by SKLEARN.SVM.LINEARSVC) is indicated by the reddish purple dotted line, and is upper bounded by $\sqrt{1/2^{D-1}}$ (blue dotted line).

Since each separability constraint $y_i\langle \boldsymbol{w}, x_i\rangle > 0$ *depends only on the projection of* $\boldsymbol{w}$ *onto* $\mathscr{S}$*, every* $\boldsymbol{w} \in \mathbb{R}^D$ *can be uniquely decomposed as*

$$\boldsymbol{w} = \boldsymbol{w}_\parallel + \boldsymbol{w}_\perp, \qquad \boldsymbol{w}_\parallel \in \mathscr{S}, \quad \boldsymbol{w}_\perp \in \mathscr{S}^\perp.$$

Since $\boldsymbol{x}_i y_i \in \mathscr{S}$, $\forall i \in [N]$*, we have* $\langle \boldsymbol{w}_\perp, x_i y_i\rangle = 0$*, and therefore* $\langle \boldsymbol{w}, x_i y_i\rangle = \langle \boldsymbol{w}_\parallel, x_i y_i\rangle$*, the separability is fully determined by* $\boldsymbol{w}_\parallel$*. Consequently, the version space admits the product decomposition*

$$\mathscr{V}\mathscr{S} = \mathscr{C} \oplus \mathscr{S}^\perp,$$

where $\mathscr{C} := \{\boldsymbol{u} \in \mathscr{S} | \langle \boldsymbol{u}, x_i y_i\rangle > 0, \ \forall i\}$ *is a polyhedral cone in the* r*-dimensional subspace* $\mathscr{S}$*.*

Therefore, the variable governing the small-margin behaviour of $\Pr[\boldsymbol{w} \in \mathscr{V}\mathscr{S}]$ *is* r*, rather than the ambient dimension* D*. In particular, as* $\gamma \to 0$*, we can still use the same* *expression as in Theorem 10, that*

$$\Pr[\boldsymbol{w} \in \mathcal{VS}] = \frac{V(\mathcal{VS})}{V(B_1)}.$$

However, as the volume of version space $V(\mathcal{VS})$ is restricted only by the subspace $\mathcal{S}$ with dimension $r \ll D$, while the orthogonal complement $\mathcal{S}^\perp$ contributes no additional constraints, it could scale much larger compared with the scenario that $r \sim D$. Thus, this indicates that the probability is likely $\Pr[\boldsymbol{w} \in \mathcal{VS}] = \Omega(\gamma^r)$, up to dimension-dependent constants.

Hence, following a similar analysis of the Monte-Carlo version space sampling method, as at the end of Section 3.3.2, the exponential dependence of $O^*(1/\gamma^D)$ should be interpreted as a worst-case counterexample, which could be substantially mitigated when the data matrix is low-rank. Particularly, we speculate that a much lower query complexity could be incurred for QVSP with low-rank datasets, scaling as $O^*(\sqrt{N/\gamma^r})$.

Nevertheless, since our goal here is to demonstrate the corrected bound, we do not fully characterise the practical effectiveness of the version space sampling method.

3.4 Quantum-enhanced perceptron learning algorithms

As shown in the last section, the use of QVSP can be severely limited in worst-case scenarios with small margins and high feature-space dimensions. In view of this, we investigate alternative quantum-enhanced approaches to address margin dependency and the practical usage of high-dimensional data. We might as well first investigate how the OQP algorithm (Kapoor et al. 2016) works successfully.

Following a classical online perceptron framework, the model continuously refines its current position in the dual space after perceiving each misclassified pair. In the end, it converges to an arbitrary point $\boldsymbol{w}_r \in \mathcal{VS}$. However, in a traditional online learning setup, training examples are accessed incrementally in a stream fashion, which leads to a worst-case query complexity bound that scales as $O(N)$, from deterministically checking all N training pairs. To provide a fair comparison with quantum query models and define analogous quantum-classical hybrid algorithms such as OQP, ibid. considered a *weak online learning* framework.

3.4.1 Quantum weak online learning

The weak online learning approach follows the same methodology as online learning, where data are acquired incrementally from an unknown training dataset of unknown distribution. However, it differs in that it assumes each example is sampled uniformly at random from a fixed dataset $\{(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_N, y_N)\}$. Although this may resemble batch learning, we emphasise that the learner does not have access to the full dataset *at once* nor have the *whole knowledge* of the dataset like in a white-box model. Instead, interaction occurs through a black-box interface that returns a training example upon request.

In the classical setting, this black-box is a uniform sampler: each example is returned with probability $1/N$, and repeated sampling of the same example is permitted. Consequently, to obtain a misclassified pair, a classical uniform sampler requires repeated sampling and verifications, which induces $O(N)$ classical queries, see Eqn.(33). This relaxed weak online learning assumption naturally extends to a quantum weak online learning setting, where the black-box is a quantum Grover's search subroutine. Training examples are accessed in superposition, after applying $O(\sqrt{N})$ quantum queries, as in Eqn.(34), a specific example (e.g., a misclassified one) can be retrieved (measured) with probability $O(1)$, which is the idea underpinning the OQP algorithm (Kapoor et al. 2016). Below we provide an illustration flowchart of (quantum) weak online learning algorithms for the perceptron learning problem (see Figure 3.2). It summarises the key idea behind the OQP algorithm.

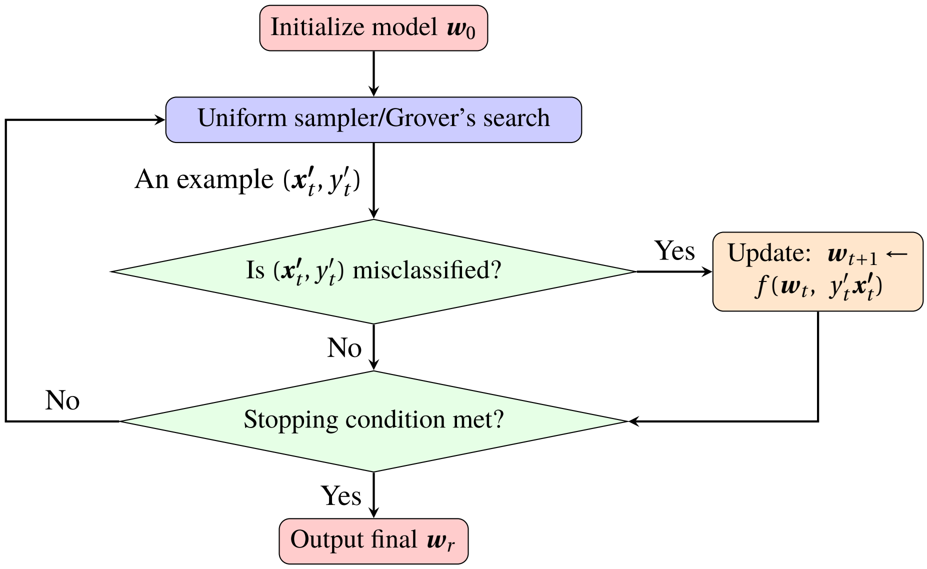

Figure 3.2: Flowchart of (quantum) weak online learning algorithm for perceptron learning: a (Grover's search subroutine) uniform sampler is used to sample a misclassified example pair, and the model is iteratively updated until convergence ($\boldsymbol{w}_r \in \mathcal{V}\mathcal{S}$).

In round t, a uniform sampler (Grover's search) returns a misclassified training example $(\boldsymbol{x}'_t, y'_t)$ with success probability at least $1-\delta$, where $\delta > 0$. The algorithm then checks whether the current hypothesis $\boldsymbol{w}_t$ misclassifies this example. If it does, the model is updated according to its corresponding rule, as $\boldsymbol{w}_{t+1} \leftarrow f(\boldsymbol{w}_t,\ \boldsymbol{x}'_t, y'_t)$, e.g., $\boldsymbol{w}_{t+1} = \boldsymbol{w}_t + y'_t\boldsymbol{x}'_t$ for online (quantum) perceptron. If the returned example pair is classified correctly, the algorithm continues to query the black-box until a misclassified pair is found. The process repeats until a stopping condition is met at round r, at which point the final hypothesis $\boldsymbol{w}_r$ lies within the version space $\mathcal{V}\mathcal{S}$.

3.4.2 Hybrid ellipsoid algorithm

As discussed in Section 2.2.2, a feasibility algorithm primitive can be a proxy for an algorithm solving the perceptron learning problem. In feasibility algorithms, each misclassified pair functions as a *separating hyperplane* that separates the current solution from the feasible region. The algorithms incrementally refine the convex feasible region containing the version space using the separating hyperplane. Although such a hyperplane is typically assumed to be provided by a *separation oracle*, in practice, one needs to implement such a separation oracle.

Under a (quantum) weak online learning setup, one can implement the separation oracle via a uniform sampler or a quantum sampler, such as Grover's search. Thus, similarly, one can gain a speedup of $O^*(\sqrt{N})$ by using a quantum approach. Considering a straightforward use of Grover's search with a feasibility algorithm, the first thing that comes to mind is likely the ellipsoid algorithm, which is the first polynomial-time algorithm for linear programs proposed by Khachiyan 1979.

With an initialisation being the centre of an isotropic ellipsoid $\mathscr{E}_0$, we identify the centre of Löwner John ellipsoids $\boldsymbol{w}_t$ in each round as a perceptron model, similarly to the idea of L. Yang et al. 2009. By acquiring a misclassified example $(\boldsymbol{x}'_t, y'_t)$, the perceptron updates its position by $\boldsymbol{w}_{t+1} = \boldsymbol{w}_t + \frac{\boldsymbol{b}_t}{D+1}$ (Algorithm 3). In each round, a centre cut-off of half an ellipsoid reduces the current space effectively while also preserving the version space in the remaining half $\{\boldsymbol{w} \mid (\boldsymbol{w} - \boldsymbol{w}_t)^T \boldsymbol{x}'_t y'_t > 0\}$. Hence, a stop criterion can be found when the volume of the convex region $V(\mathscr{E}_r) \leq V(\mathscr{V\!S})$. Due to $V(\mathscr{E}_r)/V(\mathscr{E}_0) < e^{-r/2D}$ (Grötschel et al. 2012) and $V(\mathscr{V\!S})/V(\mathscr{E}_0) \geq \frac{\gamma^{D-1}}{\pi(D-1)}$ (Eqn. (37)), hereby letting $e^{-\frac{r}{2D}} \leq \frac{\gamma^{D-1}}{\pi(D-1)}$, one can make sure $V(\mathscr{E}_r) < V(\mathscr{V\!S})$. Consequently, the hyperplane $\boldsymbol{w}_r \in \mathscr{V\!S}$ is a perfect classifier. By a simple calculation, one has $r \geq 2D(D-1)\ln\left(\frac{1}{\gamma}\right) + 2D\ln(\pi(D-1))$ and a stop criterion is thus $r = \lceil 2D^2 \ln(\frac{D}{\gamma}) \rceil$.

Therefore, we introduce a classical-quantum hybrid ellipsoid (HE) algorithm by a straightforward use of Grover's search subroutine, shown as $Qsearch$ in Algorithm 3. With $O(\sqrt{N}\log\left(\frac{D^2\ln(1/\gamma)}{\epsilon}\right))$ quantum queries and $O(D^2)$ arithmetic operations used in each round, an overall complexity scales as $O(D^2\log\left(\frac{1}{\gamma}\right)\cdot\sqrt{N}\log\left(\frac{D^2\ln(1/\gamma)}{\epsilon}\right) + D^4\log\left(\frac{1}{\gamma}\right))$, with a probability of failure at most ϵ.

3.4.3 Hybrid cutting-plane random walk algorithm

Following a similar scheme as the HE algorithm, in this section, we propose another classical-quantum hybrid algorithm, HCP-RW, based on the *cutting-plane random walk* (CP-RW) algorithm (Bertsimas et al. 2004), as the CP methods have an optimal update bounds of $O^*(D)$ (Nemirovskij et al. 1983; Jiang et al. 2020) among all feasibility algorithms. Below, we first review the geometric analysis of the classical CP-RW algorithm.

Proposed by Bertsimas et al. 2004, the CP-RW algorithm provides a method to effectively reduce the volume in $\mathbb{R}^D$. In each round, the algorithm preserves an approximate centroid $\boldsymbol{z}_t$ of the current space, which we use as a perceptron model similarly to the

Algorithm 3 HE

Input : $\{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$
Initialize :
$\boldsymbol{w} := \boldsymbol{0}$, $A := \mathbb{1}$
for $i \in \{1, \dots, \lceil 2D^2 \ln(\frac{D}{\gamma}) \rceil\}$ **do**
  for $j \in \{1, \dots, \lceil \log_{3/4}(\frac{\epsilon}{2D^2 \ln(D/\gamma)}) \rceil\}$ **do**
    $(x', y') \leftarrow Qsearch(\boldsymbol{w}, \{(\boldsymbol{x}_i, y_i)\}_{i=1}^N)$ ▷ with success probability $\geq \frac{1}{4}$.
    $\boldsymbol{b} = \frac{A\boldsymbol{x}'y'}{\sqrt{\boldsymbol{x}'^T A\boldsymbol{x}'}}$
    if $\boldsymbol{w}^T \boldsymbol{x}' y' \leq 0$ **then**
      $\boldsymbol{w} \leftarrow \boldsymbol{w} + \frac{\boldsymbol{b}}{D+1}$
      $A \leftarrow \frac{D^2}{D^2-1}(A - \frac{2}{D+1}\boldsymbol{b}\boldsymbol{b}^T)$
    end if
  end for
end for

work of Louche et al. 2015. Initialized being the centre of $\mathscr{P}_0 = B_1$, the perceptron updates its position by $\boldsymbol{z}_{t+1} \leftarrow \frac{1}{M}\sum_{j=1}^M \boldsymbol{w}'_{j,t}$ (Algorithm 4), where $\boldsymbol{w}'_{j,t}$ are uniformly sampled vectors over the remaining space $\{\boldsymbol{w} \mid (\boldsymbol{w} - \boldsymbol{z}_t)^T \boldsymbol{x}'_t y'_t > 0\}$ cut by a misclassified example $(\boldsymbol{x}'_t, y'_t)$. When the sample size N is small, sampling $2M = O(\log^2(N)) \leq O(D)$ points suffices that the volume cut-off in each round is at least 1/3 with a high probability (Bertsimas et al. 2004). From Eqn.(37), one has $V(\mathscr{V}\mathscr{S})/V(\mathscr{P}_0) \geq \frac{\gamma^{D-1}}{\pi(D-1)}$, so letting $\frac{V(\mathscr{P}_r)}{V(\mathscr{P}_0)} \leq (\frac{2}{3})^r \leq \frac{\gamma^{D-1}}{\pi(D-1)} \leq \frac{V(\mathscr{V}\mathscr{S})}{V(\mathscr{P}_0)}$ ensures $V(\mathscr{P}_r) \leq V(\mathscr{V}\mathscr{S})$ and $\boldsymbol{w}_r \in \mathscr{V}\mathscr{S}$. A direct calculation yields $r \geq (D-1)\log_{\frac{3}{2}}(1/\gamma) + \log_{\frac{3}{2}}(\pi(D-1))$, hence the stop criterion can be $r = \lceil D \log_{3/2}(D/\gamma) \rceil$.

To ensure fast sampling, we adapt the *Hit-and-Run* algorithm proposed by Smith 1984 with a uniform sampling scheme. Starting from a *warm start* distribution[9], for all $\boldsymbol{x} \in \mathscr{P}_t$, it works as follows:

Hit-and-Run

(1) Choose a line ℓ through the current point $\boldsymbol{x}$ uniformly at random.

(2) Move to a point $\boldsymbol{y}$ chosen **uniformly** from $\mathscr{P}_t \cap \ell$.

It is known that given a warm start, it has the best-known mixing bound such that it generates a stationary distribution inside of a well-rounded convex body $\mathscr{K}$ in $O^*(D^3)$ steps (Lovász 1999). Particularly, this stationary distribution is uniform since it follows a uniform sampling scheme. In the t^{th} round, starting from a current point $\boldsymbol{w} \in \mathscr{P}_t$ and giving a *membership oracle* $\mathscr{O}_{\mathscr{P}_t}$ defined on the current convex body $\mathscr{P}_t$, each step of walk costs $O(\log(1/\epsilon))$ calls to $\mathscr{O}_{\mathscr{P}_t}$ leveraging binary search. However, to implement a

9. A warm start is a starting distribution that is almost close to the stationary distribution.

membership oracle $\mathcal{O}_{\mathcal{P}_t}$ in practice, one needs to query t example pairs, where

$$\mathcal{O}_{\mathcal{P}_t}(\boldsymbol{w}) = f_{\boldsymbol{w}}(\boldsymbol{x}'_1 y'_1, \boldsymbol{z}_0) \vee f_{\boldsymbol{w}}(\boldsymbol{x}'_2 y'_2, \boldsymbol{z}_1) \vee \cdots \vee f_{\boldsymbol{w}}(\boldsymbol{x}'_t y'_t, \boldsymbol{z}_{t-1}), \tag{39}$$

taking a similar form as Eqn.(35). Here the Boolean functions are defined similarly as Eqn.(33), where $f_{\boldsymbol{w}}(\boldsymbol{x}'_j y'_j, \boldsymbol{z}_{j-1}) = 0$ iff $(\boldsymbol{w} - \boldsymbol{z}_{j-1})^T \boldsymbol{x}'_j y'_j > 0, \forall j \in [t]$. Herewith, $\mathcal{O}_{\mathcal{P}_t}(\boldsymbol{w}) = 0$ iff $\boldsymbol{w} \in \mathcal{P}_t$. Consequently, it takes $O(t \log(1/\epsilon))$ classical quires to implement the membership oracle $\mathcal{O}_{\mathcal{P}_t}$ classically. Overall, this classical cutting-plane algorithm with the Hit-and-Run walk has a classical query complexity of $O^*(D \log(1/\gamma) \cdot (N + \log^2 N \cdot D^3 \cdot r) = O^*(D \log(1/\gamma) \cdot (N + D^5 \log(1/\gamma))$.

Following a CP-RW framework, we present the HCP-RW algorithm as shown in Algorithm 4. It adopts Grover's search (shown as $Qsearch$ in Algorithm 4) under the quantum weak online learning scheme and outputs a vector $\boldsymbol{w}_r \in \mathcal{V}\mathcal{S}$ with a probability of failure at most ϵ.

Algorithm 4 HCP-RW

Input : $\{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$
Initialize :
$W := \{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_M\} \in B_1$, $\boldsymbol{z} := \boldsymbol{0}$, $S(\boldsymbol{x}) := \frac{\boldsymbol{x}-\boldsymbol{z}}{\sqrt{\mathbb{E}[(\boldsymbol{x}-\boldsymbol{z})(\boldsymbol{x}-\boldsymbol{z})^T]}}$
for $i \in \{1, \ldots, \lceil D \log_{3/2}(\frac{D}{\gamma}) \rceil\}$ **do**
  for $j \in \{1, \ldots, \lceil \log_{3/4}(\frac{\epsilon}{D \log_{3/2}(D/\gamma)}) \rceil\}$ **do**
    $(\boldsymbol{x}', y') \leftarrow Qsearch(\boldsymbol{z}, \{(\boldsymbol{x}_i, y_i)\}_{i=1}^N)$ ▷ with success probability $\geq \frac{1}{4}$.
    if $\boldsymbol{z}^T \boldsymbol{x}' y' \leq 0$ **then**
      delete $\boldsymbol{w}_i$ **from** W **if** $f_{\boldsymbol{w}}(\boldsymbol{x}' y', \boldsymbol{z}) = 1, \forall i \in [M]$.
      calculate the affine transform function S
      $W \leftarrow \{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_{2M}\} \leftarrow$ *Hit-and-Run* $(W, S, \mathcal{Q}_{\mathcal{P}})$
      $W' \leftarrow \{\boldsymbol{w}'_1, \ldots, \boldsymbol{w}'_M\} \leftarrow$ random sample M points from W
      $\boldsymbol{z} \leftarrow \frac{1}{M} \sum_{j=1}^M \boldsymbol{w}'_j$
      discard W' **from** W ▷ avoid correlations with future samples.
    end if
  end for
end for

Instead of using the classical membership oracle $\mathcal{O}_{\mathcal{P}_t}$, the HCP-RW algorithm implements a quantum membership oracle (shown as $\mathcal{Q}_{\mathcal{P}}$ in Algorithm 4) by leveraging a quantum counting algorithm as of Liao et al. 2024. Specifically, $\mathcal{Q}_{\mathcal{P}_t}$ can be implemented with a small probability of failure by using $O(\sqrt{t})$ quantum queries defined similarly with Eqn.(34)

$$F|w\rangle |x'_j y'_j\rangle |z_{j-1}\rangle |0\rangle = |w\rangle |x'_j y'_j\rangle |z_{j_1}\rangle |0 \oplus f_{\boldsymbol{w}}(\boldsymbol{x}'_j, y'_j, \boldsymbol{z}_{j-1})\rangle \,. \tag{40}$$

Taking the quantum speedups into account, the overall quantum query complexity is $O^*(D \log(1/\gamma) \cdot (\sqrt{N} + \log^2 N \cdot D^3 \cdot \sqrt{t})$, substituting $t \leq r = O(D \log(1/\gamma))$, it is asymptotically bounded by $O^*(D\log(1/\gamma)\cdot(\sqrt{N}+\log^2 N\cdot D^{3.5}\cdot\sqrt{\log(1/\gamma)} = O^*(D\log(1/\gamma)\cdot(\sqrt{N}+D^{4.5}\cdot\sqrt{\log(1/\gamma)}$. Additionally, in both HCP-RW and CP-RW algorithms, affine transforms are required to ensure the convex bodies are well-rounded; it incurs $O(D^2)$ arithmetic operations, so their arithmetic operations are bounded by $O^*(D\log(1/\gamma)\cdot(\log^2 N\cdot D^3\cdot D^2) = O^*(\log^2 N\cdot\log(1/\gamma)\cdot D^6) = O^*(D^7\log(1/\gamma))$.

From another perspective, it can also be seen clearly how quantum algorithms/subroutines can speed up solving a feasibility problem. In a similar vein, there are flourishing advancements in quantum linear programs and semidefinite programs algorithms. For example, Kerenidis et al. 2020; Apers et al. 2026 developed quantum algorithms to speed up the interior point method. In the former, the authors emphasise an efficient quantum algorithm for constructing block encodings to solve linear systems. In the latter, the authors deliberate on using Grover's search to efficiently approximate the Newton linear system matrices, and based on which, X. Li et al. 2024 proposed a quantum algorithm to approximate the Löwner–John ellipsoid.

3.4.4 Fully-quantum cutting-plane quantum walk algorithm

Realising that for the HCP-RW algorithm, a query complexity of $O^*(D^{5.5})$ is still fairly high when D is large. However, it has been proved that the cutting plane methods achieve the optimal number of calls of the separation oracle $\Omega^*(D)$ for solving the feasibility problem. Remarkably, it has also been shown by Apeldoorn et al. 2020 that quantum algorithms cannot improve this lower bound under a black-box interface. Yet we can see that using a random walk sampler incurs a high computational cost of $O^*(D^3)$.

In this section, we aim to improve the computational complexity by applying alternative quantum enhancements in the sampling process. Specifically, we propose a fully-quantum algorithm, the *cutting-plane quantum walk* (QCP-QW) algorithm in Section 3.4.4.2, adopting the quantum weak online learning framework as HCP-RW, but differs in that both the training dataset and the *model itself* in the dual (parameter) space are represented in quantum superposition.

On one hand, it preserves the strengths of the CP method, achieving a margin dependence of $O^*(D\log(1/\gamma))$. On the other hand, its algorithmic structure is inherently quantum: both the dataset and the perceptron model are encoded as quantum states, and quantum operations are done coherently. Specifically, the algorithm incorporates the *hit-and-run quantum walk* framework (Chakrabarti et al. 2023; T. Li et al. 2022), which leverages the quantum walk search algorithm introduced by Magniez et al. 2007 along with other quantum subroutines, such as quantum mean estimation and affine transformation estimation. As a result, the quantum perceptron model itself is updated continuously without destructive measurements. Compared with the HCP-RW algorithm, the sampling process is done quantumly instead of classically, which is the fundamental reason for gaining an extra speedup of $O^*(D^{1.5})$.

3.4.4.1 Szegedy's quantum walk

In the Appendix A.2, we introduce some preliminaries of classical and quantum Markov chains.

The Szegedy's quantum walk is defined based on an *ergodic* and *time-reversible* Markov chain P, it works on the Hilbert space $\mathscr{H} = \mathbb{C}^N \otimes \mathbb{C}^N$ with basis states $|\boldsymbol{x}\rangle|\boldsymbol{y}\rangle$ for $\boldsymbol{x}, \boldsymbol{y} \in \Omega$. The Szegedy's quantum walk operator $W(P) := R_{\mathscr{B}} R_{\mathscr{A}}$, where $R_{\mathscr{I}} = 2\Pi_{\mathscr{I}} - \mathbb{1}$ is the reflection about the $\mathscr{I}$ subspace, with subspaces $\mathscr{A} = \text{span}\{|\boldsymbol{x}\rangle|\boldsymbol{0}\rangle : \boldsymbol{x} \in \Omega\}$ and $\mathscr{B} = \text{span}\{|\boldsymbol{0}\rangle|\boldsymbol{x}\rangle : \boldsymbol{x} \in \Omega\}$. It was shown by Szegedy 2004; Magniez et al. 2007 that for an *irreducible* and time-reversible Markov chain P, the quantum walk operator $W(P)$ has a unique eigenvalue-1 eigenvector

$$|\pi\rangle = |\pi\rangle' |\boldsymbol{0}\rangle \in \mathscr{H},$$

in the subspace $\mathscr{A} \cap \mathscr{B}$, where $|\pi\rangle'$ is a quantum sample [10] of the stationary distribution $\boldsymbol{\pi}$ of P, given by

$$|\pi\rangle' = \sum_{\boldsymbol{x}\in\Omega} \sqrt{\pi_{\boldsymbol{x}}} |\boldsymbol{x}\rangle.$$

The eigenvalues of $W(P)$ with non-zero imaginary parts are exactly $e^{\pm 2\pi i\theta_j}$, where

$$\cos\theta_j = |\lambda_j|,$$

with $j \in [M]$, $M \leq N-1$, on the subspace $\mathscr{A} + \mathscr{B}$. These $2M$ eigenvalues satisfy $\theta_j \in (0, \pi/2)$ and $0 < |\lambda_j| < 1$ for P. In the following, we consider quantum walks acting on the subspace $\mathscr{A} + \mathscr{B}$, since their action on the orthogonal complement is trivial (eigenvalues ± 1).

A quantum-walk-update for P_t is given by a unitary operator $U(P_t)$ that satisfies, for all $\boldsymbol{x} \in \Omega$ and fixed $\boldsymbol{0} \in \Omega$,

$$U(P_t)|\boldsymbol{x}\rangle|\boldsymbol{0}\rangle = |\boldsymbol{x}\rangle|\boldsymbol{p}_x\rangle = |\boldsymbol{x}\rangle \sum_{\boldsymbol{y}\in\Omega} \sqrt{p_{xy}} |\boldsymbol{y}\rangle. \tag{41}$$

The quantum walk operator for the t^{th} Markov chain W_t can be implemented as

$$W_t := W(P_t) = U(P_t)^{\dagger} S U(P_t) R_{\mathscr{A}} U(P_t)^{\dagger} S U(P_t) R_{\mathscr{A}}, \tag{42}$$

where S denotes the swap operator.

3.4.4.2 Algorithm constriction and analysis

Below, we provide a general description of our QCP-QW algorithm with pseudocode (Algorithm 5) and an illustration picture (Fig. 3.3).

— **High-level:** The QCP-QW algorithm follows the quantum weak online learning framework as illustrated in Figure 3.2, similar to the HCP-RW algorithm. Specifi-

10. A quantum sample corresponding to a classical probability distribution f can is defined as $|f\rangle := \sum_{\boldsymbol{x}\in\Omega} \sqrt{f(\boldsymbol{x})}|\boldsymbol{x}\rangle$, with $\sum_{\boldsymbol{x}\in\Omega} |f(\boldsymbol{x})|^2 = 1$.

cally, the inner loop employs Grover's search ($Qsearch$ in Algorithm 5) to ensure the probability of finding a separating hyperplane is at least 1/4. Relying on the CP method as detailed in Section 3.4.3, the outer loop performs $O^*(D\log(1/\gamma))$ updates to guarantee the final solution $|\pi_r\rangle$ lies inside the version space with small error.

— **Middle-level:** Apply the non-destructive quantum mean estimation circuit U_t^{mean} on the state $|\tilde{\pi}_t\rangle$ to acquire the multivariate mean $\boldsymbol{z}_{t+1}$ of $\tilde{\boldsymbol{\pi}}_{t+1}$. Similarly, apply the non-destructive affine estimation circuit U_t^{aff} on $M = O(D)$ copies of $|\tilde{\pi}_t\rangle$ to obtain the empirical covariance matrix S_{t+1} of $\tilde{\boldsymbol{\pi}}_{t+1}$.

— **Low-level:** Apply the affine transformation function g_t to the states $|\tilde{\pi}_t\rangle$ to ensure the convex body $\mathscr{P}_t$ is well-rounded.[11] Then run the quantum-walk-evolution unitary $\tilde{U}_{t,m}$ to generate the distribution $|\tilde{\pi}_{t+1}\rangle$ via

$$|g_t\tilde{\pi}_{t+1}\rangle = \tilde{U}_{t,m}|g_t\tilde{\pi}_t\rangle .$$

Algorithm 5 QCP-QW

Input: $\{(\boldsymbol{x}_i, y_i)\}_{i\in[N]}$
Initialize:
  $|\pi_0\rangle|0\rangle^{\otimes ac} := \sum_{w\in B_{1,\epsilon}} \sqrt{\frac{1}{|B_{1,\epsilon}|}}|w\rangle|0\rangle^{\otimes ac}$
  $\boldsymbol{z}_0 := \boldsymbol{0}, \quad S_0 = \mathbb{1}, \quad g_0(\boldsymbol{w}) := S_0(\boldsymbol{w} - \boldsymbol{z}_0)$
for $i = 1, \ldots, \lceil D\log_{3/2}(D/\gamma)\rceil$ **do**
  for $j = 1, \ldots, \lceil \log_{3/4}(\frac{\epsilon}{D\log_{3/2}(D/\gamma)})\rceil$ **do**
    $(\boldsymbol{x}', y') \leftarrow Qsearch(\boldsymbol{z}_0, \{(\boldsymbol{x}_i, y_i)\}_{i\in[N]})$ ▷ success probability $\geq 1/4$
    if $\boldsymbol{z}^T\boldsymbol{x}'y' \leq 0$ **then**
      (1) Update $|g_t\tilde{\pi}_{t+1}\rangle = \tilde{U}_{t,m}|g_t\tilde{\pi}_t\rangle$ and then invert to obtain $|\tilde{\pi}_{t+1}\rangle$.
      (2) Apply U_{t+1}^{mean} on the state $|\tilde{\pi}_t\rangle$ to acquire the
       multivariate mean $\boldsymbol{z}_{t+1}$ of $\tilde{\boldsymbol{\pi}}_{t+1}$ non-destructively.
      (3) Apply U_{t+1}^{aff} on $M = O(D)$ copies of $|\tilde{\pi}_t\rangle$ to acquire the
       covariance matrix S_{t+1} of $\tilde{\boldsymbol{\pi}}_{t+1}$ non-destructively.
    end if
  end for
end for

Since the classical Hit-and-Run (Smith 1984) is defined on continuous spaces, we adapt the analysis tool of the continuous-space hit-and-run quantum walk framework, similar to Chakrabarti et al. 2023 and T. Li et al. 2022. For a uniform distribution $\boldsymbol{\pi}_t$ over $\mathscr{P}_t$, its quantum sample is

$$|\pi_t\rangle' = \int_{\mathscr{P}_t} \sqrt{\frac{1}{V(\mathscr{P}_t)}}|\boldsymbol{x}\rangle\, \mathrm{d}\boldsymbol{x},$$

11. A convex body is well-rounded if it contains a ball of radius r and is contained in a ball of radius R, such that $R/r = O(\sqrt{D})$.

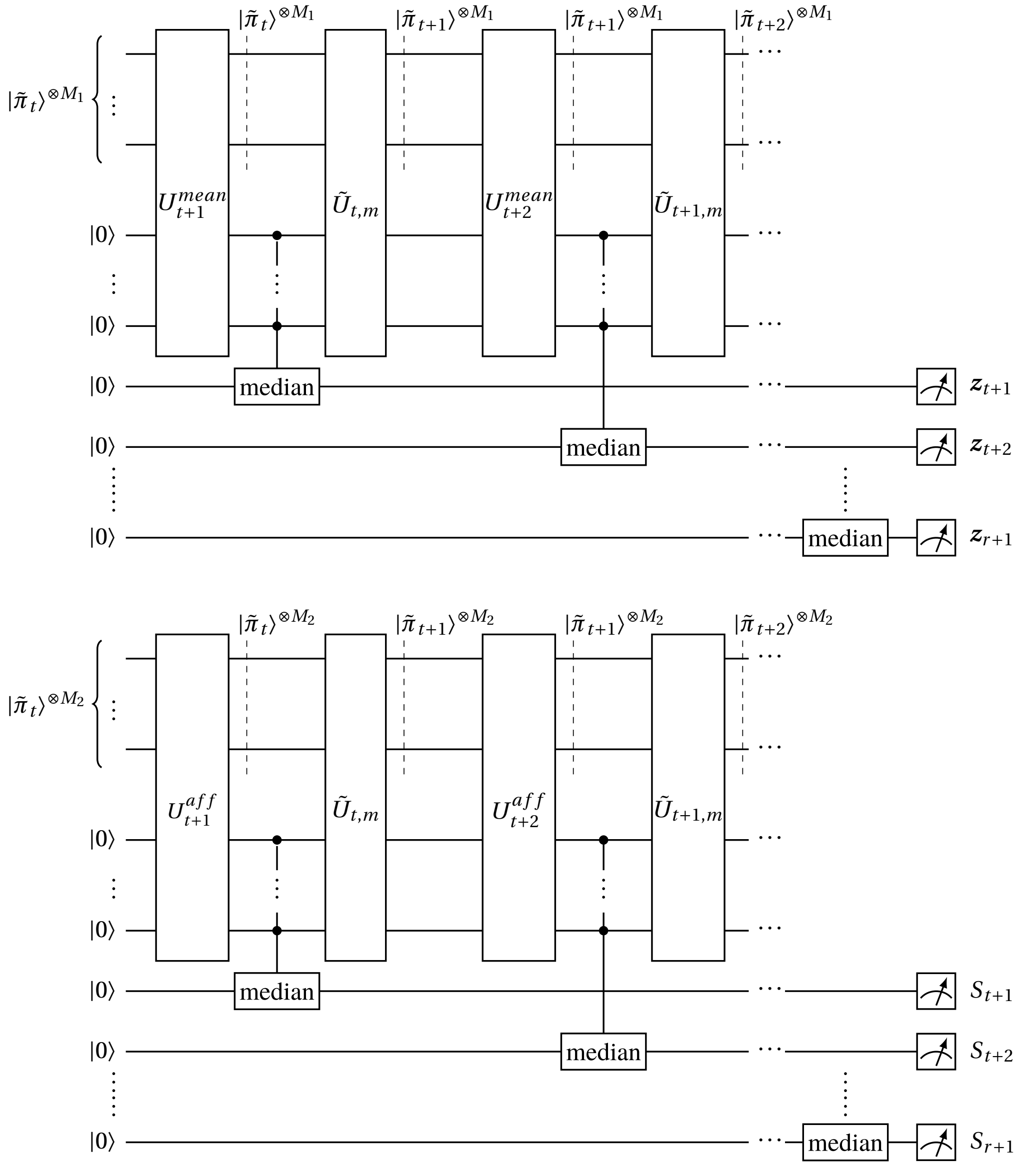

Figure 3.3: An illustration of the QCP-QW algorithm (assuming well-rounded). The upper one acts on M_1 copies of samples, and U^{mean} is the non-destructive mean estimation circuit. The lower one acts on M_2 copies, with U^{aff} denoting the circuit for non-destructively estimating the covariance matrix. Here the $\tilde{U}_{t,m}$ is the quantum-walk-evolution unitary that drives state from $|\tilde{\pi}_t\rangle$ to $|\tilde{\pi}_{t+1}\rangle$, and median denotes taking the median over multiple copies' results.

where

$$|\pi_t\rangle = \int_{\mathscr{P}_t} \sqrt{\pi_t(x)}\,|\boldsymbol{x}\rangle \otimes |\psi_x\rangle \,\mathrm{d}\boldsymbol{x}, \quad \text{with} \quad |\psi_x\rangle := \int_{\mathscr{P}_t} \sqrt{p_{xy}}\,|\boldsymbol{y}\rangle \,\mathrm{d}\boldsymbol{y},$$

defined on the support $\mathscr{P}_t \subseteq \mathbb{R}^D$.

For digital implementation, we adopt the ϵ-discretisation suggested by Chakrabarti et al. 2023, which provides a detailed analysis of discretisation errors. Specifically, each coordinate of any vector inside the convex body $\mathcal{K} \subseteq \mathbb{R}^D$ is represented using $O(\log(1/\epsilon))$ bits. This finite set of vectors $\mathcal{K}_\epsilon$ is called an ϵ-net of $\mathcal{K}$. Assume access to an initial quantum state

$$|\pi_0\rangle = |\pi_0\rangle' |\mathbf{0}\rangle$$

over $\Omega_0 = B_{1,\epsilon}$. Since $\|\boldsymbol{w}\| \leq 1$, there are $(1/\epsilon)^D$ possible values defined over a grid with spacing ϵ, so the quantum sample can be represented with $O(D\log(1/\epsilon))$ qubits. Hence, the quantum sample $|\pi_t\rangle'$ can be expressed as

$$|\pi_t\rangle' = \sum_{\boldsymbol{w}\in\mathcal{P}_{t,\epsilon}} \sqrt{\frac{1}{|\mathcal{P}_{t,\epsilon}|}} |\boldsymbol{w}\rangle .$$

Theorem 12 (Hit-and-Run Quantum Walk)**.** *Let P_t be an ergodic symmetric Markov chain defined by the* ***Hit-and-Run*** *algorithm as in Section 3.4.3 with stationary uniform distribution $\boldsymbol{\pi}_t$ on the support $\mathcal{P}_t$. Suppose we have an initial distribution*

$$|\pi_t\rangle = |\pi'_t\rangle |\mathbf{0}\rangle ,$$

the following properties hold:
- ***Mixing time:*** $d_{\mathrm{TV}}(P_t^\tau \cdot \boldsymbol{\pi}_{t+1}, \boldsymbol{\pi}_t) \leq \epsilon$.
- ***Warmness:*** $\|\boldsymbol{\pi}_{t+1}/\boldsymbol{\pi}_t\| = O(1)$, *where* $\|\boldsymbol{\pi}/\boldsymbol{\sigma}\| := \int_{\mathbb{R}^D} \frac{\pi(\boldsymbol{x})}{\sigma(\boldsymbol{x})}\pi(\boldsymbol{x})\,\mathrm{d}\boldsymbol{x}$.
- ***Overlap:*** $|\langle\pi_t|\pi_{t+1}\rangle| = \int_{\mathbb{R}^D} \sqrt{\pi_t(\boldsymbol{x})\pi_{t+1}(\boldsymbol{x})}\, d\boldsymbol{x} = \Omega(1)$.

Then, we can obtain a state $|\tilde{\pi}_{t+1}\rangle$ via a quantum-walk-evolution unitary $\tilde{U}_{t,m}$ such that

$$|\tilde{\pi}_{t+1}\rangle = \tilde{U}_{t,m} |\pi_t\rangle$$

with

$$\| |\tilde{\pi}_{t+1}\rangle - |\pi_{t+1}\rangle \| \leq \epsilon,$$

using $O\big(\sqrt{\tau}\log(1/\epsilon)\big)$ calls to the quantum walk operators W_t.

Proof. See Appendix A.3.1 □

Lemma 13. *In our QCP-QW algorithm, the overlap condition $|\langle\pi_t|\pi_{t+1}\rangle| = \Omega(1)$ is met.*

Proof. To show the overlap, we notice that as a result of Bertsimas et al. 2004, the volume cut-off $\geq 1/3$ with high probability in each round, so equivalently the volume left is at least $1/3$. Therefore,

$$|\langle\pi_t|\pi_{t+1}\rangle| = \int_{\mathbb{R}^D} \sqrt{\frac{1}{V(\mathcal{P}_t)V(\mathcal{P}_{t+1})}}\mathrm{d}\boldsymbol{x} = \frac{V(\mathcal{P}_{t+1})}{\sqrt{V(\mathcal{P}_t)V(\mathcal{P}_{t+1})}} \geq \sqrt{\frac{1}{3}} = \Omega(1).$$

□

Lemma 14. *In our QCP-QW algorithm, the warmness conditions $\|\boldsymbol{\pi}_{t+1}/\boldsymbol{\pi}_t\| = O(1)$ is met.*

Proof. $\|\boldsymbol{\pi}_{t+1}/\boldsymbol{\pi}_t\| = \int_{\mathbb{R}^D} \frac{\pi_{t+1}(\boldsymbol{x})}{\pi_t(\boldsymbol{x})}\pi_{t+1}(\boldsymbol{x})\mathrm{d}\boldsymbol{x} = \int_{\mathscr{P}_{t+1}} \frac{\frac{1}{V(\mathscr{P}_{t+1})}}{\frac{1}{V(\mathscr{P}_t)}} \frac{1}{V(\mathscr{P}_{t+1})}\mathrm{d}\boldsymbol{x} = \frac{V(\mathscr{P}_t)}{V(\mathscr{P}_{t+1})} \leq 3 = O(1).$ $\square$

From Theorem 12, one can obtain a state $|\tilde{\pi}_{t+1}\rangle = \tilde{U}_{t,m}|\pi_t\rangle$ with $\| |\tilde{\pi}_{t+1}\rangle - |\pi_{t+1}\rangle \| \leq \epsilon_1$ by invoking $\{R_t, R_t^\dagger, R_{t+1}, R_{t+1}^\dagger\}$ $O(\log\frac{1}{\epsilon_1})$ times, with $O(\sqrt{\tau}\log(1/\epsilon_1))$ calls to the quantum walk operators. Furthermore, by the result of Wocjan et al. 2008, Corollary 1, that given $|\pi_0\rangle$, one is able to prepare a state through $|\tilde{\pi}_r\rangle = \tilde{U}_{r-1}\cdots\tilde{U}_0|\pi_0\rangle$, such that $\||\tilde{\pi}_r\rangle - |\pi_r\rangle\| \leq r\epsilon_1$, by invoking the unitaries from $\{R_{t\in[r]}, R_{t\in[r]}^\dagger\}$ at most $O(r\log\frac{1}{\epsilon_1})$ times, with at most $O(r\sqrt{\tau}\log(1/\epsilon_1))$ calls of CONTROLLED-$W_{t\in[r]}$ gates. Let $r\epsilon_1 = \epsilon$, one can obtain $\||\tilde{\pi}_r\rangle - |\pi_r\rangle\| = O(\epsilon)$ with at most $O(r\sqrt{\tau}\log(r/\epsilon))$ calls of CONTROLLED-$W_{t\in[r]}$ gates.

It is known that the Hit-and-Run algorithm has the best-known mixing time bounded by $\tau = O^*(D^3)$ given a warm start condition, and the distribution space is well-rounded (Lovász 1999). Since we are using a uniform sampling scheme as described in Section 3.4.3 **Hit-and-Run**, its stationary distribution is uniform. Assuming that the subroutines required by Algorithm 5 can be implemented as described, after r updates with misclassified examples, one can obtain an approximate uniform distribution $|\tilde{\pi}_r\rangle \in \mathcal{VS}$ that satisfies $\||\tilde{\pi}_r\rangle - |\pi_r\rangle\| = O(\epsilon)$ with $O^*(rD^{1.5})$ calls of CONTROLLED-$W_{t\in[r]}$ gates. Moreover, since $|\pi_r\rangle \in \mathcal{VS}$ is a quantum state, one can directly use it to classify quantum states test data by leveraging techniques such as the *swap test* (Buhrman et al. 2001), where a probability of measurement can estimate the overlap of two quantum states.

3.4.4.3 Feasibility

Non-destructive mean estimation After preparing a uniform superposition via $|\tilde{\pi}_t\rangle = \tilde{U}_{t-1,m}|\tilde{\pi}_{t-1}\rangle$, we need to approximate the centroid $\boldsymbol{z}_t$ of the distribution $\tilde{\boldsymbol{\pi}}_t$. While measuring multiple copies of $|\tilde{\pi}_t\rangle$ is the most straightforward method, it would collapse the quantum state and require substantial resources. To circumvent this, we note that non-destructive amplitude estimation techniques (Aram W. Harrow et al. 2020; Cornelissen and Hamoudi 2023) (see Appendix A.3.2), together with recent results on multivariate mean estimation (Apeldoorn 2021; Cornelissen, Hamoudi, and Jerbi 2022), provide tools that may be useful for this task.

The former guarantees that $M_1 = O^*(1)$ copies of $|\tilde{\pi}_{t-1}\rangle$ can be recovered with high probability after quantum phase estimation (Brassard et al. 2000), while the latter provides a multivariate mean estimation procedure that can be implemented using $O^*(D)$ applications of the quantum walk unitary $\tilde{U}_{t-1,m}$, yielding an estimate with constant ℓ_2-norm error.

Taken together, these results suggest that a non-destructive multivariate mean-estimation subroutine U_t^{mean} may be realizable using $O^*(1)$ copies of $|\tilde{\pi}_{t-1}\rangle$, with each copy undergoing $O^{(D)}$ applications of $\tilde{U}_{t-1,m}$ while being recovered with high probability. Under this interpretation, since each $\tilde{U}_{t-1,m}$ involves $O^*(D^{1.5})$ calls to CONTROLLED-$W(P_{t\in[r]})$ gates, the resulting circuit U_t^{mean} would require $O^*(D \cdot D^{1.5}) = O^*(D^{2.5})$ calls to CONTROLLED-$W(P_{t\in[r]})$ gates.

Non-destructive affine transformation estimation To satisfy the well-roundedness condition, it is known that $M_2 = O^*(D)$ samples from a convex body $\mathscr{P}_t$ suffice to compute an affine transformation $g_t(\boldsymbol{w}) = S_t(\boldsymbol{w} - \boldsymbol{z}_t)$ that renders the body well-rounded (Rudelson 1999). Specifically, the empirical mean is defined as $\boldsymbol{z}_t := \frac{1}{M_2}\sum_{i=1}^{M_2} \boldsymbol{w}_i$, and the affine transformation matrix is $S_t = A_t^{-1/2}$, where the empirical covariance matrix A_t is

$$A_t = \frac{1}{M_2}\sum_{i=1}^{M_2}(\boldsymbol{w}_i - \boldsymbol{z}_t)(\boldsymbol{w}_i - \boldsymbol{z}_t)^T.$$

Hence, $M_2 = O^*(D)$ copies of $|\tilde{\pi}_{t-1}\rangle$ are sufficient in each round.

As suggested by Chakrabarti et al. 2023; T. Li et al. 2022, one may consider a non-destructive affine transformation estimation procedure. Here, we adopt the complexity analysis for the affine transformation subroutine as stated in these works (Chakrabarti et al. 2023; T. Li et al. 2022).

In particular, $M_2 = O^*(D)$ copies are used to estimate the covariance structure and derive the corresponding transformation S_t, which can be held in an ancillary quantum register. Then, with $O^*(1)$ applications of the non-destructive amplitude estimation circuit (Aram W. Harrow et al. 2020; Cornelissen and Hamoudi 2023), which consists of the quantum-walk-evolution unitary $\tilde{U}_{t-1,m}$, the affine transformation matrix can be estimated to high accuracy [12]. Meanwhile, a call of $\tilde{U}_{t-1,m}$ involves $O^*(D^{1.5})$ calls to CONTROLLED-$W(P_{t\in[r]})$ gates. Consequently, the circuit U_t^{aff} requires $O^*(D \cdot D^{1.5}) = O^*(D^{2.5})$ calls to CONTROLLED-$W_{t\in[r]}$ gates.

Since the quantum state evolution depends on the updated means and affine transformations for each distribution in $\tilde{\boldsymbol{\pi}}_{t\in[r]}$, our algorithm is structured such that the circuits U_{t+1}^{mean} and U_{t+1}^{aff} are interleaved between the states $|\tilde{\pi}_t\rangle$ and the quantum-walk-evolution unitary $\tilde{U}_{t,m}$, as illustrated in Figure 3.3.

Quantum-walk-update unitary Although Algorithm 5 is presented with complete quantum subroutines, we note that certain components, such as the quantum-walk-update unitary $U(P_t)$ within $W(P_t)$, can be very difficult to implement in practice, and as far as we know, it has not been made available on current quantum hardware.

Notably, a proof-of-principle realisation of Szegedy's quantum walk, along with a quantum walk search algorithm, has been proposed for experimental implementation in trapped ion systems (Dunjko et al. 2015), where the authors give generic constructions of quantum walk search components, and they numerically verified the robustness of the scheme with a small-scale simulation. However, as they use a rank-one Markov chain (all columns of P are identical and each of them is the stationary distribution), the implementation of $U(P)$ is simplified.

Recognising the challenges in implementing Szegedy's quantum walk with complex Markov chains on existing hardware, we frame our results as theoretical upper bounds in an idealised, decoherence-free setting and leave the demonstrations on near-term or fault-tolerant devices to future research. Nevertheless, circuit-level implementations of

12. We note that this complexity hinges on the implementation details of the moment estimation step.

$U(P_t)$ have been explored for both continuous and discrete cases (Chakrabarti et al. 2023; T. Li et al. 2022), and here we only provide a brief discussion of how to implement it in the discrete case in Appendix A.3.3.

Overall, the above discussion outlines a plausible implementation strategy based on existing quantum subroutines; in the following, we analyse the resulting complexity under this interpretation.

3.4.4.4 Computational complexity analysis

Here, we analyse the query complexity and arithmetic complexity of the proposed QCP-QW algorithm. From Eqn.(42), it can be seen that one $W(P_t)$ can be implemented by constant calls of $U(P_t)$. And it is proved by Chakrabarti et al. 2023; T. Li et al. 2022, that each $U(P_t)$ can be implemented with $O^*(1)$ calls of the quantum membership oracle $\mathcal{Q}_{\mathscr{P}_t}$ with use of binary search (see Appendix A.3.3), where every $\mathcal{Q}_{\mathscr{P}_t}$ can be realized with $O^*(\sqrt{t})$ $(t \leq r)$ quantum queries of Eqn.(40).

Under the implementation described in the feasibility discussion, both U_t^{mean} and U_t^{aff} take $O^*(D^{2.5}\sqrt{r})$ quires, substituting $r = O^*(D\log(1/\gamma))$, the overall query complexity is upper bounded by $O^*(D\log(1/\gamma)\cdot(\sqrt{N}+D^{2.5}\sqrt{r})) = O^*(D\log(1/\gamma)\cdot(\sqrt{N}+D^3\sqrt{\log(1/\gamma)})$. In comparison, the Hit-and-Run HCP-RW algorithm has a query complexity of $O^*(D\log(1/\gamma)\cdot(\sqrt{N}+D^{4.5}\cdot\sqrt{\log(1/\gamma)}$ and the CP-RW algorithm has a query complexity of $O^*(D\log(1/\gamma)\cdot(N+D^5\log(1/\gamma))$ (see Section 3.4.3).

Additionally, since the quantum walk unitary $\tilde{U}_{t,m}$ is needed on each copy of $|\tilde{\pi}_t\rangle$ inside both the U_{t+1}^{mean} and U_{t+1}^{aff}, and for performing $\tilde{U}_{t,m}$ in a convex body $\mathscr{P}_t$, it has to be well-rounded. Hence, an affine transformation function g_t is needed for each round as $|g_t\tilde{\pi}_{t+1}\rangle = \tilde{U}_{t,m}|g_t\tilde{\pi}_t\rangle$. This matrix–vector product requires $O(D^2)$ arithmetic operations classically and dominates the overall arithmetic complexity of the algorithm. We assume that elementary classical arithmetic operations (such as addition and multiplication) can be implemented by reversible quantum circuits efficiently (e.g. the quantum adders or multipliers) acting on $O^*(D)$ qubits. Since each unitary gate in the logical reversible circuit can be approximated to precision ϵ using a universal gate set with polylogarithmic overhead in $1/\epsilon$ by the Solovay–Kitaev theorem. Under this implementation framework, the arithmetic complexity of our algorithm is dominated by the matrix–vector product, bounded by $O^*(D\log(1/\gamma)\cdot(D^2\cdot(D^{2.5}))) = O^*(D^{5.5}\log(1/\gamma))$. In comparison, both the Hit-and-Run CP-RW and HCP-RW algorithms (see Section 3.4.3) has $O^*(D^7\log(1/\gamma))$ arithmetic operations.

3.4.5 Comparisons of quantum-enhanced perceptron learning algorithms

To summarise, we provide a Table 3.2 to compare the query complexities of the classical and quantum-enhanced algorithms that we examined for the perceptron learning problem [13]. As we can see, all the quantum-enhanced algorithms achieve some speedups under idealised quantum computing models.

Table 3.2: A summary of algorithms and query complexities.

Algorithm	Complexity
Version Space Perceptron	$O^*(\frac{N}{\gamma^D})$
QVSP	$O^*(\sqrt{\frac{N}{\gamma^D}})$ [a]
Online Perceptron	$O^*(\frac{1}{\gamma^2}\log(1/\gamma^2)N)$
OQP	$O^*(\frac{1}{\gamma^2}\log(1/\gamma^2)\sqrt{N})$
Ellipsoid Algorithm	$O^*(D^2\log\left(\frac{1}{\gamma}\right)(N+D^4\log\left(\frac{1}{\gamma}\right)))$
HE (3)	$O^*(D^2\log\left(\frac{1}{\gamma}\right)(\sqrt{N}+D^4\log\left(\frac{1}{\gamma}\right)))$
CP-RW (Hit-and-Run)	$O^*(D\log\left(\frac{1}{\gamma}\right)(N+D^5\log\left(\frac{1}{\gamma}\right)))$ [b]
HCP-RW (4)	$O^*(D\log\left(\frac{1}{\gamma}\right)(\sqrt{N}+D^{4.5}\sqrt{\log\left(\frac{1}{\gamma}\right)}))$ [b]
QCP-QW (5)	$O^*(D\log\left(\frac{1}{\gamma}\right)(\sqrt{N}+D^3\sqrt{\log\left(\frac{1}{\gamma}\right)}))$ [c]

[a] Here we only present the corrected result of the improved-QVSP of Liao et al. 2024.

[b] For Hit-and-Run CP-RW and HCP-RW introduced in Section 3.4.3, we simplify the result using $O(\log^2 N) \le O(D)$. Otherwise, tighter bounds can be achieved by replacing an $O^*(D)$ factor in $O^*(D^5)$ and $O^*(D^{4.5})$ with $\log^2 N$.

[c] This complexity is derived under the implementation assumptions discussed in the feasibility section.

From a statistical perspective, the quantum-enhanced CP algorithms we proposed in this work (HCP-RW and QCP-QW) offer improved margin dependence compared with the OQP and the QVSP algorithms in the worst case, though it comes at the cost of higher dimensional dependence of poly$D$ relative to the OQP algorithm. However, it is worth noting that the dominant advantage concerning the margin $O^*(D\log(1/\gamma))$ is offered by classical LP rather than an intrinsic statistical advantage from quantum computation.

In terms of algorithmic design, the OQP utilise Grover's search only in one instance, sampling a superposition over the training dataset to identify a misclassified example pair. The HCP-RW improves it by adopting another quantum counting algorithm. While for the QCP-QW, more quantum subroutines are used than just Grover's search and counting. As a result, a speedup of $O^*(D^{1.5})$ is possible compared with the HCP-RW in both query and arithmetic complexities.

13. Except for the version space perceptron, which is not applicable, other algorithms' complexities are analysed under the classical/quantum weak online learning setting.

3.5 Discussion

3.5.1 Perceptron problem under complexity standpoint

From the above discussion and Table 3.2, we can see that for solving the perceptron problem using quantum methods, one can adopt different algorithms, which have their favourable regimes. For example, QVSP favours a low-rank data matrix, OQP favours a large-margin dataset, and the CP method could perform well in regimes of small margin and moderate feature dimension D.

From an algorithmic perspective, QVSP achieves a quadratic speedup over Monte-Carlo sampling by using Grover's search, which is known to be optimal for unstructured search. Nevertheless, this may not be the most effective strategy. More broadly, related lower-bound results in quantum convex optimisation provide useful context. It has been shown that solving LPs in the black-box setting requires at least $\Omega(D\sqrt{N})$ queries (Apeldoorn et al. 2020); while in the white-box model, a tight lower bound of $\Omega(\sqrt{ND})$ queries to the data matrix has also been conjectured (Apers et al. 2026). These results, together with ours, suggest the utility of viewing the version space problem within the context of convex optimisation. And this naturally motivates the open question of identifying the most efficient algorithmic framework for the perceptron problem.

From a complexity-theoretic perspective, whether the version space volume problem could be resolved efficiently using quantum superposition remains an open problem. Especially, if the goal is to reach a version space (or a distribution) without the help of example instances (or separation hyperplanes) that define it, the task resembles a quantum state preparation problem. A direct approach based on unstructured search leveraging Grover's algorithm would typically incur a cost proportional to $O^*(1/\sqrt{\pi(x)})$, where $\pi(x)$ can be exponentially small when the state space is exponentially large (Wocjan et al. 2008; Aram W. Harrow et al. 2020). A general and efficient solution to such sampling tasks could have significant complexity-theoretic implications, potentially relating to the complexity classes SZK and BQP (Aharonov and Ta-Shma 2003; Orsucci et al. 2018).

From the above discussion, we explicitly separate the two aspects of algorithmic efficiency and fundamental limitation. We would like to emphasise that our contributions presented in this work only focus on the perspective of algorithm efficiency, rather than the fundamental limitations of quantum computing.

3.5.2 Practical Limitations

Regarding algorithmic frameworks, we can see that both HCP-RW and QCP-QW rely heavily on Grover's search and its extension of quantum counting. The fully-quantum QCP-QW algorithm additionally requires quantum phase estimation, quantum-walk updates, and non-destructive amplitude estimation, making it technically more demanding. Beyond these technical differences, both approaches share some practical limitations in realising hardware-level speedups.

First, for realistic machine learning tasks, the dataset size N is typically large. Achieving the speedup of $O^*(\sqrt{N})$ over the classical $O^*(N)$ baseline requires an efficient data access scheme (oracle) or QRAM (Aaronson 2015). However, in practice, implementing such access can be computationally expensive and may be limited by hardware constraints such as data movement and I/O bandwidth. It has been shown in a perfect fault-tolerant quantum computer [14] that a 10^4 slowdown is possible in I/O bandwidth (Hoefler et al. 2023).

The second issue is that, on real devices, decoherence is unavoidable, and oracles are imperfect. Both algorithms are based on a strong oracle model (as in Eqn.(34)). It is well-known that Grover-type speedups degrade significantly in the presence of oracle imperfections (Shenvi et al. 2003; Regev et al. 2008). Additionally, both algorithms require $O^*(\sqrt{N})$ repetitions of Grover's unitary, which can result in substantial circuit depth and cause decoherence for NISQ devices.

One may attempt to solve the problem by using a fault-tolerant quantum computing framework. Asymptotically, quantum computation does provide speedup since fewer oracles will be needed on a quantum computer than on a classical computer. However, as highlighted in recent analyses of realistic hardware costs, a constant-factor gap of $\sim 10^{10}$ operation throughput between a classical GPU and a fault-tolerant quantum computer is representative (Hoefler et al. 2023).

Hence, the number of oracle calls needs to scale $\sim 10^{10}$ for a quadratic speedup of $O^*(\sqrt{N})$. This, in turn, places strong constraints on the allowable complexity of each oracle call. In particular, to achieve quantum advantage within a practical time scale (e.g., two weeks), each oracle would need to involve only $O^*(1)$ arithmetic operations. In our setting, this requirement is highly restrictive, since each oracle evaluation requires at least $\Omega(D)$ operations to compute vector inner products [15].

Taken together, these observations suggest that our results should be viewed as conceptual, asymptotic improvements, rather than immediately realisable practical speedups on either NISQ or fault-tolerant quantum platforms.

3.6 Conclusion and Outlook

In conclusion, through this work, we first point out the limitation of QVSP, which has an exponential dependence on data dimension under worst-case scenarios, and recognise that it may still perform effectively, such as in low-rank regimes. We then introduce two new quantum-enhanced perceptron learning algorithms grounded in the formulation of LPs, and we discuss their practical limitations. As such, we provide a refined understanding of the asymptotic computational complexity behaviours and potentials of some variants of quantum perceptron models. We hope that the present work provides useful insight into both the potential and the limitations of these models.

Furthermore, although current practical constraints are significant (as discussed in

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14. We refer to the hypothetical fault-tolerant quantum computer with 10^4 error-corrected logical qubits, 10 μs gate time for logical operations, all-to-all connectivity, and fully fault-tolerant two-qubit operations as investigated by Hoefler et al. 2023.

15. Although we assume a black-box oracle model, considering hardware practicality, one needs to open it.

Section 3.5.2), they do not preclude the possibility of meaningful quantum advantage in future settings. Such advantages may arise from alternative problem regimes (e.g., structured inputs), improved algorithmic frameworks achieving higher-order speedups, or advances in quantum hardware and system architectures.

Beyond the current work, we note that the classifier produced by the CP-RW method corresponds to a Bayes point machine (BPM) solution (Herbrich et al. 2001), which is known to possess strong theoretical and empirical generalisation guarantees (Minka 2001). Exploring whether similar properties can be realised within the proposed quantum-enhanced frameworks, under suitable implementation assumptions, remains an open question for future work.

Although the QCP–QW algorithm relies on a very strong oracle model and should therefore be interpreted primarily as a conceptual upper bound rather than a near-term implementable scheme (Shenvi et al. 2003; Regev et al. 2008), the approach may provide a conceptual perspective for convex optimisation problems, since standard reductions allow convex minimisation problems to be reformulated as feasibility problems (Nemirovski 1994). However, extending the present framework to general optimisation settings would require a more complete implementation of the underlying quantum subroutines, as well as satisfying all feasibility requirements outlined in Section 3.4.4.3. This remains an open direction for future work.

3.7 More on an ML perspective

3.7.1 On generalisation performance

Beyond considerations of computational complexity, another motivation for studying quantum perceptron algorithms stems from recent results on their generalisation performance.

In Section 3.2.2, we discuss the hybrid quantum perceptron (HQP) proposed by Roget et al. 2022, which improves upon the classical online perceptron bound (Eqn.(32)) and suggests that quantum approaches may offer both computational and statistical advantages. This naturally motivates a closer investigation of the generalisation properties of quantum algorithms.

However, in light of the QVSP approach, where the dependence on the ambient dimension D is effectively neglected, it becomes necessary to revisit these generalisation bounds more carefully. In particular, we reassess the generalisation error under assumptions that explicitly account for the role of the dimension. In Section 3.4, we proposed several quantum-enhanced perceptron algorithms; here we analyse their generalisation risks together with previously studied quantum perceptron models.

Beyond leave-one-out (LOO) analysis mentioned in Section 2.2.4.3, additional tools can be used to study generalisation, including PAC learning and VC-dimension-based bounds (see Section 2.2.4). In standard online learning, no distributional assumption is made, and performance is typically measured via regret or mistake bounds rather than generalisation error. However, in the weak online learning setting considered here, we assume access to a finite sample of size N drawn i.i.d. from an unknown distribution $\mathscr{D}$, which allows us to invoke PAC-style guarantees.

Since the problem is linearly separable, the hypothesis class corresponds to linear hyperplanes in $\mathbb{R}^D$. This class has VC dimension $D+1$, and thus admits standard generalisation bounds. In the realisable case, this yields a bound of order $\tilde{O}(\frac{D+1}{N})$. We now examine different algorithms:

OQP For the online quantum perceptron (OQP), the leave-one-out analysis still applies, and the generalisation bound derived earlier (see Eqn. (7)) remains valid.

QVSP For the QVSP approach, we have shown that the probability of sampling a separating hyperplane from the version space scales as γ^D, rather than $\Theta(\gamma)$ in Section 3.3.2. This implies that the effective hypothesis class size is $|\mathscr{H}| \sim 1/\gamma^D$. Since a perfect classifier exists (zero empirical risk), we can apply the consistent PAC bound as in Eqn.(25), yielding

$$\tilde{O}\left(\frac{\log(1/\gamma^D)}{N}\right) = \tilde{O}\left(\frac{D}{N}\right). \tag{43}$$

HQP The HQP algorithm maintains an outer loop over $K = \tilde{O}(1/\gamma)$ candidate classifiers. However, taking into account the correct dimension dependence, this should scale as $K = \tilde{O}(1/\gamma^D)$. Applying the same consistent PAC bound as above leads to the same order of generalisation error as QVSP, $\tilde{O}\left(\frac{\log(1/\gamma^D)}{N}\right) = \tilde{O}(\frac{D}{N})$, rather than the bound previously suggested in Eqn.(36).

Proposed quantum-enhanced algorithms For the quantum-enhanced algorithms introduced in Section 3.4, the analysis follows similar principles to classical cutting-plane methods. In this setting, the resulting solution can be interpreted in terms of a Bayes point machine classifier (see Section 3.6), which is known to exhibit strong generalisation guarantees.

As a consequence, these classifiers may achieve improved generalisation performance compared to the previously discussed approaches. We do not pursue a detailed analysis here, and instead refer interested readers to Herbrich et al. 2001; Minka 2001 for further discussion.

3.7.2 Active learning

Active learning provides a setting in which strong generalisation guarantees can be achieved with significantly fewer labelled samples. A notable example is the query-by-committee (QBC) algorithm (Freund et al. 1997), which queries labels only when disagreement arises within the version space. Under suitable assumptions (e.g., near-uniform data distributions), QBC achieves a generalisation error ϵ using $O(1/\epsilon)$ unlabelled samples and only $O(\log(1/\epsilon))$ label queries.

Subsequent works have proposed more efficient implementations and alternative strategies, including greedy and margin-based approaches, while retaining similarly favourable statistical guarantees (Tong et al. 2001; Dasgupta 2004; Dasgupta et al. 2005; Gonen et al. 2013). These results highlight that active learning can substantially improve label complexity without sacrificing generalisation performance.

In light of our earlier discussion on generalisation error, this suggests that active learning constitutes a particularly promising direction. In particular, developing quantum analogues of active learning algorithms remains an open and interesting problem, with the potential to further enhance both computational efficiency and generalisation guarantees.

4 Quantum separability via Frank-Wolfe and machine learning

Sommaire

4.1 Introduction

In quantum information, quantum entanglement has always played a crucial role in distinguishing quantum and classical physics. It is also known as a great resource in quantum communication; for example, it established the cornerstone of the quantum key distribution protocol and quantum teleportation. It is known that for a pure state (see Section 2.1.1), by performing the Schmidt decomposition (introduced in Section 4.2.3), one can efficiently determine the separability. However, it is known that certifying general mixed state (see Section 2.1.1) quantum entanglement, or its opposite, quantum separability, is an NP-hard problem even when considering only bipartite systems (see Eqn.(3)) with local dimension p_A and p_B.

Recently, it has been suggested to use classical supervised ML algorithms as binary classifiers to approximately solve the bipartite mixed state separability problem, and an approximated-FW algorithm combined with witness tests has been proposed to create large-scale (with the system dimension $p = p_A \times p_B = 7 \times 7$), high-quality entangled datasets (Casalé, Di Molfetta, Anthoine, et al. 2023). In this chapter, we first revise this part of the work and based on which we make three new contributions. First, we analyse the convergence or accuracy of the approximated-FW algorithm, which is used to approximate a nearest separable state. Second, we examine the effectiveness of the proposed entanglement witness test, which is used for creating bounded-entangled datasets. And in the end, we verify the utility of the trained ML models by performing bias tests, then we apply more advanced ML tools to achieve more accurate classification.

4.1.1 Motivation and contribution

Determining whether a general mixed quantum state is entangled or separable is an NP-hard problem. Recently, several works have applied supervised machine learning (ML) methods to address this task (Ma et al. 2018; M. Yang et al. 2019; Roik et al. 2021; Ureña et al. 2023; Asif et al. 2023). A common challenge in these approaches is the difficulty of generating large-scale, reliably labelled datasets of entangled states. Consequently, most existing studies are restricted to systems with small local dimensions, typically $2 \sim 4$ qubits ($p = 2^2 \sim 2^4$).

For systems with dimension $p \leq 6$, the Peres-Horodecki criterion (also known as the PPT criterion) provides a necessary and sufficient condition for separability. However, beyond this regime, labelling becomes significantly more challenging. Alternative approaches are therefore required, such as semidefinite programming (SDP) (M. Yang et al. 2019), or restricting attention to states with specific structures.

Some previous works have explored entanglement in both bipartite and multipartite systems (see Section 2.1.1). Ureña et al. 2023 considers only three-qubit pure states, where the possible classes reduce to separable, bipartite entangled (BE), Greenberger-Horne-Zeilinger (GHZ), and W states. Another strategy is to simplify the classification task by reducing entanglement to the BE category (Ma et al. 2018) (so fully-entanglement is not considered), in which case the PPT criterion can still be applied to reduced density matrices. Beyond supervised approaches, unsupervised ML methods have also been explored. For instance, Y. Chen et al. 2021 treats entangled states as anomalies relative to the convex set of separable states, and demonstrates accurate detection for systems of up to 10 ($p = 1024$) qubits.

In this work, we focus on supervised ML methods and aim to develop efficient methods for the separability problem of bipartite mixed states at larger scales (up to $p = 49$). To the best of our knowledge, this is among the first attempts to address this problem at such a scale. As far as we know, the closest literature to this work is by Lu et al. 2018. In the paper of ibid., the convex hull approximation (CHA) algorithm is employed to approximate the set of separable states, and a bootstrap aggregating (bagging) approach is used to increase the accuracy. However, CHA becomes computationally intractable when the local dimension grows large since it requires sampling a large amount of random separable states, and this could be the reason that the experiments are only demonstrated on 2-qubit systems ($p = 4$). Another work we are aware of that analysed learning scenarios for quantum separability with high-dimensional quantum states is the recent paper of Girardin et al. 2022. Although the authors did not provide a mechanism for entangled PPT data generation and augmentation, which is central to our task, the authors proposed a generative neural network to find the closest separable state to a given target quantum density matrix, which works up to dimension $p = 100$.

The rest of the Chapter is organised as follows. In the Section 4.2, we first give the problem formulation, notions and tools that we use later, then in Section 4.3, we introduce the background of the research with the proposed algorithm construction and method to create large-scale PPT-ENT datasets. In Section 4.4 and Section 4.5, we provide partial theoretical guarantees for the proposed algorithm and examine the effectiveness of the entanglement witness test. Section 4.6 gives new results of experiments on several ML models. Finally, in Section 4.7, we summarise this work by a discussion of its limitations and physical significance, and we point out possible future directions.

4.2 Problem formulation

4.2.1 Notations

Before we go into the details of the problem formulation, let us review some background on the mathematical structure of quantum states, and we will introduce the notations needed for this Chapter.

In the following, we adapt the notations of Dahl et al. 2007 to the *complex* space, denote by $\mathcal{T}_p$ the space of complex symmetric $p \times p$ matrices. This space has dimension $p(p+1)/2$ and is equipped with the inner product

$$\langle A, B \rangle = \mathrm{tr}(AB), \quad A, B \in \mathcal{T}_p^+.$$

The associated norm is the Frobenius norm

$$\|A\|_F = \sqrt{\sum_{i,j} a_{ij}^2}.$$

A matrix $A \in \mathcal{T}_p^+$ is said to be positive semidefinite, denoted $A \succeq 0$, if $x^\dagger A x \geq 0$ for all $x \in \mathbb{C}^p$. The set of all positive semidefinite matrices is denoted by

$$\mathcal{T}_+^p = \{A \in \mathcal{T}_p : A \succeq 0\},$$

which forms a positive semidefinite convex cone. A density matrix is a matrix $A \in \mathcal{S}_+^p$

such that $\mathrm{tr}(A) = 1$. We denote the set of density matrices by

$$\mathcal{S}_+^p = \{A \in \mathcal{T}_p^+ : \mathrm{tr}(A) = 1\},$$

where it is convex and compact, and its extreme points are the rank-one matrices of the form $A = xx^\dagger$, where $\|x\| = 1$. In particular,

$$\mathcal{S}_+^p = \mathrm{conv}\{xx^\dagger : x \in \mathbb{C}^p,\ \|x\| = 1\}.$$

4.2.2 Definition of mixed state separability

We now extend these definitions to bipartite systems. Let $p = p_A p_B$, where p_A and p_B denote the dimensions of the subsystems $\mathcal{H}_A = \mathbb{C}^{p_A}$ and $\mathcal{H}_B = \mathbb{C}^{p_B}$, respectively. A bipartite density matrix is a matrix $\rho \in \mathcal{S}_+^p$ acting on $\mathbb{C}^{p_A} \otimes \mathbb{C}^{p_B}$. A density matrix $\rho \in \mathcal{S}_+^p$ is said to be *separable* (SEP) if it can be written as a convex combination of product states as below, otherwise it is said to be *entangled* (ENT).

$$\rho = \sum_{k=1}^{m} \lambda_k A_k \otimes B_k, \tag{44}$$

where $\lambda_k \geq 0$, $\sum_{k=1}^m \lambda_k = 1$, and $A_k \in \mathcal{S}_+^{p_A}$, $B_k \in \mathcal{S}_+^{p_B}$. Here we call m the *mixing number*, as it represents how many pure states compose the mixture. We denote by $\mathcal{S}_+^{p,\otimes} = \mathrm{conv}\big(\mathcal{S}_+^{p_A} \otimes \mathcal{S}_+^{p_B}\big)$ the set of separable density matrices, which is a convex and compact subset of $\mathcal{S}_+^p$. Its extreme points are the rank-one product states of the form

$$A = (x \otimes y)(x \otimes y)^\dagger,$$

where $x \in \mathbb{C}^{p_A}$ and $y \in \mathbb{C}^{p_B}$ satisfy $\|x\| = \|y\| = 1$. In particular, we have

$$\mathcal{S}_+^{p,\otimes} = \mathrm{conv}\Big\{(x \otimes y)(x \otimes y)^\dagger : x \in \mathbb{C}^{p_A},\ y \in \mathbb{C}^{p_B},\ \|x\| = \|y\| = 1\Big\}.$$

4.2.3 Schmidt decomposition

The Schmidt decomposition (Michael A. Nielsen et al. 2010b, Theorem 2.7) provides a canonical representation of bipartite pure states. Any state $|\psi_k\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ can be written as

$$|\psi_k\rangle = \sum_{i=1}^{r \leq \min(p_A, p_B)} \lambda_{i,k} |a_i\rangle \otimes |b_i\rangle_k, \tag{45}$$

where $\lambda_{i,k} \geq 0$ are the Schmidt coefficients satisfying $\sum_{i=1}^r \lambda_{i,k}^2 = 1$, and $\{|a_i\rangle\}$, $\{|b_i\rangle_k\}$ are orthonormal sets. The number of non-zero Schmidt coefficients, known as the Schmidt rank r, characterises the structure of the state. In particular, a pure state is separable if and only if $r = 1$, and entangled otherwise. The Schmidt decomposition can be computed numerically via the singular value decomposition (SVD) of the coefficient matrix associated with $|\psi_k\rangle$ in a product basis.

4.2.4 PPT criterion

Because of its simplicity, the PPT criterion, also known as the Peres-Horodecki criterion, is one of the most popular criteria for quantum separability Peres 1996; M. Horodecki et al. 1996. A more detailed illustration is presented below.

Let ρ be a density matrix on $\mathscr{H}$. The matrix ρ is in $\mathscr{S}_+^p$ and then can be written as a block matrix composed of the matrix blocks $\rho_{i,j}$ of size $p_B \times p_B$, $\forall i, j = 1, \ldots, p_A$. The partial transpose of ρ with respect to subsystem B, denoted by $\rho^{\top_B}$, is defined as follows:

$$\rho = \begin{bmatrix} \rho_{1,1} & \cdots & \rho_{1,p_A} \\ \vdots & & \vdots \\ \rho_{p_A,1} & \cdots & \rho_{p_A,p_A} \end{bmatrix} \Rightarrow \rho^{\top_B} = \begin{bmatrix} \rho_{1,1}^\top & \cdots & \rho_{1,p_A}^\top \\ \vdots & & \vdots \\ \rho_{p_A,1}^\top & \cdots & \rho_{p_A,p_A}^\top \end{bmatrix}.$$

The PPT criterion states that if a density matrix ρ is separable, then it has positive partial transpose, i.e., $\rho^{\top_B} \geq 0$. Any separable density matrix must have a positive partial transpose. Yet, in general, this condition is not sufficient to guarantee separability, as there might be non-separable density matrices fulfilling the PPT condition. However, this condition guarantees that if the partial transpose matrix has a negative eigenvalue, the state is entangled.

Moreover, if ρ is a bipartite density matrix such that $p_A = 2$ and $p_B = 2$ or $p_B = 3$, then $\rho^{\top_B} \geq 0$ implies that ρ is separable. So, in this case, the PPT criterion becomes a necessary and sufficient condition (M. Horodecki et al. 1996). In other dimensions, this is no longer the case (Pawel Horodecki 1997). See Figure 4.1 for a schematic illustration.

4.2.5 Entanglement witness

The entanglement witness test is a quantum separability criterion based on characterising a separating hyperplane between the set of all separable density matrices and an entangled density matrix (see Figure 4.1). Such a separating hyperplane always exists since the set of separable states is convex (M. Horodecki et al. 1996). More formally, given an entangled density matrix $\rho \in \mathscr{S}_+^p$, an entanglement witness W is a Hermitian matrix in $\mathbb{C}^{p \times p}$ such that:

$$\operatorname{tr}\big(W\rho_S\big) \geq 0 \text{ for all separable } \rho_S \in \mathscr{S}_+^{p,\otimes}, \text{ and } \operatorname{tr}\big(W\rho\big) < 0. \tag{46}$$

In this case, we say that W detects the entanglement of ρ, and then for each entangled density matrix, there exists a witness that detects its entanglement M. Horodecki et al. 1996; Gühne et al. 2009. In this work, we are interested in the notion of *optimal* entanglement witness (Reinhold A. Bertlmann et al. 2005). An entanglement witness W_{opt} is optimal if there is a separable density matrix $\tilde{\rho}_S \in \mathscr{S}_+^{p,\otimes}$ such that $\langle W_{opt}, \tilde{\rho}_S \rangle = 0$. One interesting feature of an optimal entanglement witness is that it can be computed analytically and efficiently from the *nearest separable* state of an entangled density matrix ρ. In other words, if $\tilde{\rho}_S$ is a separable density matrix that satisfies $\|\tilde{\rho}_S - \rho\|_F = \min_{\rho_S \in \mathscr{S}_+^{p,\otimes}} \|\rho_S - \rho\|_F$, where $\|\cdot\|_F$ is the Frobenius norm, then the optimal entanglement

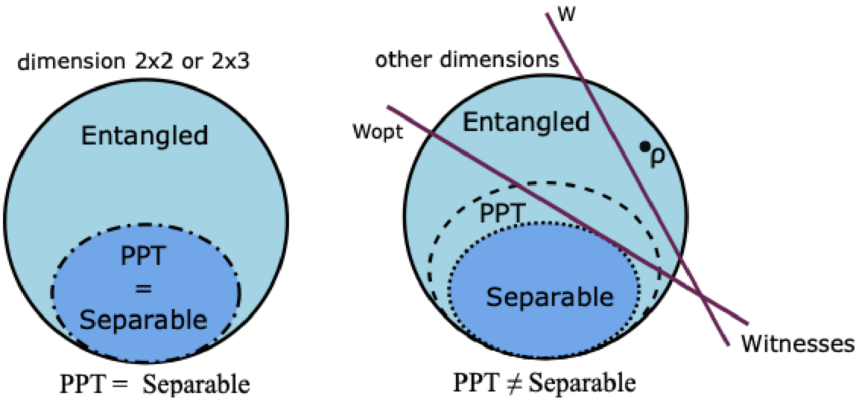


Figure 4.1: Schematic illustration of the set of separable, PPT and entangled bipartite quantum states. Left: When $p = p_A p_B \leq 6$, the PPT criterion is a necessary and sufficient condition for quantum separability, i.e., the set of separable states and the set of PPT states are the same. Right: However, when $p > 6$, the PPT criterion is only a necessary condition, such that all separable states are PPT, but there exist PPT states which are not separable (PPT-ENT or bounded entangled). If a state is not PPT (NPPT), then it is entangled. The hyperplanes W and W_{opt} are entanglement witnesses; they detect the entanglement of the entangled density matrix ρ. The entanglement witness W_{opt} defines a *tangent* hyperplane to the set of separable states, so it is optimal.

witness is defined as

$$W_{\rho,opt} = \frac{\tilde{\rho}_S - \rho - \langle \tilde{\rho}_S, \tilde{\rho}_S - \rho \rangle \mathbb{I}_p}{\|\tilde{\rho}_S - \rho\|_F}, \tag{47}$$

where $\mathbb{I}_p$ is the identity matrix, is an optimal entanglement witness Reinhold A. Bertlmann et al. 2005. It is easy to check that $\langle W_{\rho,opt}, \tilde{\rho}_S \rangle = 0$. The geometric interpretation of an optimal entanglement witness is that it defines a tangent hyperplane to the set of separable density matrices, as illustrated in Figure 4.1 (right).

4.3 A ML pipeline for quantum separability

4.3.1 Proposed strategy

As we have introduced before, a main limitation for using supervised ML to solve the large-scale entanglement problem is the absence of PPT-ENT training data. Without such examples, the learning process cannot properly characterise the boundary between separable and entangled states beyond the PPT criterion. To overcome this issue, we propose a method to generate PPT-ENT density matrices based on the concept of optimal entanglement witnesses.

Notice that if we have access to optimal entanglement witnesses of quantum states, we can always determine the states' separability by using Eqn.(46), which is a necessary and sufficient condition. However, the question we first faced is how to calculate the optimal

Algorithm 6 FW

Input: $f : \mathscr{S}_+^p \mapsto \mathbb{R}$ convex, differentiable function; $\mathscr{S}_+^{p,\otimes} \subset \mathscr{S}_+^p$ convex set
Output: $\sigma_T \in \mathscr{S}_+^{p,\otimes} \approx \arg\min_{\sigma \in \mathscr{S}_+^{p,\otimes}} f(\sigma)$
$\sigma_0 \leftarrow \sigma \in \mathscr{S}_+^{p,\otimes}$
for $t = 0$ to T **do**
  $s_t \leftarrow \arg\min_{s \in \mathscr{S}_+^{p,\otimes}} \langle \nabla f(s), s \rangle$
  $\gamma_t \leftarrow \frac{2}{t+2}$
  $\sigma_{t+1} \leftarrow (1-\gamma_t)\sigma_t + \gamma_t s_t$
end for
return σ_T

Algorithm 7 Approximated FW

Input: $\rho \in \mathscr{S}_+^p$ a density matrix
Output: $\rho_T \in \mathscr{S}_+^{p,\otimes}$ a separable density matrix approximating (48)
$\rho_0 \leftarrow |\psi_A\rangle\langle\psi_A| \otimes |\psi_B\rangle\langle\psi_B|$ ▷ random state
for $t = 0$ to T **do**
  $|\psi_t\rangle \leftarrow$ largest eigenvector of $(\rho - \rho_t)$
  $|a_1\rangle, |b_1\rangle \leftarrow$ largest Schmidt states [1] of $|\psi_t\rangle$
  $\gamma_t \leftarrow \frac{2}{t+2}$
  $\rho_{t+1} \leftarrow (1-\gamma_t)\rho_t + \gamma_t \tilde{s}_t$
end for
return ρ_T

entanglement witness for a given quantum state. Since the quantum state space is convex and positive semidefinite, the separable set is convex. Finding a point that minimises the Frobenius distance on the convex set is indeed a *constrained convex optimisation* problem. In the literature, a well-known algorithm, the Frank-Wolfe (FW) algorithm (Frank et al. 1956), also known as the conditional gradient method, has been leveraged to solve the problem (Dahl et al. 2007), which is depicted below in Algorithm 6.

Inspired by this work, we first develop a more efficient approximated FW algorithm to approximate the nearest separable state to a given density matrix. This allows us to construct optimal entanglement witnesses to some accuracy, then we label PPT-ENT states using Eqn.(46). Since the method is costly, a data augmentation strategy is proposed to generate more PPT-ENT states only using the already labelled ones and their optimal entanglement witnesses. In the end, we train ML classification models, such as kernel SVMs (see Section 2.2.3), neural networks (NNs), and XGBoost (eXtreme Gradient Boosting), to detect entanglement, where the simulation results show a considerable accuracy.

4.3.1.1 FW and an approximated FW algorithms

The Frank-Wolfe (FW) algorithm is a well-established method in machine learning for solving constrained convex optimisation problems, particularly due to its projection-free nature (Jaggi 2013). This makes it well-suited for high-dimensional settings. This approach was previously applied by Dahl et al. 2007, where the output of the FW algorithm converges to the nearest separable density matrix, and a convergence proof is provided in Chapter 2 of Bertsekas 1997.

Computing the nearest separable state to a given density matrix can be formulated as a constrained convex optimisation problem. Let $\rho \in \mathscr{S}_+^p$ be a density matrix. The nearest

1. The states corresponding to the largest Schmidt coefficient of a Schmidt decomposition.

separable state ρ^* is defined as the solution of

$$\rho^* := \underset{\sigma \in \mathcal{S}_+^{p,\otimes}}{\arg\min} f_\rho(\sigma), \tag{48}$$

where $f_\rho(\sigma) := \|\sigma - \rho\|_F^2$ [2], and $\mathcal{S}_+^{p,\otimes}$ denotes the set of separable density matrices. For minimising $f(\sigma)$, it is equivalent to minimise $f(\sigma) = \frac{1}{2}\|\sigma - \rho\|_F^2$, with aim to simply calculation, where we define $M_t = -\nabla f(\sigma_t) = \rho - \sigma_t$.

At the k-th round, the linear subproblem in line 5 of Algorithm 6 is a standard linear minimisation problem, and it reduces to

$$\underset{|a_1\rangle \in \mathcal{H}_A, |b_1\rangle \in \mathcal{H}_B}{\arg\max} \big\langle (\rho - \sigma_k)\,|a_1\rangle \otimes |b_1\rangle\,, |a_1\rangle \otimes |b_1\rangle\,\big\rangle, \tag{49}$$

A standard approach to solve this problem is to use a block coordinate ascent method, alternating optimisation over $|a_1\rangle$ and $|b_1\rangle$. However, this iterative procedure becomes computationally expensive and quickly intractable, even for moderate system sizes.

To address this issue, we propose a computationally efficient approximation of the linear subproblem (49). Our approach consists of two main steps. At t-th iteration, first, we compute

$$|\psi_k\rangle = \arg \max_{|s\rangle \in \mathcal{S}_+^p} \Big\langle (\rho - \sigma_k)\,|s\rangle\,, |s\rangle\,\Big\rangle,$$

which corresponds to the eigenvector associated with the largest eigenvalue of $(\rho - \sigma_k)$. This step ignores the separability constraint and can be efficiently performed using the power method.

Next, we project this vector onto the set of separable states using the Schmidt decomposition (see Section 4.2.3),

$$|\psi_k\rangle = \sum_{i=1}^{r \le \min(p_A, p_B)} \lambda_i\,|a_i\rangle \otimes |b_i\rangle_k\,.$$

We then construct the approximation

$$\tilde{s}_k := |a_1\rangle\langle a_1| \otimes |b_1\rangle\langle b_1|_k\,,$$

where $|a_1\rangle_k$ and $|b_1\rangle_k$ are the quantum states corresponding to the largest Schmidt coefficient $\lambda_{1,k}$. This provides the best rank-one separable approximation of the pure state $|\psi_k\rangle$, i.e., $|a_1\rangle \otimes |b_1\rangle_k = \arg\min_i \|\,|\psi_k\rangle - |a_i\rangle \otimes |b_i\rangle\,\|$. Importantly, this step does not require computing the full Schmidt decomposition; only the leading singular vectors are needed, which keeps the computational cost low. The resulting approximated FW procedure for approximating the nearest separable density matrix is summarised in Algorithm 7.

2. We write $f(\sigma)$ instead of $f_\rho(\sigma)$ when the dependence on ρ is clear from the context.

4.3.2 Creating bounded entangled datasets

Training machine learning models for large-scale entanglement classification requires a high-quality dataset covering separable (SEP), non-PPT-ENT (NPPT-ENT), and PPT-ENT (PPT-ENT) states.

Generating data for SEP and NPPT-ENT classes is relatively straightforward. Separable states can be constructed by sampling random density matrices of sizes $p_A \times p_A$ and $p_B \times p_B$, and combining them using Eqn. (44), which guarantees separability by construction. For NPPT-ENT states, we sample random density matrices in $\mathscr{S}_+^p$ and apply the PPT criterion to identify those with non-positive partial transpose. We refer to Życzkowski, Penson, et al. 2011; Aubrun 2014; Aubrun et al. 2014 for standard methods to generate random density matrices.

The main difficulty lies in constructing PPT-ENT states. To address this, we first generate random density matrices in $\mathscr{S}_+^p$ and retain only those satisfying the PPT condition. These states may still be either separable or entangled. To distinguish between them, we approximate the nearest separable state using Algorithm 7, and construct the corresponding (approximate) optimal entanglement witness using Eqn. (47).

A PPT state ρ is labelled as entangled if the computed witness $\tilde{W}_{\rho,\mathrm{opt}}$ satisfies

$$\mathrm{tr}(\rho_S \tilde{W}_{\rho,\mathrm{opt}}) \geq 0 \quad \text{for all separable } \rho_S \in \mathscr{S}_+^{p,\otimes}, \quad \text{and} \quad \mathrm{tr}(\rho \tilde{W}_{\rho,\mathrm{opt}}) < 0,$$

as in Eqn. (46). Otherwise, the state is considered separable and discarded.

Since verifying these conditions over the entire set $\mathscr{S}_+^{p,\otimes}$ is intractable, we adopt a randomised approximation. For each candidate state, we generate 10^4 separable density matrices sampled randomly from $\mathscr{S}_+^{p,\otimes}$ and check the witness conditions on this finite subset. Only states that satisfy these conditions are retained as PPT-ENT. More precisely, we sample the 10^4 separable states with a randomly generated mixing number (Eqn. (44)) $m \in [1, \ldots, 2d^2]$ by using a pseudorandom number generator (PRNG) which is realised by NUMPY.RANDOM.RANDINT.

Data augmentation As we have talked about, generating PPT-ENT density matrices is computationally expensive, especially in high dimensions. To mitigate this issue, we introduce a data augmentation scheme inspired by the robustness of bipartite entangled states (Bandyopadhyay et al. 2008).

The main idea is to perturb a given PPT-ENT state while preserving both entanglement and the PPT property. Let $\rho \in \mathscr{S}_+^p$ be a PPT-ENT density matrix. As shown in ibid., Theorem 1, there exists a region R_ρ containing ρ such that all states in this region remain PPT-ENT.

This region admits the following analytical characterisation:

$$R_\rho := \left\{ (1-\mu)\big(\nu\rho + (1-\nu)\tfrac{\mathbb{I}}{p}\big) + \mu\sigma \;:\; \nu \in]f(W_\rho), 1[,\ \mu \in [0, g(\nu, W_\rho)[,\ \sigma \in \mathscr{S}_+^p \right\}, \tag{50}$$

where σ is a density matrix representing noise, $\mathbb{I}$ is the identity matrix, and the scalar functions are defined by

$$f(W_\rho) = \frac{1}{1+\lambda_\rho p}, \qquad g(\nu, W_\rho) = \min\left[\frac{1-\nu}{p-1-\nu}, \frac{\mathrm{num}(W_\rho^+)\{\nu(1+p\lambda_\rho)-1\}}{p\,\mathrm{tr}\big(W_\rho^+\big)+\mathrm{num}(W_\rho^+)\{\nu(1+p\lambda_\rho)-1\}}\right],$$

with $\lambda_\rho = -\mathrm{tr}\big(W_\rho \rho\big)$, $\mathrm{num}(W_\rho^+)$ the number of positive eigenvalues of W_ρ, and $\mathrm{tr}\Big(W_\rho^+\Big)$ their sum. All these quantities are computed from the entanglement witness W_ρ associated with ρ.

In practice, once a PPT-ENT state ρ and its corresponding (approximated) optimal witness W_ρ are obtained during the data generation phase, we sample new states from R_ρ to augment the dataset. This allows us to efficiently generate additional PPT-ENT samples while preserving their labels.

To further increase variability, we also apply random local unitary transformations of the form

$$\rho \mapsto U\rho U^\dagger, \qquad U = U_A \otimes U_B,$$

where U_A and U_B are unitary matrices acting on $\mathscr{H}_A$ and $\mathscr{H}_B$, respectively. These transformations preserve entanglement and separability, and thus provide additional diversity without altering the class labels.

As we can see, the two steps of calculating the nearest separable states and entanglement witness tests based on 10^4 randomly sampled points on $\mathscr{S}_+^{p,\otimes}$ are both done approximately. These two elements constitute the basic approach that we used to generate high-quality PPT-ENT training data in the ML pipeline. Hence, it is necessary to provide theoretical guarantees of the validity and effectiveness of these two methods.

In the next two sections, we will give a systematic examination of the two methods. This helps validate the quality of our generated large-scale dataset and helps to justify the effectiveness of the proposed quantum entanglement detection pipeline.

4.4 Analysis of the approximated-FW algorithm

We now define the exact additive-error quantity used in approximated FW proofs and show how to upper-bound it. For the standard linear minimisation problem at iteration k, we define the value

$$\mu_{\text{sep}}\ (M_k) := \min_{x\in\mathscr{S}_+^{p,\otimes}} \big\langle \nabla f(\rho_k), x\big\rangle = \max_{x\in\mathscr{S}_+^{p,\otimes}} \langle M_k, x\rangle = \big\langle M_k, s^*\big\rangle.$$

Since the space $\mathscr{S}_+^{p,\otimes} \subseteq \mathscr{S}_+^{p}$, the $\max_{x\in\mathscr{S}_+^{p,\otimes}} \langle M_k, x\rangle \le \max_{x\in\mathscr{S}_+^{p}} \langle M_k, x\rangle$. Let $\big(\mu_1, \psi_k\big)$ be the top eigenpair of M_k (so $\mu_1 \ge \mu_2 \ge \cdots \ge \mu_p$ and $\big\||\psi_k\rangle\big\| = 1$), then one has $\mu_{sep}(M_k) \le \mu_1(M_k)$.

For the inexact output $\tilde{s}_k = |a_1\rangle\langle a_1| \otimes |b_1\rangle\langle b_1|_k$, define the true additive error be

$$\delta_k^{\text{true}} := \mu_{\text{sep}}\ (M_k) - \langle M_k, \tilde{s}_k\rangle \ge 0. \tag{51}$$

Then it yields,

$$\delta_k^{\text{true}} \leq \mu_1 (M_k) - \langle M_k, \tilde{s}_k \rangle := \delta_k^{\text{ub}}, \tag{52}$$

where the true additive error is upper-bounded by a variable δ^{ub}. As we will show later, it is identified to control the convergence behaviours.

Theorem 15 (Primal Convergence)**.** *For each $k \geq 1$, if the upper bound of the true additive error $\delta_k^{ub} \leq \frac{\gamma_k C_f \delta}{2} = \frac{2\delta}{k+2}$, where $\delta \geq 0$ is the accuracy to which the internal linear sub-problems are solved ($\delta = 0$ in the exact FW algorithm). Then the iterate ρ_k of the approximated FW algorithm satisfies,*

$$f(\rho_k) - f(\rho^*) \leq \frac{2C_f}{k+2}(1+\delta) = \frac{4(1+\delta)}{k+2} = O(\frac{1}{k}), \tag{53}$$

where $\rho^* \in \mathscr{S}_+^{p,\otimes}$ is the optimal solution.

The proof is provided in Appendix B.1.

Theorem 16 (Suboptimal-region Confinement)**.** *For each $k \geq 1$, if the upper bound of the true additive error $\delta_k^{ub} \to \eta > 0$ $(k \to \infty)$, then one has that when $k \to \infty$, the iterate ρ_k of the approximated FW algorithm satisfies,*

$$\frac{1}{2}\|\rho_k - \rho^*\|_F^2 \leq f(\rho_k) - f(\rho^*) \leq \eta + o(1), \quad \forall \rho_k \in \mathscr{S}_+^{p,\otimes}, \tag{54}$$

where $\rho^* \in \mathscr{S}_+^{p,\otimes}$ is the optimal solution. It can be concluded that the modified FW algorithm outputs a sequence $\{\rho_k\}$ that asymptotically remains confined to a neighbourhood of the optimum, with the Frobenius radius being $\sqrt{2\eta}$.

The proof is provided in Appendix B.2.

Numerical evidence To test our theorem, we ran simulations on three classes of well-known families of quantum states (isotropic, Werner, and Horodecki states).

Isotropic states

$$\rho_{\text{iso}}(q) = \frac{1-q}{d^2}\mathbb{I}_p + q\left|\phi_d^+\right\rangle\left\langle\phi_d^+\right|, \tag{55}$$

where $0 \leq q \leq 1, \mathbb{I}_p$ is the identity operator on the joint space, and $\left|\phi_d^+\right\rangle = \frac{1}{\sqrt{d}}\sum_{i=1}^d |i\rangle_1|i\rangle_2$ is the maximally entangled state of local dimension $d = p_A = p_B$. When $q \in [0, \frac{1}{d+1}]$, the state ρ_{iso} is separable. When $q \in (\frac{1}{d+1}, 1]$, the state $\rho_{iso}(q)$ is entangled.

Werner states

$$\rho_{\text{Werner}}(q) = (1-q)\frac{2}{d(d+1)}P_{\text{sym}} + q\frac{2}{d(d-1)}P_{\text{as}}, \tag{56}$$

with $P_{\text{sym}} = \frac{1}{2}\left(\mathbb{I}_p + F_p\right)$, and $P_{\text{as}} = \frac{1}{2}\left(\mathbb{I}_p - F_p\right)$, where $F_p = \sum_{i,j}^d |i\rangle\left\langle j\right|_a \otimes |j\rangle\langle i|_b$ is the flip operator. The Werner states are separable for $q \in [0, \frac{1}{2}]$, and entangled for $q \in (\frac{1}{2}, 1]$.

Horodecki states The two-qutrit state ($p_A = p_B = 3$) β_q is the family of states that follows (Paweł Horodecki et al. 1999) (cyclic convention)

$$\beta_q = \frac{1}{21}\begin{pmatrix} 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \\ 0 & \beta_- & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \beta_+ & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \beta_+ & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & \beta_- & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \beta_- & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \beta_+ & 0 \\ 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \end{pmatrix}, \tag{57}$$

where $\beta_\pm = \frac{5}{2} \pm q$, and $q \in [-2.5, 2.5]$, bur one only need to consider $q \in [0, 2.5]$. It is known that β_q is separable for $q \in [0, 0.5]$, is PPT-ENT for $q \in (0.5, 1.5]$, and is NPPT-ENT for $q \in (1.5, 2.5]$.

In the experiments, we track the variables δ^{ub} of those states in different entanglement regions, and we examine the 1/2 squared Frobenius distance $f(\rho_k) = \frac{1}{2}||\rho_k - \rho||_F^2$ (F-distance). The results confirm that δ^{ub} of those states are classified into two cases exactly as in Theorems 15 and 16, and the distance function demonstrates different behaviours correspondingly.

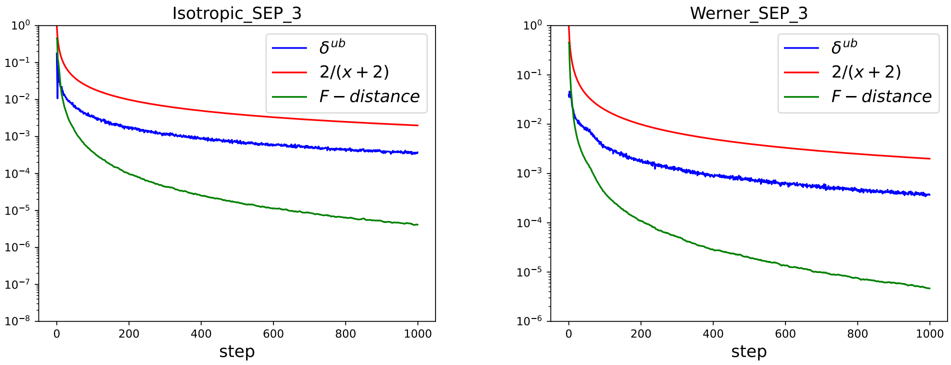
Primal Convergence : When $\delta_k^{ub} \leq \frac{\gamma_k C_f \delta}{2} = \frac{2\delta}{k+2}$, the algorithm shows primal convergence, such that for each $k \geq 1$, the iterates ρ_k of the approximated FW algorithm satisfies

$$f(\rho_k) - f(\rho^*) \leq \frac{2C_f}{k+2}(1+\delta) = \frac{4(1+\delta)}{k+2} = O(\frac{1}{k}),$$

where $\rho^* \in \mathscr{S}_+^{p,\otimes}$ is the optimal solution.

The simulation results show that, in the average sense, when the states ρ are random *separable* states from the families of isotropic and Werner states, the convergence follows Theorem 15.

In the log-plot figures (see Fig. 4.2 for d=3 and Fig. B.1 of Appendix B.1 for $d = 7$), we choose $\delta = 1$, and we take the average of 100 samples such that δ^{ub} and the F-distance are averaged over 100 experiments. From the empirical experiments, one can see that the δ_k^{ub} (blue line) non-monotonically decreases and keeps below the bound $\frac{2}{k+2}$ (red line), which indicates the convergence as in Theorem 15. In which case, the $\rho^* = \rho$ itself. Meanwhile, we can also see that the averaged F-distance (green line) is decreasing at a rate of $1/k$, where k is the number of steps (iterations) of our approximated-FW algorithm.



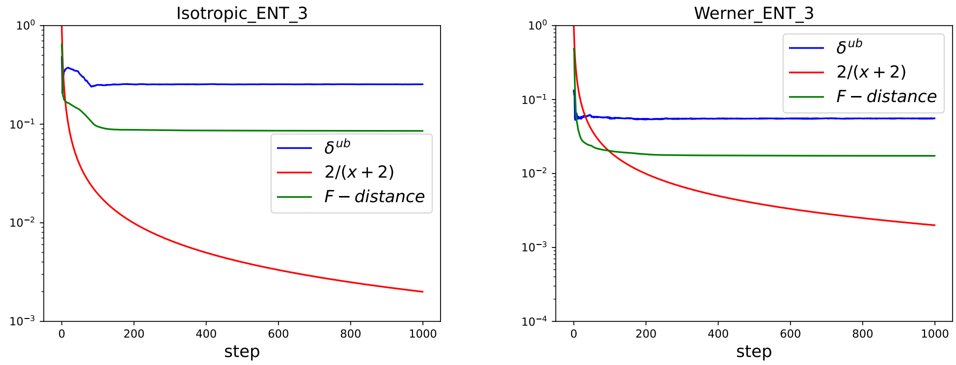
(a) Isotropic (3×3)

(b) Werner (3×3)

Figure 4.2: SEP results for isotropic and Werner states in $d = 3$.

Suboptimal-region Confinement : When $\delta_k^{ub} \to \eta > 0$, $(k \to \infty)$, the F-distance $\frac{1}{2}\|\rho_k - \rho^*\|_F^2 \le f(\rho_k) - f(\rho^*) \le \eta + o(1)$. In this case, $\limsup_{k\to\infty} f(\rho_k) - f(\rho^*) \le \eta$, and $\limsup_{k\to\infty} \left\|\rho_k - \rho^*\right\|_F \le \sqrt{2\eta}$. Asymptotically, the modified FW algorithm converges to a neighbourhood of the optimum, with the suboptimal points' Frobenius radius being $\sqrt{2\eta}$.

The simulation results show that, in the average sense, when the states ρ are random *entangled* states from the families of isotropic states, and Werner states, the output sequence behaviour follows Theorem 16. In the log-plot figures (see Fig. 4.3 for d=3 and Fig. B.2 of Appendix B.2 for $d = 7$), we choose $\delta = 1$, and we take the average of 100 samples as before. Note that when we only take a single sample, δ_k^{ub} first decreases non-monotonically and then fluctuates around a positive value without further decay. Averaging over 100 independent runs, it yields a similar trend, and shows asymptotic convergence to a positive constant. Hence, this corresponds to the case $\delta_k^{ub} \not\lesssim \frac{2}{k+2}$, but asymptotically, $\delta_k^{ub} \to \eta > 0$, which confirms hte behaviour follows Theorem 16. In which case, the output sequence of the approximated-FW algorithm is confined to a neighbourhood of the optimal separable state $\rho_k \neq \rho^*$, More precisely, the iterates remain asymptotically suboptimal, and their objective values (F-distances) satisfy, $\limsup_{k\to\infty} f(\rho_k) \le f(\rho^*) + \eta$.

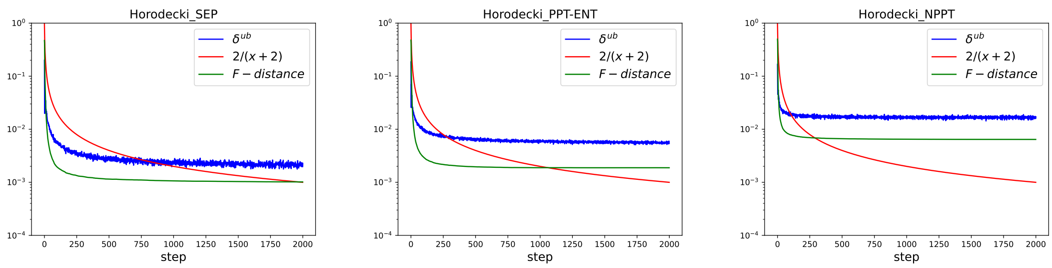


(a) Isotropic (3×3) (b) Werner (3×3)

Figure 4.3: ENT results for isotropic and Werner states in $d = 3$.

From the average performance of the experiments, one can conclude that, for the separable states of isotropic and Werner states families, the approximated FW algorithm performs as well as the standard FW (primal convergence). In contrast, in the entangled case of these two families, we see that the asymptotic approximation output lies in a suboptimal region.

The benchmarking tests on these two families indicate a high chance of effectiveness of our approximated FW algorithm. However, it is well known that there is no PPT-ENT zone for isotropic and Werner states, i.e. there is only PPT-SEP and NPPT. Hence, to evaluate the modified FW algorithm's performance in the PPT-ENT region, we consider the Horodecki-state family. In the Fig. 4.4 below, the empirical results show that for SEP, PPT-ENT, and NPPT Horodecki states, in an average sense, the modified FW algorithm follows the behaviour as in Theorem 16.



(a) SEP (b) PPT-ENT (c) NPPT

Figure 4.4: Results for Horodecki states.

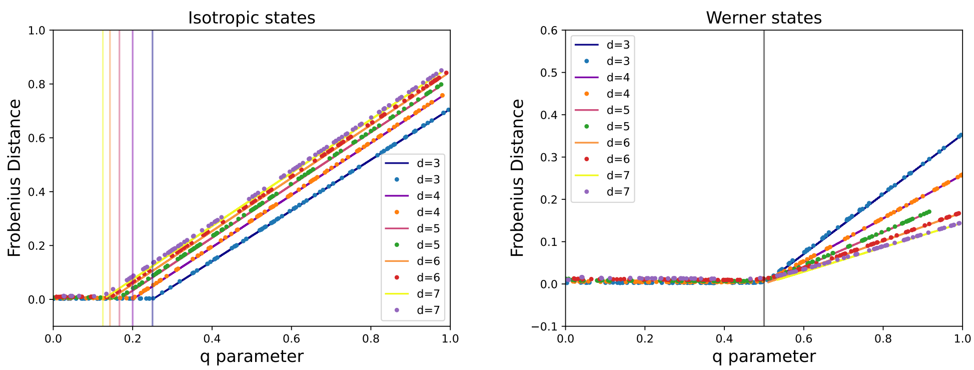
From the figures, we can see that, on average, for both separable and entangled cases of Horodecki states, the asymptotic value of ρ_k remains in a neighbourhood of the optimal.

Remarks From the above analysis and numerical experiments, we identify at least two cases in which the behaviour of the approximated FW algorithm is predictable. These cases are theoretically justified and empirically verified in an averaged sense, representing the dominant behaviours observed in practice.

However, we emphasise that these two cases do not provide a complete characterisation of all possible behaviours. In particular, we do not exclude the existence of other scenarios in which δ^{ub} is neither bounded by $O(1/k)$ nor converges to a positive constant. For example, it may diverge by oscillating around some value.

The conclusions drawn here should therefore be interpreted as a partial characterisation and empirical benchmarking of the behaviour of the approximated FW algorithm, rather than a complete classification. In particular, we have identified and analysed two representative cases, while other behaviours may coexist.

Algorithm performance benchmarking To better qualify the performance of our algorithm and have a better understanding of the approximation error as introduced in Theorem 16. We compare the empirical F-distance to its theoretical real value. For the isotropic states, the true Frobenius distance (Hilbert-Schmidt distance) is $\|\rho - \sigma^*\|_F = \frac{\sqrt{d^2-1}}{d}\left(q - \frac{1}{d+1}\right)$ (Reinhold A. Bertlmann et al. 2005). And for the Werner states, the distance has been conjectured as $||\rho - \sigma^*||_F = \frac{2}{\sqrt{d^2-1}}\left(q - \frac{1}{2}\right)$ in (Girardin et al. 2022). The following figures (see Fig. 4.5) show that the modified FW algorithm approximates the Frobenius distance well and predicts the separability turning point well (the thresholds are indicated with vertical lines).



(a) Isotropic (b) Werner

Figure 4.5: Theoretical/Conjectured Frobenius distances and numerical results from approximated FW, with local dimension $p_A = p_B = d$.

In the Fig 4.6, we ran simulations on Horodecki states over the range from separable to entangled. We indicate the SEP/PPT-ENT/NPT boundaries with vertical solid lines. It can be seen that although the SEP/ENT turning point is not as sharp as in the isotropic/Werner states, it essentially arises starting from the SEP/ENT boundary.

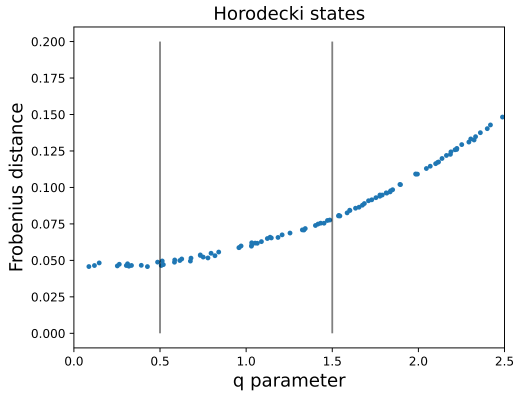


Figure 4.6: Frobenius distance of Horodecki state.

4.5 Entanglement witness test justification

As we discussed before, since there are infinitely many separable states in the set $\rho_S \in \mathscr{S}_+^{p,\otimes}$, we cannot check all these constraints. As a consequence, we adopt a simple numerical strategy by generating ten thousand separable density matrices *randomly* on the set $\mathscr{S}_+^{p,\otimes}$, checking the two assumptions, and selecting only the PPT states that satisfy the assumptions on these generated separable states.

Since the outputs of the modified-FW algorithm are approximated nearest, they either converge with a rate or are confined in a suboptimal region, the approximated-optimal witnesses can be non-witnesses that are not tangent to the set $\mathscr{S}_+^{p,\otimes}$ with some probability. Therefore, when an input data is PPT-ENT, and the output is an approximated-nearest separable state, the non-witness defines an area $\mathscr{S'}_+^{p,\otimes}$ that for all $\rho_{S'} \in \mathscr{S'}_+^{p,\otimes}$, $\mathrm{tr}(\rho_{S'}\tilde{W}_{\rho,opt}) < 0$.

Now assuming the random generated data ρ_S are *uniformly distributed* in the separable space, then the probability that an arbitrary $\rho_{i\in[N]} \in \rho_S$ falls into the the area $\mathscr{S'}_+^{p,\otimes}$ is defined by the quotient of volumes V, that $p = V(\mathscr{S'}_+^{p,\otimes})/V(\mathscr{S}_+^{p,\otimes})$, correspondingly the probability that a random separable state fall into the set $\mathscr{S}_+^{p,\otimes} - \mathscr{S'}_+^{p,\otimes} = 1-p$. the probability that all n separable states $\rho_{i\in[N]} \in \rho_S$ fall into the set $\mathscr{S}_+^{p,\otimes} - \mathscr{S'}_+^{p,\otimes}$ is thus $(1-p)^n$. In other words, a non-optimal witness has an escaping probability of $(1-p)^n$. When our approximated-FW algorithm satisfies primal error convergence or when the suboptimal region is very small, p is small. Thus, a good approximation can have a higher chance of escaping this witness test, and vice versa. To have an intuitive way to quantify the escaping probabilities, we give a few examples here: $(1-1/100)^{1,0000} \sim 10^{-44}$, $(1-1/1,000)^{1,0000} \sim 10^{-5}$, $(1-1/1,0000)^{1,0000} \sim 0.37$, and $(1-1/1,00000)^{1,0000} \sim 0.90$. From it, we can see that the smaller p is, the higher the chance of a non-witness escape.

By doing so, when the approximated nearest separable state is of high quality, its approximated witness is more likely to escape the test, and hence the PPT matrix is more likely to be labelled as PPT-ENT. And we view this as an akin-Monte-Carlo witness test, where a trade-off between accuracy and computational cost can be justified in an appropriate way. However, the real challenge here is how to guarantee a uniformly distributed $\rho_S \in \mathscr{S}_+^{p,\otimes}$.

As a counterexample, assuming the generated separable states ρ_S have a small mixing number (m is small), they will concentrate on the surfaces (faces) of the separable set. Hence, $p = Voln(\mathscr{S}'^{p,\otimes}_+)/Voln(\mathscr{S}_+^{p,\otimes})$ would be larger than the case when ρ_S is generated uniformly, since the former is calculated by projecting into a lower-dimensional space, thus the value, such that a non-witness has an escaping probability of $(1-p)^n$, can be biased-small with the same sample size n. For example, if p differs by a factor of 10 from $1/1,0000$ to $1/1,000$, then the escaping probability decreases exponentially from $(1-1/1,0000)^{1,0000} \sim 0.37$ to $(1-1/1,000)^{1,0000} \sim 10^{-5}$. Under this scenario, even if the approximated nearest separable state is of high quality, the non-witnesses can rarely escape. And as a consequence, even though they are PPT-ENT, they will be labelled as SEP and abandoned.

One may reckon it as a valid approach, since we do not rely on this method to generate SEP states, and the remaining PPT-ENT states with their witnesses passing the empirical tests are well-labelled. However, the practical issue exists; it will cause great difficulty in generating PPT-ENT data where every approximated witness with error is discarded. And this effect can get magnified with some types of data, for which the approximated FW algorithm has relatively large errors on (e.g. when the Frobenius radius $\sqrt{2\eta}$ is large). Or under certain situations, when the approximated-nearest separable state is relatively close to the SEP/ENT boundary (which could be the case in some regions where the PPT-ENT layer is very thin).

In conclusion, it seems essential to sample the separable set as uniformly as possible, but it is indeed tricky to generate true uniform samples since the separable set lives in high-dimensional space, and the high-dimensional convex set has non-trivial geometry. Thus, every sampling generator unavoidably introduces some bias. However, to avoid strongly biased tests, one should avoid sampling only small- or large-mixing-number states.

While the rank-1 separable states are on the vertices, it is known that the more interior states generally require a high mixing number, where Carathéodory's theorem guarantees that any separable state can be expressed as a convex sum of at most $m = p^2 = d^4$ product states. However, this is a worst-case bound, and a mixture of $O(d^2)$ pure-product states according to the Haar measure as in Eqn.(44) produces a full-rank state almost surely, where the $\mathrm{rank}(\rho) = \min\{d^2, m\}$. As a consequence, we think the mixture of $m \le 2d^2$ pure-product states should be able to explore the interior of the separable set, and we sample the mixing number from 1 to $2d^2$ to balance computational cost.

So, we apply a simple randomised strategy to sample the $1,0000$ separable states with a randomly generated mixing number between $[1,\ldots,2d^2]$ by using a pseudorandom number generator (PRNG) which is realised by NUMPY.RANDOM.RANDINT. To the best of our knowledge, this is the first practical and efficient method that can provide general PPT-ENT density matrices. Note that some examples of PPT-ENT states can be found in the literature, see, e.g., (Halder et al. 2019; Sindici et al. 2018). However, they are very specific and restricted to particular configurations (specific dimension requirements, for example), which renders them unsuitable for learning a good decision rule.

Numerical benchmarking To test the effectiveness and practicality of our witness-test method, we choose the known states with explicit entanglement regions, i.e. isotropic states, Werner states, and Horodecki states as our benchmarks, and examine the error rate of misclassification in local dimension $p_A = p_B = d$ from 3 to 10 examples each family, where the parameter q is uniformly sampled (see Eqns.(55), (56), (57)). With $|\rho_S| = n = 1{,}0000$ being the size of the separable set, we randomly generate $2d^2$ mixing numbers between $[1,\dots,2d^2]$ by using a PRNG, and balanced the ten-thousand samples such that for every mixing number there are $\sim \frac{1{,}0000}{2d^2}$ samples.

In the following, we present the results in the form of empirical mean ($\pm$ std) averaged over 5 repeated experiments to validate the stability. For reproducibility, we fixed samples isotropically, Werner and Horodecki states (100 samples each) in their corresponding dimensions by fixing a random seed (101). The results are presented below, where the mean and std is over 5 experiments with different randomly generated separable states.

n	Horodecki $SEP \to ENT$	Horodecki $PPT\text{-}ENT \to SEP$
10000	0.040 ± 0.017 (314s)	0.932 ± 0.035 (321s)
d	Isotropic $SEP \to ENT$	Werner $SEP \to ENT$
3	0.000 ± 0.000 (316s)	0.000 ± 0.000 (342s)
4	0.000 ± 0.000 (722s)	0.012 ± 0.004 (659s)
5	0.000 ± 0.000 (1597s)	0.032 ± 0.019 (1410s)
6	0.000 ± 0.000 (3164s)	0.066 ± 0.005 (2686s)
7	0.000 ± 0.000 (6094s)	**0.092** ± **0.012** (7673s)
8	0.000 ± 0.000 (10419s)	0.070 ± 0.013 (11159s)
9	0.000 ± 0.000 (22032s)	0.016 ± 0.010 (19926s)
10	0.000 ± 0.000 (36707s)	0.000 ± 0.000 (34385s)

Table 4.1: Mean and standard deviation over 5 independent experiments with the size of the separable set $n = 10000$, and randomly generated mixing number between $[1,\dots,2d^2]$. The time in the brackets is the average time taken to check 100 states in the corresponding local dimension.

From the experiments, we can see that there are no SEP isotropic states mislabelled as ENT, and for Werner states, the mislabelling rates are low (up to ~ 0.09 in $d = 7$). From the results of Horodecki states with the PPT-ENT region, an error rate of ~ 0.93 shows that a large portion of PPT-ENT states are labelled as SEP and thus discarded, while there is still a ratio of ~ 0.07 PPT-ENT states that are correctly labelled and used for creating the training dataset. This justified our method for labelling, which is highly selective and preserves utility at the same time.

It is well known that the separability problem is NP-hard, and what we do here is indeed trade accuracy for computational cost. Sometimes, a higher error in correct labels could be allowed, which is often the case for noisy datasets, and one can resort to noise-robust classification models, such as soft-margin SVMs, as we will show later. In this case, the tests can be less conservative, and the size $|\rho_S|$ can be further reduced. To give a benchmark, we reduce the size to $5{,}000$ and $2{,}500$, and we present the results in Tables C.1, C.2 of Appendix C.1. We observed that the errors increase as the size decreases. In the experiments, we notice that the witness tests for size $n = 1{,}0000$ are very slow in higher local dimensions ($\sim 3s$ per state for $d = 3$, ~ 1 min per state for $d = 7$, and ~ 6 min per state for $d = 10$.). To address this issue, in the following section, we present a data augmentation method to efficiently generate enough (1,000) PPT-ENT states for training.

4.6 Experiments

Vector Representation of Density Matrices Following Lu et al. 2018, we adopt the generalised Gell-Mann matrices (GGM) to obtain a vector representation of density matrices (Reinhold A Bertlmann et al. 2008; Kimura 2003). These matrices generalise the Pauli matrices (for qubits) and the Gell-Mann matrices (for qutrits), corresponding to dimensions 2×2 and 3×3, respectively.

For a p-dimensional Hilbert space $\mathscr{H}$, there are p^2-1 GGM, defined as follows:

— $\frac{p(p-1)}{2}$ symmetric matrices: $S_{i,j} = E_{i,j} + E_{j,i}$, for $1 \le i < j \le p$,

— $\frac{p(p-1)}{2}$ antisymmetric matrices: $A_{i,j} = -iE_{i,j} + iE_{j,i}$, for $1 \le i < j \le p$,

— $(p-1)$ diagonal matrices:

$$D_k = \sqrt{\frac{2}{k(k+1)}}\left(\sum_{i=1}^{k} E_{i,i} - kE_{k+1,k+1}\right), \quad 1 \le k < p,$$

where $E_{i,j}$ denotes the $p\times p$ matrix with entry (i,j) equal to 1 and zeros elsewhere.

The set $\{G_l\} = \{S_{i,j}\} \cup \{A_{i,j}\} \cup \{D_k\}$ forms an orthogonal basis of traceless Hermitian matrices (Reinhold A Bertlmann et al. 2008). Any density matrix $\rho \in \mathscr{S}_+^p$ admits the Bloch representation

$$\rho = \frac{1}{p}\mathbb{I} + \sum_{l=1}^{p^2-1} \beta_l G_l,$$

where $\beta_l = \operatorname{tr}(G_l\rho)$, and the vector $\beta = (\beta_1, \ldots, \beta_{p^2-1}) \in \mathbb{R}^{p^2-1}$ is referred to as the Bloch vector. In the training of all the models, we adapted the GGM representation so that the input density matrices are of dimension p^2-1.

Dataset information We construct two fully labelled datasets of 6,000 bipartite density matrices following the procedure described in Section 4.3.2. Each dataset is a collection of pairs of input density matrices and labels indicating whether the corresponding density matrix is separable or entangled, and contains SEP, PPT-ENT, and NPPT density matrices with 2,000 examples each, where they are separated again into training sets and test sets, with 1,000 examples each.

For the generation of random density matrices, we build upon and adapt functions from the TOQITO library (Russo 2021), an open-source Python library for quantum information that provides many of the same features as the excellent MATLAB toolbox QETLAB (Johnston 2016). The datasets are generated in the same manner but differ in dimension.

In this work, we restrict our attention to the symmetric setting $p_A = p_B = d$, which simplifies the analysis compared to the general case. In particular, it avoids mixed-dimension scenarios, e.g., 4×9 (a qubit and a qutrit), while still capturing the essential difficulty of the problem.

We first consider the case $d = 3$, which is the smallest dimension where the PPT criterion is no longer sufficient to fully characterise separability. This regime has been studied by Lu et al. 2018 using the bagging convex hull approximation (BCHA) method. We then consider $d = 7$, corresponding to density matrices acting on $\mathbb{C}^{d^2} \otimes \mathbb{C}^{d^2} = \mathbb{C}^{2401}$, where entanglement detection becomes significantly more challenging.

In this regime, an efficient algorithm is essential, as has been shown by Girardin et al. 2022, where the authors utilise generative NNs to produce the nearest separable state. In our experiments, we aim to provide a comparison with the previous works (Lu et al. 2018; Girardin et al. 2022) on 3×3 and 7×7 using the proposed approximated FW algorithm, both in classification accuracy and algorithmic efficiency.

4.6.1 Optimal training sizes

We make more simulations to find optimal training sizes. Specifically, we also trained our SVMs in sizes 100, 500, and 1,000 for separable and entangled, respectively, and we also assigned a ratio (0, 0.25, 0.5, 0.75, 1) to represent how much PPT-ENT are used out of ENT data. We compare the average training scores of each model using 5-fold cross-validation on different sizes, and the results are shown in Table 4.2.

Table 4.2: Training scores of SVMs on local dimensions $d = 3$ (left) and $d = 7$ (right) random mixed states with 100/500/1,000 training examples for SEP and ENT per class. Both polynomial and RBF kernels are examined.

PPT (%)	Scores (100)	Scores (500)	Scores (1,000)	Scores (100)	Scores (500)	Scores (1,000)
0%	**0.870 (±0.014)**	**0.981 (±0.001)**	**0.995 (±0.001)**	0.624 (±0.034)	0.619 (±0.011)	0.626 (±0.004)
	without augmentation			Without augmentation		
25%	0.819 (±0.020)	0.976 (±0.002)	0.991 (±0.001)	0.663 (±0.016)	0.661 (±0.004)	0.666 (± 0.002)
50%	0.810 (±0.033)	0.974 (±0.002)	0.992 (±0.002)	0.753 (±0.014)	0.749 (±0.008)	0.747 (± 0.003)
75%	0.817 (±0.041)	0.978 (±0.001)	0.993 (±0.002)	0.831 (±0.023)	0.843 (±0.006)	0.842 (± 0.002)
100%	0.784 (±0.030)	0.978 (±0.002)	0.993 (±0.002)	**0.936 (±0.010)**	**0.930 (±0.006)**	**0.931 (± 0.001)**
	with augmentation			With augmentation		
25%	0.843 (±0.031)	0.983 (±0.005)	0.989 (±0.002)	0.620 (±0.006)	0.624 (±0.002)	0.628 (±0.003)
50%	0.794 (±0.012)	0.971 (±0.005)	0.990 (±0.002)	0.712 (±0.011)	0.728 (±0.002)	0.732 (±0.003)
75%	0.775 (±0.016)	0.974 (±0.003)	0.991 (±0.003)	0.788 (±0.002)	0.819 (±0.003)	0.825 (±0.006)
100%	0.765 (±0.024)	0.975 (±0.004)	0.992 (±0.003)	0.835 (±0.016)	0.893 (±0.008)	0.916 (±0.003)
Average	0.810	0.977	0.992	0.751	0.763	0.768

From the tables, one can see that increasing the training data size from 100 to 500 to 1,000 brings stable improvement for the local dimension 3 dataset. While for the local dimension 7 dataset, we notice that the test accuracy does not change much when enlarging the size from 100 to 500, and even less when the size increases from 500 to 1,000. Hence, we reckon that the training size of 1,000 per SEP/ENT data should suffice.

4.6.2 Classification accuracy-SVMs, NNs

We now evaluate the accuracy performance of the classifiers.

SVMs For SVMs, noticing that the drop in accuracy from local dimension 3 to 7 is mainly due to the misclassification of NPPT data, as shown below in Table 4.3 for $d = 3$, and Table 4.4 for $d = 7$.

Table 4.3: Classification accuracy of SVM on 3-dimensional qudit dataset with 1,000 examples per class. For $d = 3$, we notice that polynomial kernels perform better than RBF kernels.

PPT (%)	SEP (3)	PPT-ENT (3)	NPPT (3)	Total (3)	Total 2 (3)
0%	0.995 (± 0.000)	0.969 (± 0.002)	0.997 (± 0.000)	0.987 (± 0.001)	0.982 (± 0.001)
			without augmentation		
25%	0.994 (± 0.001)	0.985 (± 0.004)	0.998 (± 0.001)	0.992 (± 0.001)	0.989 (± 0.002)
50%	0.995 (± 0.001)	0.991 (± 0.002)	0.999 (± 0.001)	0.995 (± 0.001)	0.993 (± 0.001)
75%	0.995 (± 0.000)	0.994 (± 0.002)	1.000 (± 0.000)	0.996 (± 0.001)	0.995 (± 0.001)
100%	0.994 (± 0.000)	0.994 (± 0.001)	1.000 (± 0.000)	0.996 (± 0.000)	0.994 (± 0.001)
			with augmentation		
25%	0.993 (± 0.001)	0.990 (± 0.005)	0.998 (± 0.001)	0.994 (± 0.001)	0.992 (± 0.002)
50%	0.994 (± 0.001)	0.998 (± 0.001)	1.000 (± 0.000)	0.997 (± 0.001)	**0.996 (± 0.001)**
75%	0.992 (± 0.003)	0.999 (± 0.001)	1.000 (± 0.000)	0.997 (± 0.001)	**0.996 (± 0.002)**
100%	0.992 (± 0.001)	1.000 (± 0.000)	1.000 (± 0.000)	0.997 (± 0.000)	**0.996 (± 0.001)**

Table 4.4: Classification accuracy of SVM on 7-dimensional random mixed (generated similarly with the training data) states with 1,000 test examples per class. For $d = 7$, both polynomial and RBF kernels are selected via grid search. The average (± std) is taken over 5 independent experiments.

PPT (%)	SEP (7)	PPT-ENT (7)	NPPT (7)	Total (7)	Total 2 (7)
0%	0.596 (± 0.023)	0.780 (± 0.110)	0.662 (± 0.058)	0.679 (± 0.025)	0.688 (± 0.066)
			without augmentation		
25%	0.740 (± 0.001)	0.996 (± 0.000)	0.425 (± 0.004)	0.720 (± 0.002)	0.868 (± 0.001)
50%	0.867 (± 0.002)	0.977 (± 0.002)	0.261 (± 0.004)	0.702 (± 0.001)	0.922 (± 0.001)
75%	**0.903 (± 0.012)**	**0.957 (± 0.008)**	0.199 (± 0.015)	0.686 (± 0.004)	**0.930 (± 0.002)**
100%	0.944 (± 0.000)	0.910 (± 0.000)	0.133 (± 0.001)	0.662 (± 0.000)	0.927 (± 0.000)
			with augmentation		
25%	0.997 (± 0.001)	0.646 (± 0.039)	0.029 (± 0.005)	0.557 (± 0.014)	0.822 (± 0.019)
50%	1.000 (± 0.000)	0.168 (± 0.016)	0.000 (± 0.000)	0.389 (± 0.005)	0.584 (± 0.008)
75%	1.000 (± 0.000)	0.007 (± 0.004)	0.000 (± 0.000)	0.336 (± 0.001)	0.504 (± 0.002)
100%	1.000 (± 0.000)	0.000 (± 0.000)	0.000 (± 0.000)	0.333 (± 0.000)	0.500 (± 0.000)

This can be solved simply by introducing the PPT criteria together with our classifiers. In the two tables, we also present the accuracies taken over SEP and PPT-ENT data only in the columns Total 2 (d). It can be seen that after applying the PPT criterion, the accuracy is increased up to 0.996 for local dimension 3 (see Table 4.3 above) and up to 0.930 for local dimension 7 (see Table 4.4 above).

The results indicate that using labelled PPT-ENT density matrices in the training process could improve the classification accuracy, and it suggests that further improvements could be achieved for SVMs by incorporating multiple entanglement detection criteria.

NNs From the results in (Casalé, Di Molfetta, Anthoine, et al. 2023, Table 3), we can see the accuracy for the Neural Networks in local dimension 3 is high (~ 0.98), while in local dimension 7 it drops to (~ 0.8). So we also tried to add more structure to the NNs for learning the data in $d = 7$. First, we normalise the training data. Then we devised a deeper NN using 7 hidden layers, with 2d-d-1024-512-128-32-1 neurons in each layer, with ReLU activation, batch normalisation, and Adam with a learning rate of 0.001 as before. We also tried to increase the training time (longer epoch) or use a larger batch size with ReLU/GeLU activation functions, but we did not observe any significant classification accuracy improvement comparing with the results in (ibid., Table 3), the new additional NN results are shown in the Table C.3 of Appendix C.2.

4.6.3 More advanced ML method-XGBoost

So far, the best classification accuracy is achieved by SVMs combined with the PPT criterion, where in local dimension $d = 3$ it is up to 0.996, and in $d = 7$ it is up to 0.930 without data augmentation. While we notice that the performances are not ideal when using the PPT-ENT data augmentation in $d = 7$ (up to 0.822, as shown in Table 4.4), which we will further discuss in the Section 4.7.1. Here, we explore the performances of other machine learning models other than SVMs and NNs. A potential candidate is the decision tree classification. Here, we use the XGBoost algorithm (Adam-Bourdarios et al. 2015) with parallelised boosted decision trees, and we achieve the (SEP/PPT-ENT) test accuracy up to ~ 0.98 for $d = 3$, ~ 0.95 (without augmentation) and ~ 0.94 (with augmentation) for $d = 7$. Here we present the test accuracy result below, Table 4.5.

Table 4.5: Classification accuracy of XGBoost methods on local dimensions $d = 3$ (left) and $d = 7$ (right) random mixed states with 1,000 training examples per class.

PPT (%)	SEP (3)	PPT-ENT (3)	NPPT (3)	SEP (7)	PPT-ENT (7)	NPPT (7)
0%	0.967 (±0.001)	1.000 (±0.000)	1.000 (±0.000)	0.914 (±0.003)	0.971 (±0.006)	0.937 (±0.003)
without augmentation				without augmentation		
25%	**0.967 (±0.001)**	**1.000 (±0.001)**	**1.000 (±0.000)**	0.921 (±0.006)	0.969 (±0.004)	0.936 (±0.006)
50%	0.962 (±0.001)	1.000 (±0.000)	1.000 (±0.000)	0.919 (±0.004)	0.969 (±0.003)	0.928 (±0.005)
75%	0.964 (±0.002)	1.000 (±0.000)	1.000 (±0.000)	0.930 (±0.003)	0.970 (±0.004)	0.925 (±0.003)
100%	0.965 (±0.003)	1.000 (±0.000)	1.000 (±0.000)	**0.940 (±0.005)**	**0.968 (±0.004)**	**0.922 (±0.004)**
with augmentation				with augmentation		
25%	0.965 (±0.003)	1.000 (±0.000)	1.000 (±0.000)	0.931 (±0.004)	0.959 (±0.006)	0.911 (±0.006)
50%	0.962 (±0.004)	1.000 (±0.000)	1.000 (±0.000)	0.948 (±0.004)	0.934 (±0.007)	0.880 (±0.008)
75%	0.961 (±0.003)	1.000 (±0.000)	1.000 (±0.000)	0.967 (±0.002)	0.909 (±0.010)	0.827 (±0.017)
100%	0.960 (±0.001)	1.000 (±0.000)	1.000 (±0.000)	0.991 (±0.003)	0.818 (±0.030)	0.687 (±0.031)

4.6.4 Bias tests

Learning behaviour with respect to k In this experiment, we train our SVMs and NNs using randomly generated density matrices according to Haar measurement that are based on sampling k independent pure states $\{|\psi_i\rangle\}$ on $\mathbb{C}^p$ (Życzkowski, Penson, et al. 2011), and considering their uniform mixture, $\rho = \frac{1}{k}\sum_{i=1}^{k} |\psi_i\rangle\langle\psi_i|$, which can also be generated by the induced measures that take partial trace over k of the composite $p \times k$ dimensional system (Zyczkowski and Sommers 2001).

It is well-known that when $k \to \infty$, the state converges to the maximally mixed state $\mathbb{I}_p/p$. It has been shown that for any $\epsilon > 0$, the following holds if the local dimension d is *large enough*. There is a threshold $cd^3 \leq s_0(d) \leq Cd^3 log^2 d$, where $C, c > 0$ are universal constants. When the environment dimension k is smaller than $(1-\epsilon)s_0$, the random induced state is entangled with overwhelming probability. When the environment dimension exceeds $(1+\epsilon)s_0$, the random induced state is separable with overwhelming probability (Aubrun et al. 2014; Aubrun 2014). It has been shown that the maximally mixed state $\mathbb{I}_p/p$ is at the centre of the separable set, and there are infinitely many extreme points (rank-1 pure state) that constitute the separable convex hull. There exists a maximal inscribed separable ball centred at the maximally mixed state, such that all the states inside the ball are separable. Specifically, in (Gurvits et al. 2002, Corollary 3), the authors proved that if a state ρ satisfies $||\rho - \mathbb{I}_p/p||_F \leq \frac{1}{\sqrt{p(p-1)}}$, then it is separable. And this confirmed the conjecture (Życzkowski, Paweł Horodecki, et al. 1998) that a bipartite state of a composite system with overall dimension p is separable if its purity $\mathrm{tr}(\rho^2) \leq \frac{1}{p-1}$. However, it is known that this is only a sufficient criterion but not a necessary one, so there can be separable states outside of the maximally inscribed ball.

As we discussed above, we trained our models on the separable states generated randomly as in Eqn.(44), and we generate each mixed state (PPT/NPPT) based on a uniform mixture of k independent pure states according to Haar measure. Then we tested the trained SVMs and NNs on $1,000$ test density matrices, each class is generated by the method, which suggests that our models can be biased. To know how biased our model is, we test the same trained models on other families of states. Specifically, we test the trained SVMs and NNs on the families of isotropic, Werner, and two-qutrit Horodecki states (see Eqns.(55), (56), (57)) and calculate the accuracies based on comparison with the true labels of each class.

In the bias tests, we confirmed that the accuracy of the soft margin kernel SVMs drops drastically for those families of states, which confirmed that the SVMs are indeed strongly biased. The results are shown in Tables C.5, C.6 of Appendix C.4. Additionally, we also run similar experiments using NN models, and the results are similar to SVMs, which are strongly biased. The NN results are shown Tables C.7, C.8 of Appendix C.4. However, these biased results are expected given that the separable convex hull in high dimensions can be extremely difficult to predict, and the generalisation ability can be naturally restricted by the *no free lunch* theory (Wolpert et al. 2002).

To address our concerns, we make two combined training datasets of size $2,000$ in local dimension $d = 3$ and $d = 7$, respectively. In $d = 3$, the SEP/ENT (PPT ratio ranging from $0 \sim 1$) data are of size $1,000$ each, composed of 400, 200, 200, 200 for the original test data, isotropic, Werner, and Horodecki states; in $d = 7$, the SEP/ENT (PPT ratio ranging from $0 \sim 1$) data are of size 1,000 each, composed of 600, 200, 200, for the original test data, isotropic and Werner states. We then train SVMs on them. During the experiments, we found that the models were hard to converge, so we set the maximum number of iterations for each training run to 2,000. The test accuracies show significant improvements for isotropic, and Werner states (Tables 4.6) compared with the results obtained with the original training set (400/200/200/200 per SEP/ENT).

Table 4.6: Classification accuracy (1000 test data each class) for isotropic states (left) and Werner states (right) with local dimensions $d = 3$ and $d = 7$ with the models trained on the combined dataset.

PPT (%)	SEP (3)	NPPT (3)
	without augmentation	
0%	1.000 (± 0.000)	0.974 (± 0.005)
25%	**1.000 (± 0.000)**	**0.975 (± 0.003)**
50%	1.000 (± 0.000)	0.970 (± 0.008)
75%	1.000 (± 0.000)	0.974 (± 0.009)
100%	1.000 (± 0.000)	0.962 (± 0.022)
	with augmentation	
0%	1.000 (± 0.000)	0.971 (± 0.004)
25%	1.000 (± 0.000)	0.966 (± 0.008)
50%	1.000 (± 0.000)	0.967 (± 0.005)
75%	1.000 (± 0.000)	0.974 (± 0.005)
100%	1.000 (± 0.000)	0.974 (± 0.005)
PPT (%)	**SEP (7)**	**NPPT (7)**
	without augmentation	
0%	**1.000 (± 0.000)**	**0.993 (± 0.003)**
25%	1.000 (± 0.000)	0.989 (± 0.010)
50%	1.000 (± 0.000)	0.992 (± 0.002)
75%	0.997 (± 0.003)	0.990 (± 0.003)
100%	0.989 (± 0.006)	0.992 (± 0.003)
	with augmentation	
0%	1.000 (± 0.000)	0.990 (± 0.001)
25%	0.984 (± 0.003)	0.992 (± 0.002)
50%	0.970 (± 0.003)	0.989 (± 0.002)
75%	0.931 (± 0.006)	0.989 (± 0.003)
100%	0.915 (± 0.003)	0.985 (± 0.003)

PPT (%)	SEP (3)	NPPT (3)
	without augmentation	
0%	0.999 (± 0.002)	0.993 (± 0.007)
25%	0.996 (± 0.003)	0.994 (± 0.011)
50%	0.997 (± 0.003)	0.997 (± 0.003)
75%	0.995 (± 0.007)	0.998 (± 0.003)
100%	0.991 (± 0.010)	1.000 (± 0.000)
	with augmentation	
0%	**0.999 (± 0.003)**	**0.995 (± 0.004)**
25%	0.996 (± 0.006)	0.993 (± 0.008)
50%	0.998 (± 0.003)	0.997 (± 0.003)
75%	0.994 (± 0.008)	0.999 (± 0.002)
100%	0.995 (± 0.004)	0.997 (± 0.004)
PPT (%)	**SEP (7)**	**NPPT (7)**
	without augmentation	
0%	0.913 (± 0.040)	1.000 (± 0.000)
25%	**0.925 (± 0.060)**	**0.999 (± 0.001)**
50%	0.869 (± 0.003)	1.000 (± 0.000)
75%	0.863 (± 0.001)	1.000 (± 0.000)
100%	0.884 (± 0.053)	1.000 (± 0.000)
	with augmentation	
0%	0.888 (± 0.008)	1.000 (± 0.000)
25%	0.857 (± 0.003)	1.000 (± 0.000)
50%	0.818 (± 0.010)	0.976 (± 0.048)
75%	0.773 (± 0.013)	1.000 (± 0.000)
100%	0.761 (± 0.006)	1.000 (± 0.000)

For the Horodecki states, however, we observe that SVMs, whether trained on the original dataset or the combined dataset, perform poorly. (see Table C.9 in Appendix C.3). We then train SVMs on only the Horodecki states, but we notice that the models don't converge and still perform poorly if we set a maximum number of iterations to 2000. We also evaluate the performance of the trained SVMs on 1,000 original test data points. In

$d = 3$, the accuracy is up to 0.972. However, the performances are much worse (≤ 0.573) in the $d = 7$ case (see Table C.10 in Appendix C.3).

To have a fair comparison, we discard the Horodecki states and train kernel SVMs only on the SEP/ENT (PPT ratio ranging from $0 \sim 1$) data, which are of size 1,000 each, composed of 600 original test data, 200 isotropic states, and 200 Werner states. We then create a test dataset with the same composition as the training data and present the performance below (Table 4.7). It can be seen that the performance in d=3 is still good (up to 0.988), while the test performances improve for $d = 7$ (up to 0.693) compared with the results in Table C.10. We think this could still be improved by more appropriate choices of learning models, composition of the training dataset, and training set sizes, which we leave for future work.

Table 4.7: Classification accuracy of SVMs trained on 3- and 7-dimensional combined training data (600/200/200 per SEP/ENT) with 1,000 combined test examples (600/200/200 per SEP/ENT) per class.

PPT (%)	SEP (3)	PPT-ENT (3)	NPPT (3)	Total (3)
0%	0.975 (± 0.016)	0.817 (± 0.019)	0.949 (± 0.010)	0.913 (± 0.006)
		without augmentation		
25%	0.976 (± 0.014)	0.882 (± 0.018)	0.965 (± 0.009)	0.941 (± 0.006)
50%	0.984 (± 0.001)	0.913 (± 0.013)	0.977 (± 0.008)	0.958 (± 0.006)
75%	0.971 (± 0.015)	0.947 (± 0.012)	0.987 (± 0.005)	0.968 (± 0.002)
100%	0.958 (± 0.021)	0.967 (± 0.009)	0.992 (± 0.002)	0.972 (± 0.005)
		with augmentation		
25%	0.956 (± 0.018)	0.932 (± 0.024)	0.983 (± 0.008)	0.957 (± 0.006)
50%	0.957 (± 0.014)	0.979 (± 0.015)	0.996 (± 0.004)	0.977 (± 0.003)
75%	0.976 (± 0.014)	0.981 (± 0.012)	0.996 (± 0.002)	0.984 (± 0.004)
100%	**0.972 (± 0.019)**	**0.995 (± 0.003)**	**0.999 (± 0.001)**	**0.988 (± 0.006)**
PPT (%)	SEP (7)	PPT-ENT (7)	NPPT (7)	Total (7)
0%	0.835 (± 0.015)	0.285 (± 0.008)	0.587 (± 0.007)	0.612 (± 0.002)
		without augmentation		
25%	0.782 (± 0.013)	0.402 (± 0.021)	0.645 (± 0.015)	0.642 (± 0.006)
50%	0.758 (± 0.017)	0.453 (± 0.025)	0.677 (± 0.013)	0.657 (± 0.005)
75%	0.731 (± 0.017)	0.505 (± 0.037)	0.715 (± 0.011)	0.673 (± 0.009)
100%	**0.685 (± 0.009)**	**0.599 (± 0.016)**	**0.758 (± 0.014)**	**0.693 (± 0.006)**
		with augmentation		
25%	0.695 (± 0.014)	0.586 (± 0.024)	0.738 (± 0.010)	0.686 (± 0.005)
50%	0.811 (± 0.188)	0.317 (± 0.388)	0.579 (± 0.223)	0.608 (± 0.103)
75%	0.947 (± 0.001)	0.000 (± 0.000)	0.397 (± 0.000)	0.517 (± 0.001)
100%	0.945 (± 0.003)	0.000 (± 0.000)	0.397 (± 0.000)	0.516 (± 0.001)

Here, we also present the Inference/Training time[3] of all the models we have examined

3. Timed using a machine with Intel Core i7 CPU, 2.21 GHz with 6 cores and 16 GB RAM.

in the Chapter. From the table, it can be seen that SVMs can take a long time for training (up to 7 hours), while NNs are faster (up to 30 min) but not very accurate on a larger scale $d = 7$. Meanwhile, XGBoost are both fast and accurate. We note that the training time of the models we evaluated in this work could still be potentially improved by choosing more appropriate training set sizes or compositions, or by applying more advanced hyper-parameter finding methods other than Grid-search, which we leave for future work.

Table 4.8: Inference/Training time (in seconds) of SVM/NN/XGBoost models on 3/7-dimensional random mixed states with 1,000 test examples per class. All the models are trained on 1000 SEP and 1000 ENT data. The SVM$_c$ are trained on the combined dataset of (600/200/200 original/isotropic/Werner states per class).

PPT (%)	SVM(3)	SVM$_c$(3)	NN(3)	XGB(3)	SVM(7)	SVM$_c$(7)	NN(7)	XGB(7)
0%	1/413	1/473	0.2/22	0.3/11	83/26844	54/10205	7/1190	0.6/463
without augmentation								
25%	1/387	1/469	0.2/22	0.3/10	87/21948	56/15895	7/1141	0.7/458
50%	1/388	1/464	0.2/22	0.3/10	88/20379	55/15242	7/1151	0.6/447
75%	1/416	1/452	0.2/22	0.3/10	89/19752	55/14904	7/1149	0.6/430
100%	1/455	1/445	0.2/22	0.3/10	89/27156	55/14676	7/1160	0.6/432
with augmentation								
25%	1/571	1/415	0.2/22	0.3/10	78/24857	51/14433	8/1877	0.5/452
50%	1/541	1/441	0.2/22	0.3/10	79/18157	60/20604	8/1825	0.5/446
75%	1/546	1/404	0.2/21	0.3/10	75/14628	64/20966	7/1763	0.5/440
100%	1/573	1/367	0.2/22	0.3/10	73/16181	64/18891	7/1824	0.5/434

4.7 Discussion and conclusions

4.7.1 Limitations

From our study, we noticed there are some limitations of our methods that can be further improved. For instance, first, as discussed in Section 4.4, based on the analysis and numerical experiments of the approximated FW algorithm, we identify two cases in which its behaviour is predictable. However, the conclusions should be interpreted as a partial characterisation and empirical benchmarking of the behaviour of the approximated FW algorithm, where other behaviours may coexist. A natural direction for addressing this limitation is to derive stronger theoretical guarantees for the approximation error of the algorithm. For instance, one may attempt to rule out non-convergent behaviours of δ^{ub}, or establish convergence or accuracy guarantees for the remaining cases, thereby providing a more complete characterisation of all possible behaviours.

Second, as has been mentioned in Section 4.5, the approximated FW algorithm outputs a nearest separable state with approximations, and thus the optimal witness $\tilde{W}_{\rho,opt}$

calculated using Eqn.(47) should be interpreted as an approximated result. Then, if it is indeed a non-witness, the data augmentation can introduce some errors since it assumes the use of a real witness in the closed-form expression of Eqn.(50). Therefore, the situation could be possible that although the input state ρ is labelled correctly as a PPT-ENT data by the witness test as in Section 4.5, the augmented data obtained from it can be SEP instead of PPT-ENT, and we think this may explain why in general the models with augmented data do not perform as well as the ones without. Therefore, one can characterise the accuracy of the augmentation method and explore other possibilities to improve it. By addressing this point, one may expect to achieve a better ML performance while preserving the fast PPT-ENT data generation via the augmentation scheme.

4.7.2 Future work

Another practical direction could be investigating a more uniform-like sampling strategy on the separable set, which we used for the entanglement witness test. And it's worth mentioning that there are some uniform sampling strategies in high-dimensional space, such as the hit-and-run walk and the ball walk, with the running time in POLY(P).

Meanwhile, directly from the experimental results, one can see that the classification accuracy of the SVMs combining with the PPT criterion, or XGBoost models are good for both $d = 3$ (0.996 for SVM of Table 4.3) and $d = 7$ (0.95 for XGBoost of Table4.5), when both the training data and the test data are generated the same way based on random matrices of Haar measure. This indicates the usefulness of our trained models and the high quality of the generated PPT-ENT training dataset.

In quantum information, it is an important task to certify entanglement (Friis et al. 2019), where people are generally interested in questions about the probability of finding a random entangled state, or computing the average entanglement of a mixed quantum state. For studying these questions, it is important to define a certain measure (Zyczkowski and Sommers 2001; Życzkowski, Paweł Horodecki, et al. 1998; Zyczkowski 1999). Here in this work, we present a resource-efficient ML pipeline under the induced measures.

We expect that one can use the witness test method equipped with the modified-FW algorithm that we proposed to certify bounded entanglement. However, as one may question, for testing quantum entanglement in physical experiments, our classifiers need the complete characterisation (the density matrix) of the tested quantum states, which is known to be achievable by quantum tomography (Salles et al. 2008). Still, the cost is huge, which is exponential in the number of qubits.

Hence, we view a strategy of using only partial information of quantum states as a next step in our methods. Under this case, one can still follow the same ML pipeline to label the training dataset, but only partial features will be used for training the ML models. Existing approaches include using measurement probabilities (Roik et al. 2021) or expectation values of selected observables (Ureña et al. 2023). Extending these ideas to higher-dimensional systems and autonomously generated datasets, as in our work, remains an open challenge.

Furthermore, the dataset generated in this work could also be leveraged to extract features for alternative entanglement detection methods, such as those based on Bell inequalities. For instance, CHSH-type observables (Ma et al. 2018) or measures such as the relative entropy of coherence (Asif et al. 2023) could be incorporated within the proposed framework.

In summary, we think that the ML pipeline we proposed and tested in this work can be used as simple prototypes for training and testing autonomously generated data. Future improvements could focus on refining the data generation process, conducting a more thorough error analysis, and developing more efficient and experimentally feasible detection methods.

5 Conclusion and further perspective

Sommaire

5.1 Quantum for ML

5.1.1 Other quantum perceptron algorithms

During the development of the main results in Chapter 3 of this thesis, we also explored several smaller directions and related sub-problems that were not fully completed. These lead to a number of open questions connected to the perceptron problem.

For example, we investigated an alternative strategy to approach the perceptron problem from a linear algebraic perspective, which is detailed in Section 5.1.1.1. In addition to the cutting plane method, we also explored the possibility of quantum speedups in other approaches, such as the volumetric centre method, which is discussed in Section 5.1.1.2.

5.1.1.1 A potential variant of online perceptron algorithm

After noticing the connection between searching for a feasible region (the perceptron problem) in the dual space and finding feasible solutions of a linear system of equations, we consider the possibility of designing alternative algorithms to solve this problem.

Formally, we aim to find a feasible solution to $A\boldsymbol{w} = \boldsymbol{b}$, where A is an $N \times D$ data matrix of the dataset, with entries A_{ij} (the i-th row, j-th column) representing the value of the j-th variable associated with the i-th data point, i.e., $A_{ij} = y_i x_{ij}$. The vector $\boldsymbol{b}$ is an arbitrary N-dimensional vector with strictly positive components. However, choosing $\boldsymbol{b} > 0$ randomly does not guarantee that the system admits a solution.

According to the Rouché-Capelli theorem, a necessary and sufficient condition for the system to have a solution is that the rank of the augmented matrix satisfies $r(A|\boldsymbol{b}) = r(A)$. This implies that $\boldsymbol{b}$ must lie in the column space of A, i.e., it must be a linear combination of the columns of A. Therefore, one strategy is to perform elementary column operations represented by an invertible matrix E, and construct a column of AE whose components are all positive. We may then choose $\boldsymbol{b}$ to be the j-th column $(AE)_j$. Solving $A\boldsymbol{w} = \boldsymbol{b}$ yields a valid solution $\boldsymbol{w}$.

To construct such a positive column, we consider the matrix $K = AA^T$, which takes the form of a kernel matrix with entries $K_{ij} = y_i y_j \boldsymbol{x}_i \cdot \boldsymbol{x}_j$. The diagonal entries satisfy $K_{ii} = 1$, while the off-diagonal entries are bounded by $|K_{i \neq j}| < 1$. One can then attempt to generate a positive column vector by performing elementary column operations E' on K, which basically constructs a linear combination of the columns of K such that all components become strictly positive. Concretely, starting from an initial column of K, one iteratively adds other columns whose entries are positive at indices where the current vector is negative, thereby reducing the magnitude of the negative components. By repeating this process, one attempts to eliminate all negative entries and obtain a vector with strictly positive components. However, this procedure does not necessarily converge in general, and such a positive combination exists only under additional feasibility conditions.

One may then choose $\boldsymbol{b}$ by letting $\boldsymbol{b} = (KE')_j$. In this case, the linear system $K\boldsymbol{w}' = AA^T\boldsymbol{w}' = A\boldsymbol{w} = \boldsymbol{b}$ admits a solution. After solving for $\boldsymbol{w}'$, the desired vector $\boldsymbol{w}$ can be recovered via $\boldsymbol{w} = A^T\boldsymbol{w}'$.

It is also known that if $r(A|\boldsymbol{b}) = r(A)$, then the system admits either a unique solution or infinitely many solutions, depending on the number of free parameters $k = D - r(A)$, where $r(A) \leq \min\{D, N\}$. If $k = 0$, the solution is unique; if $k > 0$, the solution space is infinite-dimensional and can be parameterised by k free variables. In either case, quantum linear system algorithms, such as the HHL algorithm, can in principle be applied to obtain a solution.

However, the overall complexity of this approach is limited by the classical procedure required to perform the elementary column operations. One may consider using Grover's search to accelerate the identification of negative entries in the vector during the iterative process. To fully validate this approach, it is necessary to establish convergence guarantees together with a rigorous analysis of the algorithmic complexity.

5.1.1.2 Vaidya's cutting plane method

While random walk methods generate samples from the current convex body and query their centroid at each iteration, the sampling process can be computationally expensive. Alternative approaches instead compute query points analytically, such as the volumetric center (P. M. Vaidya 1989), analytic center (Atkinson et al. 1995), and hybrid center (Lee, Sidford, and Wong 2015).

More recently, Jiang et al. (2020) achieved complexity $O\left(ND\log(\kappa) + D^3\log(\kappa)\right)$ using a multi-layered data structure, where κ here is a condition number measuring how "wide" or "well-conditioned" the feasible region is, typically the ratio of outer to inner radius. Since separation oracle complexity has a lower bound $\Omega(D\log(\kappa))$ (Nemirovskij et al. 1983), these cutting plane methods (P. M. Vaidya 1989; Bertsimas et al. 2004; Lee, Sidford, and Wong 2015; Jiang et al. 2020) achieve optimal query complexity. Moreover, under the Online Matrix-Vector conjecture, solving the feasibility problem requires $\Omega(D^3\log(\kappa))$

time (Jiang et al. 2020), implying optimal runtime.

In the volumetric centre method (P. M. Vaidya 1989), let $S = \{x : A\boldsymbol{x} \geq \boldsymbol{b}\} \subseteq P$ be the feasible region, where P is an initial polytope with a volumetric centre easy to compute, e.g. a D-simplex. Define

$$H(\boldsymbol{x}) = \sum_{i=1}^{N} \frac{\boldsymbol{a}_i \boldsymbol{a}_i^\top}{(\boldsymbol{a}_i^\top \boldsymbol{x} - b_i)^2}, \quad F(\boldsymbol{x}) = \frac{1}{2} \ln \det(H(\boldsymbol{x})), \tag{58}$$

where a_i being the i-th row of the data matrix A. The volumetric centre is defined as $\boldsymbol{z} = \arg\min_{\boldsymbol{x} \in P} F(x)$. The leverage scores are

$$\sigma_i(\boldsymbol{x}) = \frac{\boldsymbol{a}_i^\top H(\boldsymbol{x})^{-1} \boldsymbol{a}_i}{(\boldsymbol{a}_i^\top \boldsymbol{x} - b_i)^2}. \tag{59}$$

The gradient and approximate Hessian are

$$\nabla F(\boldsymbol{x}) = -\sum_{i=1}^{N} \sigma_i(\boldsymbol{x}) \frac{\boldsymbol{a}_i}{\boldsymbol{a}_i^\top \boldsymbol{x} - b_i}, \quad \nabla^2 F(\boldsymbol{x}) \approx Q(\boldsymbol{x}) = \sum_{i=1}^{N} \sigma_i(\boldsymbol{x}) \frac{\boldsymbol{a}_i \boldsymbol{a}_i^\top}{(\boldsymbol{a}_i^\top \boldsymbol{x} - b_i)^2}. \tag{60}$$

A Newton-type method converges in $O(D)$ iterations, using the volumetric centre (or an approximation $\boldsymbol{z}$) as the query point. At each iteration, if $\min_i \sigma_i(\boldsymbol{z}) \geq \varepsilon$, a separating hyperplane is added; otherwise, a constraint is removed. The update is

$$\boldsymbol{z} \leftarrow \boldsymbol{z} - 0.18\, Q(z)^{-1} \nabla F(\boldsymbol{z}). \tag{61}$$

The main computational cost arises from calculating $H(x)$ and inverting it, which requires $O(ND^2)$ and $O(D^3)$ time, respectively (reducible to $O(D^{2.373})$ via fast matrix multiplication).

Quantum considerations The bottleneck lies in computing $H(\boldsymbol{x})^{-1}$. One may apply the HHL algorithm (Aram W Harrow et al. 2009) to solve linear systems and obtain $|\boldsymbol{x}\rangle = A^{-1} |\boldsymbol{b}\rangle$. In particular,

$$\sigma_i(\boldsymbol{z}) = \frac{\langle a_i | H(\boldsymbol{z})^{-1} | a_i \rangle}{(\boldsymbol{a}_i^\top \boldsymbol{z} - b_i)^2}. \tag{62}$$

Given quantum states $|\boldsymbol{a}_i\rangle$, HHL can be used to compute $H(\boldsymbol{z})^{-1} |a_i\rangle$. Additionally, quantum inner product estimation (Xiong et al. 2024) enables matrix multiplication in time $O\left(\epsilon^{-1} \log(MND)\right)$, which may also be applied to the calculations of this problem.

5.1.1.3 Quantum perceptron algorithms for general LPs

As discussed in the conclusion of Chapter 3, convex minimisation problems can be reduced to feasibility problems (Nemirovski 1994), suggesting the possibility of extending our QCP-QW algorithm to more general optimisation tasks. However, our quantum speedup mainly improves the uniform sampling procedure (e.g., for estimating approximate centroids), rather than the underlying optimisation framework.

Since access models determine how much information can be retrieved, the achievable quantum speedup depends strongly on the model. In black-box settings, cutting plane methods admit a quadratic speedup via Grover search, reducing the cost of implementing the separation oracle. This leads to quantum algorithms with $\tilde{O}(\sqrt{ND})$ row-query complexity. However, such approaches do not reduce the iteration complexity of $\tilde{O}(D)$, and therefore could have a fundamental limit in reaching the lower bound of $\Omega(\sqrt{ND})$ row queries for solving LPs (Van Apeldoorn et al. 2017; Apers et al. 2026).

In contrast, in the white-box setting, the full problem structure is explicitly available. The methods discussed in the previous sections, including Vaidya's cutting plane method and the proposed linear-algebraic variant, fall into this category, as they require direct access to the data matrix and perform explicit linear-algebraic operations. In this setting, interior-point methods (IPMs) provide a powerful approach to solving LPs. Modern IPMs achieve $\tilde{O}(\sqrt{D})$ iterations (Lee and Sidford 2014; Lee and Sidford 2019), and the fastest classical implementations run in $\tilde{O}(ND + D^{2.5})$ time (Van Den Brand et al. 2021).

Very recently, a quantum IPM has been proposed (Apers et al. 2026), achieving a further quadratic improvement in the dependence on N with $\tilde{O}(\sqrt{N}\,\mathrm{poly}(D))$ row queries. This matches the lower bound in N, and it remains an open problem to improve the dependence on the dimension D.

5.1.2 Multi-armed bandits

From our research on the online perception and quantum weak online learning framework, we found that online learning is very interesting, as it requires exploring the best strategy to learn as it goes. This motivates us to explore more online algorithms and their possible quantum correspondence. Specifically, the stochastic multi-armed bandit (MAB) problem greatly drew our attention, as it is very actively used in real life and has a strongly developed theoretical background in history.

The problem is defined as follows. The stochastic MAB problem is a framework for sequential decision-making under uncertainty. Consider K arms, where each arm $i \in \{1,\dots,K\}$ is associated with an unknown reward distribution ν_i with mean μ_i. At each round $t = 1,2,\dots$, the learner selects an arm A_t and observes a stochastic reward drawn from ν_{A_t} (usually assume bounded value, sub-Gaussian, etc). The goal is to learn from past observations to make better decisions over time, balancing exploration and exploitation. Two central problems in the MAB literature are *regret minimisation* and *best arm identification* (BAI). Below, we introduce them briefly with some landmarks in the development of both classical and quantum, and point out some possible directions.

Regret minimisation Let $\mu^* = \max_i \mu_i$ denote the optimal expected reward. The cumulative regret after n rounds is defined as

$$R_n = \sum_{t=1}^{n} (\mu^* - \mu_{A_t}).$$

The objective is to design a policy such that the expected regret $\mathbb{E}[R_n]$ grows as slowly as possible, ensuring that the learner performs close to the optimal strategy over time.

The regret minimisation problem in a standard setting has been studied thoroughly. A well-known algorithm called UCB1 was proposed by Auer, Cesa-Bianchi, and Fischer 2002, which achieves logarithmic regret in the distribution-dependent setting. Subsequently, Audibert et al. 2009 established a minimax lower bound of $\Omega(\sqrt{nK})$ and proposed a randomised algorithm called minimax optimal strategy in the stochastic case (MOSS) matching this bound. These results indicate that optimal regret rates are essentially tight.

At the quantum side, the regret minimisation problem of stochastic linear bandit (SLB) has been studied by Wan et al. 2023, where rewards are accessed via queries to quantum oracles for each arm Using the quantum Monte Carlo technique (Montanaro 2015) combined with the upper confidence bound (ULB) strategy, the authors propose a quantum MAB that achieves regret of $O(\text{poly}(\log T))$, yielding an exponential improvement over the classical $\Omega(\sqrt{T})$ regret.

Best arm identification In contrast, the stochastic BAI focuses on pure exploration. The goal is to identify the optimal arm $i^* = \arg\max_i \mu_i$ using as few samples as possible. The learner sequentially selects arms and, after a stopping time τ, outputs an estimate $\hat{i}$. Under a *fixed confidence* constraint, for a given $\delta \in (0, 1)$, the algorithm is said to be δ-correct if $\mathbb{P}(\hat{i} \neq i^*) \leq \delta$. The performance is measured by the number of samples required to achieve this confidence. A key complexity measure is $H = \sum_{i \neq i^*} \frac{1}{\Delta_i^2}, \quad$ where $\Delta_i = \mu^* - \mu_i$. This quantity characterises the intrinsic difficulty of distinguishing the optimal arm from suboptimal ones, and governs the optimal sample complexity up to logarithmic factors.

In the classical setting, the upper confidence bound exploration (UCB-E) algorithm has been proposed by Audibert et al. 2010, with the number of samples required being $\tilde{O}(H)$ ($\tilde{O}$ refers to big-O notation up to logarithmic factors), matching the lower bound.

In the quantum setting, recent works of Casalé, Di Molfetta, Kadri, et al. 2020; D. Wang et al. 2021 show that a quadratic improvement of $\tilde{O}(\sqrt{H})$ is possible. In particular, Casalé, Di Molfetta, Kadri, et al. 2020 proposed a quantum algorithm based on Grover's search that reduces the number of queries. D. Wang et al. 2021 proposed a fixed-confidence quantum algorithm using variable-time amplitude amplification (VTAA) (Ambainis 2010) and estimation(VTAE) (Chakraborty et al. 2018) and proved it achieved the quantum query lower bound.

Remarks It seems that the stochastic MAB problem has been extensively studied under both problem setups, with many algorithms achieving performance matching known lower bounds. In addition, quantum algorithms have been proposed that offer improved performance under certain models.

Despite these advances, a wide range of problems remains open in more general settings. For instance, one may consider more complex reward structures, such as correlated arms, which are also known as structured bandits (Kleinberg 2004; Kleinberg et al. 2008; Bubeck et al. 2008), or problems with increased complexity, such as combinatorial bandits (W. Chen et al. 2013; S. Chen et al. 2014). Other formulations, including adversarial bandits (Auer, Cesa-Bianchi, Freund, et al. 2002), introduce additional challenges. For many of these settings, the development of corresponding quantum methods is still largely unexplored.

In the following, we introduce a recently proposed concept in which the arms are distributed over a graph, and review several recent works related to this framework.

5.1.2.1 Graph bandit

Proposed by T. Zhang et al. 2023, the graph bandit can be generally interpreted as a setting where the set of vertices of the graph is arms with probability reward functions, such that the action of pulling an arm is restricted to the arm's connectivity. The regret minimisation problem defined on it can be reformulated into a reinforcement learning (RL) problem under an (infinite horizon, non-episodic, undiscounted) Markov decision process (MDP), with known deterministic transitions, for which it is difficult to define a quantum analogy.

As we have introduced briefly in 2.1.5.2, quantum walk (QW) models are universal and are preferable for designing algorithms for search and graph-related problems. Recently, a quantum spatial best-arm identification (QSBAI) problem has been proposed and studied by Yamagami et al. 2025. Analogous to the graph bandit, where the vertices are arms, and the dynamics on it are restricted by the graph's connectivity. The authors applied the quantum bandit framework of Casalé, Di Molfetta, Kadri, et al. 2020 to solve the problem. In this work, they exchanged Grover's diffusion operator in standard Grover's search (see Figure 2.6) with a QW, since the former usually requires full connectivity of the graph, where QW is more versatile. In this line of research, it might be interesting to consider incorporating QW into VTAA and VTAE algorithms as have been used by (D. Wang et al. 2021) to develop a fixed-confidence QSBAI algorithm, in which case, the query complexity may depend on the graph's parameters.

Another direction is highly related to quantum reinforcement learning, which can be reformulated as a graph bandit regret minimisation problem. More recently, there are also a few works in this field, such as by H. Zhong et al. 2023 and by Ambainis et al. 2025. Yet, the MDPs are a very big field of research and generally require a good understanding of graph theory.

5.2 ML for quantum

In Chapter 4, we discussed an application of CaQd, where classical ML techniques are employed to address the quantum separability problem. More broadly, quantum information theory and quantum physics provide a rich setting for applying ML methods, as the data are typically high-dimensional and exhibit complex structure. In particular, in the presence of experimental noise, ML methods offer a practical trade-off between accuracy and computational cost.

As discussed in Section 4.7.2, our entanglement detection approach aims to provide a preliminary step towards constructing large-scale datasets of mixed quantum states under induced measurements. We hope this can motivate the development of methods that rely only on partial information or selected features of quantum states, which are more practical in experimental settings and can be used for alternative entanglement detection methods.

More generally, we expect that ML methods can play an important role in other challenging problems in quantum physics. Promising directions include predicting phase transitions (Carrasquilla and Melko 2017) and applying ML techniques to simulate open quantum system dynamics (Hartmann et al. 2019), both of which are known to be computationally demanding.

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ANNEXES

A Perceptron appendices

A.1 Asymptotic approximation

Here we provide more detailed analysis regarding the asymptotic approximation for the incomplete beta function of Lemma 9, where $\Pr\left[\{\boldsymbol{w} | \measuredangle(\boldsymbol{w}, \boldsymbol{u}) < \arcsin\gamma\}\right] = \frac{1}{2}\mathrm{I}_{\gamma^2}(\frac{D-1}{2}, \frac{1}{2})$.

We can validate Lemma 9 under boundary conditions first. When $\gamma = 0$ (inseparable), $I_0(a, b) = 0$, indicating the probability of finding a correct classifier is zero; when $\gamma = 1$ (two classes are maximally separated), $I_1(a, b) = \frac{1}{2}$, the probability of finding a correct classifier is $\frac{1}{2}$. These results are consistent with the actual classification situation.

On the other hand, $\frac{1}{2}\mathrm{I}_{\gamma^2}(\frac{D-1}{2}, \frac{1}{2}) = \frac{1}{2}\frac{\mathrm{B}(\gamma^2; \frac{D-1}{2}, \frac{1}{2})}{\mathrm{B}(\frac{D-1}{2}, \frac{1}{2})}$, where $\mathrm{B}(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$ denotes the beta function and $\mathrm{B}(x; a, b)$ denotes the incomplete beta function. Expressing it as a continued fraction expansion, $\mathrm{B}(x; a, b) = \frac{x^a(1-x)^b}{a(1+\frac{d_1}{1+}\frac{d_2}{1+}\frac{d_3}{1+}\frac{d_4}{1+}\cdots)}$, with coefficients $d_{2m+1} = -\frac{(a+m)(a+b+m)x}{(a+2m)(a+2m+1)}$, and $d_{2m} = \frac{m(b-m)x}{(a+2m-1)(a+2m)}$. Thus, in the limit of $\gamma \to 0$,

$$\mathrm{B}(\gamma^2; \frac{D-1}{2}, \frac{1}{2}) = \frac{2\gamma^{D-1}\sqrt{(1-\gamma^2)}}{(D-1)(1-O(\gamma^2))} \sim \frac{2\gamma^{D-1}}{(D-1)}.$$

When $D \geq 2$ is a constant, one has $\mathrm{B}(\frac{D-1}{2}, \frac{1}{2}) \leq \pi$, by property of gamma function. Accordingly,

$$\Pr\left[\{\boldsymbol{w} | \measuredangle(\boldsymbol{w}, \boldsymbol{u}) < \arcsin\gamma\}\right] \sim \frac{\gamma^{D-1}}{(D-1)} / \mathrm{B}(\frac{D-1}{2}, \frac{1}{2}) \geq \frac{\gamma^{D-1}}{\pi(D-1)}. \tag{63}$$

In the limit of $D \to \infty$, one can apply Stirling's approximation $\mathrm{B}(a, b) \sim \Gamma(b)a^{-b}$, such that

$$\Pr\left[\{\boldsymbol{w} | \measuredangle(\boldsymbol{w}, \boldsymbol{u}) < \arcsin\gamma\}\right] \sim \frac{\gamma^{D-1}}{(D-1)} / \sqrt{\frac{2\pi}{D-1}} \sim \frac{\gamma^{D-1}}{\sqrt{2\pi(D-1)}}. \tag{64}$$

A.2 Classical and Quantum Markov Chains

Let $P = (p_{xy})_{\boldsymbol{x},\boldsymbol{y}\in\Omega}$ denote the stochastic transition matrix of a Markov chain with finite state space Ω, where $|\Omega| = N$. The matrix element p_{xy} represents the transition probability from state $\boldsymbol{x}$ to $\boldsymbol{y}$, satisfying $\sum_y p_{xy} = 1$ for all $\boldsymbol{x} \in \Omega$. If P is *ergodic*, which implies irreducibility, the Perron–Frobenius theorem guarantees a unique eigenvalue-1 eigenvector $\boldsymbol{\pi}$ such that $\boldsymbol{\pi}^T P = \boldsymbol{\pi}^T$. The vector $\boldsymbol{\pi} = (\pi_{\boldsymbol{x}})_{\boldsymbol{x}\in\Omega}$ is called the stationary distribution. *Time-reversibility* means $P = P^*$, while *symmetry* means $P = P^T$; in the symmetric case, the stationary distribution is uniform.

The spectral gap of P is defined as $\delta(P) = 1 - |\lambda_1|$, where the eigenvalues satisfy $1 = \lambda_0 > |\lambda_1| \geq \ldots \geq |\lambda_{N-1}| \geq 0$. For a classical Markov chain, the mixing time $\tau(P)$ of an initial distribution $\boldsymbol{\rho}$ is defined by the condition $d_{TV}(P^\tau \boldsymbol{\rho}, \boldsymbol{\pi}) \leq \epsilon$ for some $1 > \epsilon > 0$, where d_{TV} is the total variation distance. It is known that the mixing time is upper bounded by the inverse of the spectral gap, i.e., $\tau(P) = O(1/\delta)$.

The quantum analogue of the spectral gap, called the phase gap, is defined as

$$\Delta(P) = 2\theta_1, \quad \text{with } \cos\theta_j = |\lambda_j|, \quad j \in [0, \cdots, N-1]$$

The phase gap represents the minimal angular distance of 1 to the other eigenvalues on the complex plane, and satisfies

$$\Delta(P) \geq 2\sqrt{\delta(P)}.$$

A.3 Quantum speedup

A.3.1 Proof of Theorem 12

For proving Theorem 12, we first need the following Lemmas 17, 18, and 19.

Lemma 17 (Effective spectral gap for warm start)**.** *Chakrabarti et al. 2023, Proposition 4.2, T. Li et al. 2022, Lemma 5 For an ergodic time-reversible Markov chain P_t, if the state $|\pi_{t+1}\rangle$ is a warm start of the stationary sates $|\pi_t\rangle$ which mixes up to a $d_{TV} \leq \epsilon$ after τ steps, then $|\pi_{t+1}\rangle = \left|\pi_{t+1,\, good}\right\rangle + |e_{t+1}\rangle$, where $\left|\pi_{t+1,\, good}\right\rangle$ lies in the subspace spanned by the fast-mixing eigenvectors $\left|\psi_j^{\pm}\right\rangle$ of $W(P_t)$, such that its eigenvalue $e^{\pm 2\pi i\theta_j}$ satisfies $\theta_j = 0$ or $\theta_j = \Omega(\sqrt{1/\tau})$. And the state $|e_{t+1}\rangle$ is spanned by those slow-mixing eigenvectors whose $0 < \theta_i < O(\sqrt{1/\tau})$, has its ℓ_2 norm bounded by $\| |e_{t+1}\rangle \| \leq \sqrt{\epsilon}$.*

Lemma 18 (Grover's $\frac{\pi}{3}$ fix point search)**.** *Wocjan et al. 2008, Lemma 1 Let $|\pi_t\rangle$, $|\pi_{t+1}\rangle$ be two arbitrary quantum states with overlap $|\langle \pi_t|\pi_{t+1}\rangle|^2 \geq p$ for some $0 < p \leq 1$ is a constant. Given $|\pi_t\rangle$, one is able to prepare a state through $|\tilde{\pi}_{t+1}\rangle = U_{t,m}|\pi_t\rangle$, such that $\| |\tilde{\pi}_{t+1}\rangle - |\pi_{t+1}\rangle \| \leq \epsilon_1$ for any $\epsilon_1 > 0$, by choosing $m = O(\log(\log 1/\epsilon_1))$, which invokes the unitaries from $\{R_t, R_t^\dagger, R_{t+1}, R_{t+1}^\dagger\}$ at most $M = O(\log 1/\epsilon_1)$ times. Here the form of unitaries $U_{t,m}$ is defined recursively, such that $U_{t,0} = \mathbb{1}$, and $U_{t,i+1} = U_{t,i} \cdot R_t \cdot U_{t,i}^\dagger \cdot R_{t+1} \cdot U_{t,i}$. The unitaries R_t, R_{t+1} operators are selective phase shifts that $R_t = e^{\frac{i\pi}{3}}\Pi_t + \Pi_t^\perp$, where Π_t is the orthogonal projector onto $\mathrm{span}\{|\pi_t\rangle\}$.*

Proof. The state $|\pi_t\rangle$ is a uniform superposition of states on the convex body $\mathscr{P}_t$. It can be decomposed in the subspace $\mathscr{H}_{\pi_t} = \mathrm{span}\{|\pi_{t+1}\rangle, |\pi_{t+1}^\perp\rangle\}$ as $|\pi_t\rangle = \sin\theta_{t+1}|\pi_{t+1}\rangle + \cos\theta_{t+1}|\pi_{t+1}^\perp\rangle$, where

$$|\pi_{t+1}\rangle = \frac{\Pi_{\mathscr{P}_{t+1}}|\pi_t\rangle}{\|\Pi_{\mathscr{P}_{t+1}}|\pi_t\rangle\|} = \frac{\sum_{\boldsymbol{x}\in\mathscr{P}_{t+1}} |\boldsymbol{x}\rangle\langle\boldsymbol{x}| \sum_{\boldsymbol{x}'\in\mathscr{P}_t} \sqrt{\pi_{t,\boldsymbol{x}'}}|\boldsymbol{x}'\rangle|\boldsymbol{p}_x'\rangle}{\sqrt{\langle\pi_t|\Pi_{\mathscr{P}_{t+1}}|\pi_t\rangle}} = \frac{\sum_{\boldsymbol{x}\in\mathscr{P}_{t+1}} \sqrt{\pi_{t,\boldsymbol{x}}}|\boldsymbol{x}\rangle|\boldsymbol{p}_x\rangle}{\sqrt{\sum_{\boldsymbol{x}\in\mathscr{P}_{t+1}} \pi_{t,\boldsymbol{x}}}},$$

and its orthogonal complement

$$|\pi^{\perp}_{t+1}\rangle = \frac{(\mathbb{1}-\Pi_{\mathscr{P}_{t+1}})|\pi_t\rangle}{\|(\mathbb{1}-\Pi_{\mathscr{P}_{t+1}})|\pi_t\rangle\|} = \frac{\sum_{\boldsymbol{x}\notin\mathscr{P}_{t+1}}|\boldsymbol{x}\rangle\langle\boldsymbol{x}|\sum_{\boldsymbol{x}'\in\mathscr{P}_t}\sqrt{\pi_{t,\boldsymbol{x}'}}|\boldsymbol{x}'\rangle|\boldsymbol{p}'_x\rangle}{\sqrt{\langle\pi_t|(\mathbb{1}-\Pi_{\mathscr{P}_{t+1}})|\pi_t\rangle}} = \frac{\sum_{\boldsymbol{x}\notin\mathscr{P}_{t+1}}\sqrt{\pi_{t,\boldsymbol{x}}}|\boldsymbol{x}\rangle|\boldsymbol{p}_x\rangle}{\sqrt{\sum_{\boldsymbol{x}\notin\mathscr{P}_{t+1}}\pi_{t,\boldsymbol{x}}}}.$$

A transition from $|\pi_t\rangle$ to $|\pi_{t+1}\rangle$ can be realized by the Grover's $\pi/3$ amplitude amplification algorithm (Lov K Grover 2005), such that there is a rotation $U_{t,m}$ yields $|\langle\pi_{t+1}|U_{t,m}|\pi_t\rangle|^2 \geq 1-(1-p)^M$. From Wocjan et al. 2008, Corollary 1, to make sure $\||\tilde{\pi}_{t+1}\rangle - |\pi_{t+1}\rangle\| \leq \epsilon_1$, one needs $M = O(\frac{\log\frac{1}{\epsilon_1}}{\log\left(\frac{1}{1-p}\right)})$ uses of R_t, R_{t+1} and theirs inverse. □

Lemma 19. *Let P_t be an ergodic and symmetric Markov chain, the state $|\pi_{t+1}\rangle$ be a warm start of the stationary uniform state $|\pi_t\rangle$ which mixes up to a $d_{TV} \leq \epsilon_3$ after τ steps. To implement R_{t+1}, one needs one quantum query as defined in Eqn.(34) with one ancillary qubit. To approximately implement R_t up to an amplitude error $O(\sqrt{\epsilon_3})$, a number of ancillary qubits $ac = O(\log\sqrt{\tau}\log(1/\sqrt{\epsilon_3}))$ suffice with $O(\sqrt{\tau}\log\frac{1}{\sqrt{\epsilon_3}})$ invocations of* CONTROLLED-*W_t gates.*

Proof. It can be easily verified that if we have a unitary P_{t+1}, that $P_{t+1}|\pi_{t+1}\rangle|0\rangle = |\pi_{t+1}\rangle|0\rangle$ and $P_{t+1}|\pi^{\perp}_{t+1}\rangle|0\rangle = |\pi^{\perp}_{t+1}\rangle|1\rangle$, then R_{t+1} can be constructed by letting

$$R_{t+1} = P^{\dagger}_{t+1}(\mathbb{1}\otimes(e^{\frac{i\pi}{3}}|0\rangle\langle 0| + |1\rangle\langle 1|))P_{t+1} \tag{65}$$

as shown by ibid. From the action of P_{t+1}, one can see that P_{t+1} is effectively a quantum query that is similar to the Eqn.(34), such

$$P_{t+1}|\boldsymbol{w}\rangle|\boldsymbol{x}'_{t+1}y'_{t+1},\boldsymbol{z}_t\rangle|0\rangle = |\boldsymbol{w}\rangle|\boldsymbol{x}'_{t+1}y'_{t+1},\boldsymbol{z}_t\rangle|0\oplus f_{\boldsymbol{w}}(\boldsymbol{x}'_{t+1}y'_{t+1},\boldsymbol{z}_t)\rangle,$$

where $f_{\boldsymbol{w}}(\boldsymbol{x}'_{t+1}y'_{t+1},\boldsymbol{z}_t) = 1$ iff $(\boldsymbol{w}-\boldsymbol{z}_t)^T\boldsymbol{x}'_{t+1}y'_{t+1} \leq 0$ ($\boldsymbol{w}\in\mathscr{P}_{t+1}$). Therefore, R_{t+1} can be implemented simply with one quantum query and one ancillary qubit.

The realisation of R_t is more computationally expensive. Here, we adopt the quantum walk search scheme proposed by Magniez et al. 2007. The phase estimation (Cleve et al. 1998) circuit $PE(W_t)$, composed of CONTROLLED-W_t gates, can be leveraged to realise unitaries as R_t. It is further analysed in Wocjan et al. 2008, Corollary 2, that for some $\epsilon_2 > 0$, there is a quantum circuit V_t that acts on $\mathbb{C}^N\otimes\mathbb{C}^N\otimes(\mathbb{C}^2)^{\otimes ac}$. It uses the $PE(W_t)$ circuit c times with ac ancillary qubits, where $a = O(\log\frac{1}{\Delta})$, $c = O(\log\frac{1}{\sqrt{\epsilon_2}})$). A demonstration of the $PE(W_t)$ circuit is shown in Figure A.1 below.

The realisation of V_t invokes the CONTROLLED-W_t gates at most $O(\frac{1}{\Delta}\log(1/\sqrt{\epsilon_2}))$ times, where $\Delta = 2\theta_1$ as introduced in Section 3.4.4.1 is the phase gap. The V_t circuit has the following properties,

$$\begin{aligned} V_t|\pi_t\rangle|0\rangle^{\otimes ac} &= |\pi_t\rangle|0\rangle^{\otimes ac}, \\ V_t|\psi^{\pm}_k\rangle|0\rangle^{\otimes ac} &= \sqrt{1-\epsilon_2}|\psi^{\pm}_k\rangle|\chi_k\rangle + \sqrt{\epsilon_2}|\psi^{\pm}_k\rangle|0\rangle^{\otimes ac}, \end{aligned} \tag{66}$$

where $\langle 0^{\otimes ac}|\chi_k\rangle = 0$. The eigenstates $|\pi_t\rangle$ and $|\psi^{\pm}_k\rangle$ lie in the subspace $\mathscr{H}_{\pi_t}$, with $W(P_t)|\pi_t\rangle = |\pi_t\rangle$ and $W(P_t)|\psi^{\pm}_k\rangle = e^{\pm 2\pi i\theta_k}|\psi^{\pm}_k\rangle$ for all $k\in[M]$. Essentially, the cir-

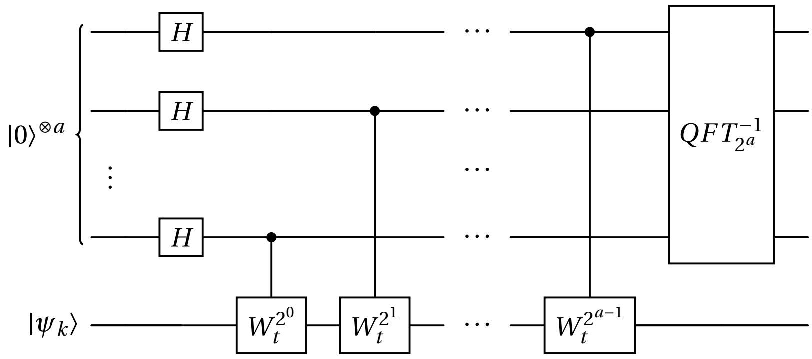


Figure A.1: Phase estimation circuit $PE(W_t)$, where H is the Hadamard gate, QFT^{-1} denotes the inverse quantum Fourier transform circuit, and $W_t = U(P_t)^\dagger SU(P_t)R_{\mathscr{A}}U(P_t)^\dagger SU(P_t)R_{\mathscr{A}}$.

cuit V_t differentiates the stationary state $|\pi_t\rangle$ and its orthogonal eigenstates $|\psi_k^{\pm}\rangle$ up to some error $O(\sqrt{\epsilon_2})$. Similarly to R_{t+1}, one can construct a circuit

$$\tilde{R}_t = V_t^\dagger(\mathbb{1}\otimes(e^{\frac{i\pi}{3}}|0\rangle\langle 0|^{\otimes ac} + (\mathbb{1} - |0\rangle\langle 0|^{\otimes ac})))V_t, \tag{67}$$

that $\|\tilde{R}_t|\psi_k^{\pm}\rangle|0\rangle^{\otimes ac} - R_t|\psi_k^{\pm}\rangle|0\rangle^{\otimes ac}\| \le 2\sqrt{\epsilon_2}$.

From Lemma 18, it can be seen that a transition from $|\pi_t\rangle$ to $|\tilde{\pi}_{t+1}\rangle$ can be achieved by a sequence of $\tilde{U}_{t,m} = \cdots R_{t+1}\tilde{R}_t R_{t+1}|\pi_t\rangle$, and the state after applying the first R_{t+1} as in Eqn.(65) will be in the space of span$\{|\pi_{t+1}\rangle, |\pi_{t+1}^{\perp}\rangle\}$. Then for implementing $\tilde{R}_t$ composed of the phase estimation circuit $PE(W_t)$, one need to make sure that $|\pi_{t+1}\rangle$ has a large overlap with the good state $|\pi_{t+1,\,\mathrm{good}}\rangle$, which is spanned by $|\psi_j^{\pm}\rangle$ with $\theta_j = \Omega(\sqrt{1/\tau}) = \Omega(\Delta')$. This is guaranteed by Lemma 17 as a warm start condition is satisfied. Therefore, for implementing $\tilde{R}_t$, it is sufficient to implement a circuit V_t that effectively differentiates states $|\psi_j^{\pm}\rangle$ and $|\pi_t\rangle$. Furthermore, as proved in Lemma 17, $\||e_{t+1}\rangle\| \le \sqrt{\epsilon_3}$, one can implement $\tilde{R}_t$ such that $\|\tilde{R}_t|\phi\rangle - R_t|\phi\rangle\| \le 2\sqrt{\epsilon_3}$ for any state $|\phi\rangle$ appears in the Grover's $\pi/3$ rotation from $|\pi_{t+1}\rangle$ to $|\pi_t\rangle$. Therefore, it takes $ac = O(c\log\frac{1}{\Delta'}) = O(c\log\sqrt{\tau})$ ancillary to implement the circuit V_t and it invokes the CONTROLLED-W_t gate at most $O(\frac{1}{\Delta'}\log(1/\sqrt{\epsilon_3})) = O(\sqrt{\tau}\log(1/\sqrt{\epsilon_3}))$ times. □

Proof. From the above Lemmas 17, 18, 19, one can see that the total error of preparing state $|\tilde{\pi}_{t+1}\rangle$ through $|\tilde{\pi}_{t+1}\rangle = \tilde{U}_{t+1,m}|\pi_t\rangle$ is bounded by $O(\epsilon_1 + (\frac{1}{p}\log\frac{1}{\epsilon_1})\sqrt{\epsilon_3})$. Taking $\epsilon_1 = O(\epsilon)$, $\epsilon_3 = O(\epsilon^2)$, one can obtain $\||\tilde{\pi}_{t+1}\rangle - |\pi_{t+1}\rangle\| = O(\epsilon)$ by using $O(\sqrt{\tau}\log(1/\epsilon))$ calls to the quantum walk operators. Theorem 12 is then proved. □

A.3.2 Non-destructive Amplitude Estimation

Theorem 20. *Aram W. Harrow et al. 2020, Theorem 18 Given state $|\psi\rangle$ and reflections $R_\psi = 2|\psi\rangle\langle\psi| - I$ and $R = 2P - I$, and any $\eta > 0$, there exists a quantum algorithm that outputs $\bar{a}$, an approximation to $a = \langle\psi|P|\psi\rangle$, so that*

$$|\bar{a} - a| \leq 2\pi \frac{a(1-a)}{M} + \frac{\pi^2}{M^2} \tag{68}$$

with probability at least $1-\eta$ *and* $O(\log(1/\eta)M)$ *uses of* R_ψ *and* R. *Moreover, the algorithm restores the state* $|\psi\rangle$ *with probability at least* $1-\eta$.

It takes use of the amplitude estimation circuit $(F_M^{-1} \otimes I)\Lambda_M(Q)(F_M \otimes I)$ proposed by Brassard et al. 2000 (as shown in Figure A.2) with $Q = -R_\psi R$. Here the output $\tilde{a} = \sin^2(\pi \frac{y}{M})$ is an approximation of $a = \langle\psi|P|\psi\rangle$, and the $\bar{a}$ in Theorem 20 is the median of multi-copy estimations of $\tilde{a}$.

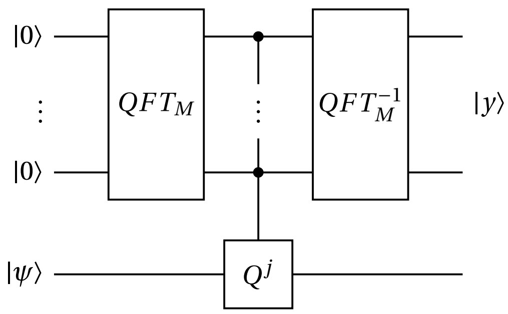


Figure A.2: Amplitude estimation circuit (Brassard et al. 2000), where F_M is the quantum Fourier transform, the CONTROLLED-Q^j operator in the middle is $\Lambda_M(Q) = \sum_{j=0}^{M-1} |j\rangle\langle j| \otimes Q^j$

Fitting in our QCP-QW algorithm, at the t^{th} round, one has the initial state $|\tilde{\pi}_t\rangle$ and the aim is to estimating the mean $\boldsymbol{z}_t$ and affine transformation matrix S_t of distribution $|\tilde{\pi}_t\rangle = |\psi\rangle$. The unitary Q can be applied by letting $Q = -R_{\tilde{\pi}_t} R = \tilde{U}_{t-1,m}(\mathbb{1} - 2|\tilde{\pi}_{t-1}\rangle\langle\tilde{\pi}_{t-1}|)\tilde{U}_{t-1,m}^\dagger R = (\mathbb{1} - 2|\tilde{\pi}_t\rangle\langle\tilde{\pi}_t|)R$. It can be easily seen that here the operation $\mathbb{1} - 2|\tilde{\pi}_{t-1}\rangle\langle\tilde{\pi}_{t-1}|$ can be realized similarly to the selective phase shift defined in Lemma 18, by letting $R'_{t-1} = e^{i\pi}\Pi_{t-1} + \Pi_{t-1}^\perp = \mathbb{1} - 2\Pi_{t-1}$. From Lemma 19, it can be shown that the R'_{t-1} gate can be implemented with $O^*(D^{1.5})$ calls to the CONTROLLED-W_{t-1} gates. Moreover, it is proved in Theorem 12, the quantum-walk-evolution unitary $\tilde{U}_{t-1,m}$ can be realised using $O^*(\sqrt{\tau}) = O^*(D^{1.5})$ calls to the CONTROLLED-W_{t-1} gates. Hence, the unitary $R_{\tilde{\pi}_t}$ can be implemented with $O^*(D^{1.5})$ calls to the CONTROLLED-W_{t-1} gates. We note that the observable P needs to be designed carefully for different observables.

A.3.3 Quantum-Walk-Update Operator Implementation

Lemma 21 (Transition probability of the Hit-and-Run walk). *Lovász 1999, Lemma 3 For the* ***Hit-and-Run*** *with uniform sampling scheme defined as in Section 3.4.3. If the current point of Hit-and-Run is* $\boldsymbol{x}$, *then the density function of the distribution of the next point* $\boldsymbol{y}$ *is*

$$f_{\boldsymbol{x}}(\boldsymbol{y}) = \frac{2}{D\pi_D} \frac{1}{\ell(\boldsymbol{x},\boldsymbol{y})\|\boldsymbol{x}-\boldsymbol{y}\|^{D-1}}, \tag{69}$$

where $\pi_D = \frac{\pi^{D/2}}{\Gamma(1+D/2)}$ *is the volume* B_1 *unit ball, and* $\ell(\boldsymbol{x}, \boldsymbol{y})$ *is the chord length through* $\boldsymbol{x}$ *and* $\boldsymbol{y}$.

Lemma 22. *The hit-and-run quantum-walk-update unitary* $U(P_t)$ *can be achieved with* $O^*(1)$ *uses of membership oracles* $\mathcal{O}_{\mathcal{P}_t}$ *and* $O^*(D)$ *gate complexity, in both discrete and continuous quantum computers.*

Proof. The implementation of the hit-and-run quantum-walk-update operator has been illustrated in detail for both continuous and discrete cases by Chakrabarti et al. 2023; T. Li et al. 2022. In the discrete case (see Eqn.(41)),

$$U(P_t)|\boldsymbol{x}\rangle|\boldsymbol{0}\rangle = |\boldsymbol{x}\rangle\,|\boldsymbol{p}_x\rangle = |\boldsymbol{x}\rangle \sum_{\boldsymbol{y}\in\mathcal{P}_{t,\epsilon}} \sqrt{p_{xy}}\,|\boldsymbol{y}\rangle\,,$$

and in the continuous case,

$$U(P_t)|\boldsymbol{x}\rangle|\boldsymbol{0}\rangle = \int_{\mathcal{P}_t} \sqrt{p_{xy}}|\boldsymbol{x}\rangle|\boldsymbol{y}\rangle \mathrm{d}\boldsymbol{y},$$

with p_{xy} denoting the x, y element of the stochastic transition matrix P_t. Here we only describe briefly how the unitary can be constructed in the discrete case, and interested readers can refer to the paper (Chakrabarti et al. 2023) for a construction in continuous quantum computing. As we mentioned in Section 3.4.4.2, we take the $\epsilon\text{-}descritiztion$ representation, such that our quantum states are defined in the $\epsilon\text{-}net$ of $\mathcal{P}_{t,\epsilon} \in \mathbb{R}^D$, thus $O^*(D)$ qubits are needed for representing quantum samples in the space.

At the t^{th} round, the walk starts from a given current point $\boldsymbol{x}$ that belongs to the *current position register* $\in \mathcal{H}_{curr}$. For choosing a direction uniformly, one can construct a D-dimensional Gaussian distribution to obtain a uniform distribution of all directions on the S^{D-1} sphere. This can be done using $O^*(D)$ ancillary *direction registers* $\in \mathcal{H}_{dir}$, and since the Gaussian distribution can be efficiently integrable, one can leverage the Grover-Rudolph method (L. Grover et al. 2002) to prepare the states. Then, for a direction $\boldsymbol{u}$, the line $\ell = \boldsymbol{x} + t\boldsymbol{u}$ uniquely defines a chord of $\mathcal{P}_t \cap \ell$ with its length $\ell(t_1, t_2)$, where $t \in [t_1, t_2]$ is a scalar and t_1, t_2 are the two endpoints. It is known that classically, the endpoints and the chord length can be determined within an error ϵ by using $O(\log(1/\epsilon))$ membership oracles $\mathcal{O}_{\mathcal{P}_t}$. Then, for the next proposed point, a new state $|\boldsymbol{y}\rangle = \frac{1}{\sqrt{|t_2-t_1|}} \sum_{t\in[t_1,t_2]} |\boldsymbol{x} + t\boldsymbol{u}\rangle$ of a *new position register* $\in \mathcal{H}_{new}$ can be constructed with controlled operations on $\boldsymbol{x} \in \mathcal{H}_{curr}$. Uncomputing the direction registers, the desired state of $|\boldsymbol{x}\rangle \sum_{\boldsymbol{y}\in\mathcal{P}_{t,\epsilon}} \sqrt{p_{xy}}\,|\boldsymbol{y}\rangle$ as in Eqn.(41), can be acquired. After change of variables, it can be verified that the probability p_{xy} of moving from $\boldsymbol{x}$ to $\boldsymbol{y}$ is the same as $f_{\boldsymbol{x}}(\boldsymbol{y})$ of Eqn.(69). Overall, it is evident that the hit-and-run quantum-walk-update unitary $U(P_t)$ can be achieved with $O^*(1)$ uses of membership oracles and $O^*(D)$ gate complexity.

□

B Analysis of approximated-FW algorithm

B.1 Primal Convergence

In this section, we provide the convergence proof of Theorem 15. We follow the same idea as in Jaggi 2013, A. Primal Convergence, and show that the primal error depends on the following Lemma 23 on the improvement in each iteration.

Lemma 23. *Let the standard duality gap of FW algorithm at round k be $g(\rho_k) := \max_{s\in\mathscr{S}_+^{p,\otimes}} \langle \nabla f(\rho_k), \rho_k - s\rangle = \langle \nabla f(\rho_k), \rho_k - s_k^*\rangle$, and let C_f be the curvature constant of f. One has*

$$f(\rho_{k+1}) \le f(\rho_k) - \gamma_k \left(g(\rho_k) - \delta_k^{true}\right) + \frac{\gamma_k^2}{2} C_f, \tag{70}$$

where C_f of the convex and differentiable function $f : \mathbb{R}^n \to \mathbb{R}$, with respect to a compact domain $\mathscr{D}$ is formally defined as

$$C_f := \sup_{x,s\in\mathscr{D},\gamma\in[0,1],y=x+\gamma(s-x)} \frac{2}{\gamma^2}\left(f(y) - f(x) - \langle y - x, \nabla f(x)\rangle\right).$$

It is known that if ∇f is L-Lipschitz continuous, then ibid.

$$C_f \le \sup_{x,y\in\mathscr{D}} \frac{2}{\gamma^2}\frac{L}{2}\|y - x\|^2 = \sup_{x,y\in\mathscr{D}} L\|y - x\|^2.$$

In our case, the compact domain $\mathscr{D} = \mathscr{S}_+^{p,\otimes}$, and $L = 1$ can be verified directly $\left|\left|\nabla f(\rho_1) - \nabla f(\rho_2)\right|\right|_F = ||\rho_1 - \rho_2||_F$. So $\|y - x\|_F^2 = \mathrm{tr}(y^2) + \mathrm{tr}(x^2) - 2\,\mathrm{tr}(yx) \le 2$, for PSD matrices x, y. Hence $C_f = 2$.

Proof. Define the true duality gap at the k round,

$$g_k^{\mathrm{true}} = \langle \nabla f(\rho_k), \rho_k - \tilde{s}_k\rangle = \langle \nabla f(\rho_k), \rho_k - s_k^*\rangle - \delta_k^{\mathrm{true}} = g(\rho_k) - \delta_k^{\mathrm{true}}.$$

Then, from the definition of the curvature constant C_f of our convex function f,

$$f(\rho_{k+1}) = f(\rho_k + \gamma_k(\tilde{s}_k - \rho_k)) \le f(\rho_k) + \gamma_k\langle \tilde{s}_k - \rho_k, \nabla f(\rho_k)\rangle + \frac{\gamma_k^2}{2} C_f = f(\rho_k) - \gamma_k g_k^{\mathrm{true}} + \frac{\gamma_k^2}{2} C_f.$$

Set the FW update with step-size γ_k such that $\rho_{k+1} = \rho_k + \gamma_k(\tilde{s}_k - \rho_k)$. Applying the definition of δ_k^{true} and rearranging, one obtains

$$f\left(\rho_{k+1}\right) \leq f\left(\rho_k\right) - \gamma_k g_k^{true} + \frac{\gamma_k^2}{2} C_f = f\left(\rho_k\right) - \gamma_k \left(g\left(\rho_k\right) - \delta_k^{\text{true}}\right) + \frac{\gamma_k^2}{2} C_f.$$

□

Theorem 24. *For each $k \geq 1$, if the upper bound of the true additive error $\delta_k^{ub} \leq \frac{\gamma_k C_f \delta}{2} = \frac{2\delta}{k+2}$, where $\delta \geq 0$ is the accuracy to which the internal linear sub-problems are solved ($\delta = 0$ in the exact FW algorithm). Then the iterate ρ_k of the approximated FW algorithm satisfies,*

$$f\left(\rho_k\right) - f\left(\rho^*\right) \leq \frac{2C_f}{k+2}(1+\delta) = \frac{4(1+\delta)}{k+2} = O(\frac{1}{k}), \tag{71}$$

where $\rho^* \in \mathscr{S}_+^{p,\otimes}$ is the optimal solution.

Proof. Writing $h(\rho_k) := f(\rho_k) - f(\sigma^*)$ for the (unknown) primal error at any point ρ_k, where $\rho^* \in \operatorname{argmin}_{\sigma \in \mathscr{S}_+^{p,\otimes}} f(\sigma)$. By the convexity of the function f, it can be concluded that $g\left(\rho_k\right) \geq h(\rho_k)$, and thus

$$h(\rho_{k+1}) \leq h(\sigma_k) - \gamma_k \left(g\left(\rho_k\right) - \delta_k^{\text{true}}\right) + \frac{\gamma_k^2}{2} C_f \leq \left(1-\gamma_k\right) h(\rho_k) + \gamma_k \delta_k^{\text{true}} + \frac{\gamma_k^2}{2} C_f.$$

We now substitute the FW step-size $\gamma_k = \frac{2}{k+2}$. Notice that if $\delta_k^{true} \leq \delta_k^{ub} \leq \frac{\gamma_k C_f \delta}{2} = \frac{2\delta}{k+2}$, then

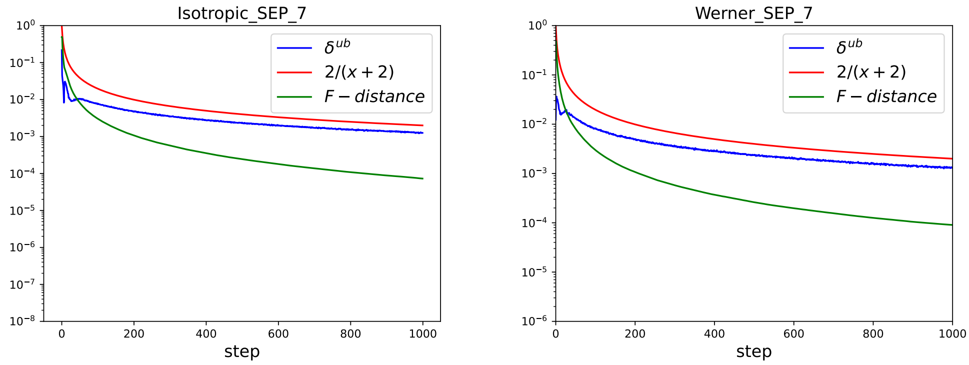
$$h(\rho_{k+1}) \leq \left(1-\gamma_k\right) h(\rho_k) + \frac{\gamma_k^2 C_f \delta}{2} + \frac{\gamma_k^2}{2} C_f = \left(1-\gamma_k\right) h(\rho_k) + \frac{\gamma_k^2 C_f}{2}(1+\delta) = \left(1-\gamma_k\right) h(\rho_k) + \gamma_k^2 C,$$

where $C = \frac{C_f(1+\delta)}{2}$. Now we reach the same condition as Jaggi 2013, Eqn.(4). Using a similar argument of induction, one can have for each $k \geq 1$, the iterate ρ_k of the approximated FW algorithm satisfies

$$f\left(\rho_k\right) - f\left(\rho^*\right) \leq \frac{2C_f}{k+2}(1+\delta) = \frac{4(1+\delta)}{k+2} = O(\frac{1}{k}),$$

where $\rho^* \in \mathscr{S}_+^{p,\otimes}$ is the optimal solution. □

Here we present the simulation results for SEP isotropic and Werner states in $d = 7$.



(a) Isotropic (7×7)

(b) Werner (7×7)

Figure B.1: SEP results for isotropic and Werner states in $d = 7$.

B.2 Suboptimal-region Confinement

In this section, we give the proof of Theorem 16, and show that the output sequence is confined in an area with its Frobenius radius bounded. First, we introduce a Lemma 25, which is used for the proof.

Lemma 25. *Function $f(\cdot)$ is* 1*-strongly convex, and precisely, one has*

$$f(y) - f(x) - \langle \nabla f(x), y - x \rangle = \frac{1}{2} \|y - x\|_F^2. \tag{72}$$

Proof. From the definition, one has $f(x) + \langle \nabla f(x), y - x \rangle = \frac{1}{2}\|x - \rho\|^2 + \langle x - \rho, y - x \rangle$. Subtracting this from $f(y)$, yields

$$f(y) - (f(x) + \langle \nabla f(x), y - x \rangle) = \frac{1}{2}\|y - \rho\|^2 - \frac{1}{2}\|x - \rho\|^2 - \langle x - \rho, y - x \rangle.$$

Substituting that $\|y - \rho\|^2 = \|x - \rho\|^2 + 2\langle x - \rho, y - x \rangle + \|y - x\|^2$, it can be shown that

$$\begin{aligned}
&f(y) - (f(x) + \langle \nabla f(x), y - x \rangle) \\
&= \frac{1}{2}\left(\|x - \rho\|^2 + 2\langle x - \rho, y - x \rangle + \|y - x\|^2\right) - \frac{1}{2}\|x - \rho\|^2 - \langle x - \rho, y - x \rangle \\
&= \frac{1}{2}\|y - x\|^2.
\end{aligned}$$

□

Theorem 26. *For each $k \geq 1$, if the upper bound of the true additive error $\delta_k^{ub} \to \eta > 0$ $(k \to \infty)$, then one has that when $k \to \infty$,*

$$\frac{1}{2}\|\rho_k - \rho^*\|_F^2 \leq f(\rho_k) - f(\rho^*) \leq \eta + o(1), \quad \forall \rho_k \in \mathscr{S}_+^{p,\otimes}, \tag{73}$$

where $\rho^* \in \mathscr{S}_+^{p,\otimes}$ is the optimal point. It can be concluded that the modified FW algorithm outputs a sequence $\{\rho_k\}$ that asymptotically remains confined to a neighbourhood of the optimum, with the Frobenius radius being $\sqrt{2\eta}$.

Proof. Combining Lemma 23 and the inequality $\delta_k^{true} \le \delta_k^{ub}$, one has

$$f(\rho_{k+1}) \le f(\rho_k) - \gamma_k \left(g(\rho_k) - \delta_k^{\text{true}}\right) + \frac{\gamma_k^2}{2} C_f \le f(\rho_k) - \gamma_k \left(g(\rho_k) - \delta_k^{\text{ub}}\right) + \frac{\gamma_k^2}{2} C_f.$$

Rearranging and summing over terms,

$$\begin{aligned}\sum_{k=0}^{\infty} \gamma_k \left(g(\rho_k) - \delta_k^{ub}\right) &\le \sum_{k=0}^{\infty} \left(f(\rho_k) - f(\rho_{k+1})\right) + \sum_{k=0}^{\infty} \frac{\gamma_k^2}{2} C_f \\ &= f(\rho_0) - f(\rho_\infty) + \sum_{k=0}^{\infty} \gamma_k^2 \le f(\rho_0) - f(\rho^*) + \sum_{k=0}^{\infty} \gamma_k^2 \\ &= C_1' + \frac{2\pi^2}{3} - 4 = C_1 \quad (0 < C_1 < \infty),\end{aligned}$$

where the last line is because $\sum_{k=0}^{\infty} \gamma_k^2 = 4 \sum_{k=0}^{\infty} \frac{1}{(k+2)^2} = 4(\sum_{k'=1}^{\infty} \frac{1}{k'^2} - 1) = 4(\frac{\pi^2}{6} - 1)$. Hence,

$$\sum_{k=0}^{\infty} \gamma_k \left(g(\rho_k) - \delta_k^{ub}\right) \le C_1 < \infty.$$

Notice that the summation over γ_k diverges, as $\sum_{k=0}^{\infty} \gamma_k = 2 \sum_{k=0}^{\infty} \frac{1}{k+2} = 2(H(\infty) - 1) = \infty$, where $H(n)$ is the n^{th} Harmonic number. This implies that $\lim_{k\to\infty} g(\rho_k) - \delta_k^{ub} = 0$. Therefore, if $\delta_k^{ub} \to \eta > 0$ $(k \to \infty)$, and $g(\rho_k) - \delta_k^{ub} \to 0$ $(k \to \infty)$, then it yields $g(\rho_k) \to \eta > 0, \ (k \to \infty)$.

From the definition,

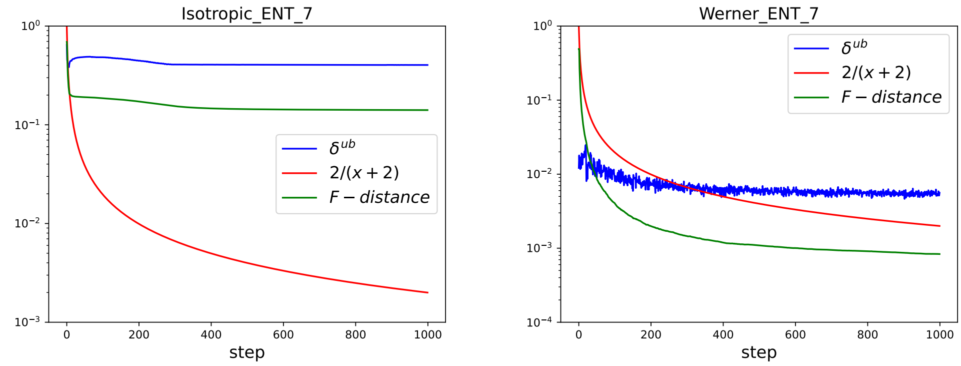
$$f(\rho_k) - f(\rho^*) \le g(\rho_k) \to \eta, \quad \text{equivalently, } f(\rho_k) - f(\rho^*) \le \eta + o(1), \ (k \to \infty). \tag{74}$$

Hence, one has $\forall \epsilon > 0, \exists K$, such that $\forall k > K: \ f(\rho_k) - f(\rho^*) \le \eta + \epsilon$. Since the duality gap $g(\rho^*) = 0$ for the optimal state (nearest separable density matrix) ρ^*, and we know the sequence $\{g(\rho_k)\}$ converges to an positive constant η, so it can be concluded the asymptotic value $\lim_{k\to\infty} \rho_k$ is not the optimal point, a.k.a, it is not the nearest separable state.

Furthermore, it has been proved that the term $\langle \nabla f(\rho^*), y - \rho^* \rangle = \langle \rho^* - \rho, y - \rho^* \rangle \ge 0$, $\forall y \in \mathscr{S}_+^{p,\otimes}$. Substituting it to Lemma 25, it yields, $f(\rho_k) - f(\rho^*) \ge \frac{1}{2} \|\rho_k - \rho^*\|_F^2$, $\forall \rho_k \in \mathscr{S}_+^{p,\otimes}$. Combining with Eqn.(74), one has that when $k \to \infty$,

$$\frac{1}{2} \|\rho_k - \rho^*\|_F^2 \le f(\rho_k) - f(\rho^*) \le \eta + o(1).$$

Therefore $\|\rho_k - \rho^*\|_F \le \sqrt{2\eta + o(1)}$, $o(1) \to 0$. It can be concluded $\limsup_{k\to\infty} \left\|\rho_k - \rho^*\right\|_F \le \sqrt{2\eta}$. Herewith, we can see that when $\delta_k^{ub} \to \eta > 0$ asymptotically, the modified FW algorithm outputs a sequence $\{\rho_k\}$ that asymptotically remains confined to a neighbourhood of the optimum, with the Frobenius radius being $\sqrt{2\eta}$. $\square$



(a) Isotropic (7×7) (b) Werner (7×7)

Figure B.2: ENT results for isotropic and Werner states in $d = 7$.

C Experimental results

C.1 Additional witness tests

Here we present the witness test results when reducing the size of the randomly sampled separable set $|\rho_S| = n$ to 5000 and 2500. In the experiments for $n = 5000$, in local dimension $d = 3$, it takes $\sim 2s$ per state; and $\sim 3\,\text{min}$ per state for $d = 10$. And in the experiments for $n = 2500$, it takes $\sim 1s$ per state in $d = 3$, and $\sim 2\,\text{min}$ per state for $d = 10$. From the experiments, we can confirm the expected increased error rate with reduced size n. For both cases, there are no SEP isotropic states mislabelled as ENT; for Werner states, the mislabelling rates are still low (up to 0.126 for $n = 5000$, and up to 0.156 for $n = 2500$, both appear in $d = 7$). For PPT-ENT Horodecki states, the ratios of ~ 0.89 and ~ 0.83 indicate that this method is still selective, where there are ~ 0.11 and ~ 0.17 PPT-ENT states that are selected and used for creating the training dataset.

n	Horodecki $SEP \to ENT$	Horodecki $PPT\text{-}ENT \to SEP$
5000	0.056 ± 0.018 (192s)	0.888 ± 0.028 (353s)
d	Isotropic $SEP \to ENT$	Werner $SEP \to ENT$
3	0.000 ± 0.000 (176s)	0.000 ± 0.000 (183s)
4	0.000 ± 0.000 (369s)	0.012 ± 0.008 (356s)
5	0.000 ± 0.000 (741s)	0.070 ± 0.011 (751s)
6	0.000 ± 0.000 (1433s)	0.094 ± 0.021 (1452s)
7	0.000 ± 0.000 (2678s)	**0.126** ± **0.018** (2751s)
8	0.000 ± 0.000 (5184s)	0.086 ± 0.008 (5209s)
9	0.000 ± 0.000 (10261s)	0.036 ± 0.020 (10053s)
10	0.000 ± 0.000 (17211s)	0.002 ± 0.004 (17049s)

Table C.1: Mean and standard deviation over independent experiments with the size of the separable set $n = 5000$, and randomly generated mixing number between $[1, \ldots, 2d^2]$. The time in the brackets is the average time taken to check 100 states in the corresponding local dimension.

n	Horodecki $SEP \to ENT$	Horodecki $PPT\text{-}ENT \to SEP$
2500	0.106 ± 0.0280 (103s)	0.826 ± 0.0350 (102s)
d	Isotropic $SEP \to ENT$	Werner $SEP \to ENT$
3	0.000 ± 0.000 (98s)	0.000 ± 0.000 (130s)
4	0.000 ± 0.000 (186s)	0.032 ± 0.008 (210s)
5	0.000 ± 0.000 (585s)	0.086 ± 0.016 (514s)
6	0.000 ± 0.000 (604s)	0.138 ± 0.017 (881s)
7	0.000 ± 0.000 (1111s)	**0.156** ± **0.008** (2044s)
8	0.000 ± 0.000 (2685s)	0.104 ± 0.010 (3440s)
9	0.000 ± 0.000 (5598s)	0.054 ± 0.016 (6708s)
10	0.000 ± 0.000 (11116s)	0.006 ± 0.0120 (16166s)

Table C.2: Mean and standard deviation over independent experiments with the size of the separable set $n = 2500$, and randomly generated mixing number between $[1, \ldots, 2d^2]$. The time in the brackets is the average time taken to check 100 states in the corresponding local dimension.

C.2 SVM and neural network classifiers

nonlinear SVM with polynomial and Gaussian kernels, scikit-learn SVM implementation Pedregosa et al. 2011. The hyperparameters of the SVM and the kernels were determined by 5-fold cross-validation. Code and trained models are available at . . .

For neural networks, two hidden layers with $2d$ - d neurons, where d is the dimension of the input. used ReLU activation, batch normalisation, and Adam with a learning rate of 0.001.

Table C.3: Classification accuracy of NNs on 3 and 7-dimensional random mixed states with 1,000 test examples per class. All the NNs here are trained on 1,000 training examples per class.

PPT (%)	SEP (3)	PPT-ENT (3)	NPPT (3)	Total (3)	Total2 (3)
0%	0.979 (±0.003)	0.988 (±0.010)	0.998 (±0.002)	0.988 (±0.005)	0.983 (±0.006)
			without augmentation		
25%	0.982 (±0.002)	0.959 (±0.039)	0.987 (±0.017)	0.976 (±0.018)	0.970 (±0.020)
50%	0.973 (±0.003)	0.990 (±0.009)	0.998 (±0.002)	0.987 (±0.004)	0.982 (±0.005)
75%	0.960 (±0.024)	0.995 (±0.004)	1.000 (±0.000)	0.985 (±0.007)	0.978 (±0.011)
100%	0.950 (±0.042)	0.989 (±0.013)	0.998 (±0.003)	0.979 (±0.013)	0.970 (±0.019)
			with augmentation		
25%	0.980 (±0.006)	0.993 (±0.003)	0.999 (±0.001)	0.991 (±0.002)	0.987 (±0.002)
50%	0.979 (±0.004)	0.997 (±0.002)	0.999 (±0.000)	0.992 (±0.002)	0.988 (±0.003)
75%	0.979 (±0.003)	0.992 (±0.007)	0.998 (±0.002)	0.990 (±0.002)	0.986 (±0.003)
100%	0.979 (±0.005)	0.989 (±0.020)	0.997 (±0.006)	0.988 (±0.007)	0.984 (±0.008)
PPT (%)	SEP (7)	PPT-ENT (7)	NPPT (7)	Total (7)	Total2 (7)
0%	0.467 (±0.100)	0.812 (±0.061)	0.735 (±0.068)	0.671 (±0.016)	0.639 (±0.025)
			without augmentation		
25%	0.516 (±0.093)	0.795 (±0.081)	0.711 (±0.090)	0.674 (±0.027)	0.655 (±0.009)
50%	0.621 (±0.134)	0.690 (±0.147)	0.626 (±0.127)	0.646 (±0.048)	0.656 (±0.015)
75%	0.472 (±0.126)	0.836 (±0.090)	0.775 (±0.093)	0.694 (±0.021)	0.654 (±0.024)
100%	0.549 (±0.132)	0.774 (±0.125)	0.713 (±0.129)	0.679 (±0.043)	0.661 (±0.018)
			with augmentation		
25%	0.666 (±0.211)	0.588 (±0.250)	0.529 (±0.232)	0.594 (±0.093)	0.627 (±0.031)
50%	0.555 (±0.090)	0.800 (±0.069)	0.723 (±0.069)	0.693 (±0.017)	0.677 (±0.014)
75%	0.688 (±0.136)	0.616 (±0.224)	0.559 (±0.195)	0.621 (±0.095)	0.652 (±0.045)
100%	0.605 (±0.082)	0.751 (±0.072)	0.676 (±0.077)	0.677 (±0.023)	0.678 (±0.008)

C.3 Additional results- SVMs

Table C.4: Classification accuracy of SVM on the 3-dimensional qudit dataset with 100 examples per class.

PPT (%)	SEP (3)	PPT-ENT (3)	NPPT (3)	Total (3)	Total 2 (3)
0%	0.923 (±0.0119)	0.634 (±0.0437)	0.806 (±0.0526)	0.787 (±0.0276)	**0.870 (±0.0141)**
			Without augmentation		
25%	0.934 (±0.0198)	0.635 (±0.0349)	0.776 (±0.0375)	0.781 (±0.0227)	0.819 (±0.0198)
50%	0.919 (±0.0083)	0.732 (±0.0259)	0.853 (±0.0434)	0.834 (±0.0202)	0.810 (±0.0332)
75%	0.916 (±0.0116)	0.738 (±0.0257)	0.853 (±0.0461)	0.836 (±0.0211)	0.817 (±0.0407)
100%	0.914 (±0.0096)	0.764 (±0.0669)	0.873 (±0.0437)	0.850 (±0.0338)	0.784 (±0.0302)
			With augmentation		
25%	0.930 (±0.0225)	0.703 (±0.0633)	0.828 (±0.0423)	0.820 (±0.0399)	0.843 (±0.0308)
50%	0.915 (±0.0154)	0.701 (±0.0646)	0.832 (±0.0696)	0.816 (±0.0419)	0.794 (±0.0120)
75%	0.909 (±0.0078)	0.810 (±0.0640)	0.901 (±0.0559)	0.874 (±0.0386)	0.775 (±0.0164)
100%	0.908 (±0.0102)	0.757 (±0.0463)	0.843 (±0.0572)	0.836 (±0.0315)	0.765 (±0.0241)

C.4 Bias test results

SVMs We observe drastic drops in classification accuracy for the isotropic, Werenr, and the Horodecki states, which indicate that our SVM models are strongly biased. However, for Isotropic and Werenr, the classification accuracy is still > 0.5 for separable states in local dimension 3. For the Horodecki state, there is no separable state that is classified correctly, but all entangled states are classified correctly, which indicates an inside-shrunk boundary. In $d = 3$, we note that the accuracy on ENT is generally better than it is on SEP.

Table C.5: Classification accuracy of SVM on isotropic states (left) and Werner states (right) with local dimension $d = 3,\ 7$ and $1,000$ test examples per class.

PPT (%)	SEP (3)	NPPT (3)	PPT (%)	SEP (3)	NPPT (3)
0%	0.698 (± 0.003)	1.000 (± 0.000)	0%	0.795 (± 0.000)	0.864 (± 0.002)
	without augmentation			without augmentation	
25%	**0.720 (± 0.013)**	**1.000 (± 0.000)**	25%	0.769 (± 0.018)	0.905 (± 0.019)
50%	0.700 (± 0.031)	1.000 (± 0.000)	50%	0.773 (± 0.018)	0.934 (± 0.018)
75%	0.703 (± 0.032)	1.000 (± 0.000)	75%	0.760 (± 0.020)	0.946 (± 0.009)
100%	0.698 (± 0.001)	1.000 (± 0.000)	100%	0.753 (± 0.004)	0.955 (± 0.003)
	with augmentation			with augmentation	
25%	0.585 (± 0.146)	1.000 (± 0.000)	25%	0.701 (± 0.118)	0.934 (± 0.036)
50%	0.567 (± 0.023)	1.000 (± 0.000)	50%	0.730 (± 0.013)	0.957 (± 0.021)
75%	0.575 (± 0.058)	1.000 (± 0.000)	75%	0.711 (± 0.027)	0.962 (± 0.028)
100%	0.564 (± 0.028)	1.000 (± 0.000)	100%	0.696 (± 0.012)	0.984 (± 0.007)

PPT (%)	SEP (7)	NPPT (7)	PPT (%)	SEP (7)	NPPT (7)
0%	0.211 (± 0.030)	0.494 (± 0.247)	0%	0.079 (± 0.090)	0.893 (± 0.214)
	without augmentation			without augmentation	
25%	0.281 (± 0.002)	0.000 (± 0.000)	25%	0.270 (± 0.000)	0.455 (± 0.001)
50%	0.298 (± 0.001)	0.000 (± 0.000)	50%	0.284 (± 0.001)	0.437 (± 0.001)
75%	0.304 (± 0.002)	0.000 (± 0.000)	75%	0.288 (± 0.001)	0.432 (± 0.002)
100%	0.310 (± 0.000)	0.000 (± 0.000)	100%	0.292 (± 0.000)	0.425 (± 0.000)
	with augmentation			with augmentation	
25%	0.332 (± 0.002)	0.000 (± 0.000)	25%	0.312 (± 0.004)	0.410 (± 0.001)
50%	0.357 (± 0.001)	0.000 (± 0.000)	50%	0.333 (± 0.000)	0.387 (± 0.002)
75%	0.377 (± 0.002)	0.000 (± 0.000)	75%	0.348 (± 0.002)	0.365 (± 0.004)
100%	0.391 (± 0.000)	0.000 (± 0.000)	100%	0.365 (± 0.002)	0.351 (± 0.001)

Table C.6: Classification accuracy of SVMs on 3-dimensional Horodecki states with 1,000 test examples per class. For d=3, only polynomial kernels are selected by the Grid-search. The average and std are taken over 5 experiments each (5 different kernel parameters).

PPT (%)	SEP (3)	PPT-ENT (3)	NPPT (3)
0%	0.000 (± 0.0000)	1.000 (± 0.000)	1.000 (± 0.000)
without augmentation			
25%	0.000 (± 0.0000)	1.000 (± 0.0000)	1.000 (± 0.0000)
50%	0.000 (± 0.0000)	1.000 (± 0.0000)	1.000 (± 0.0000)
75%	0.000 (± 0.0000)	1.000 (± 0.0000)	1.000 (± 0.0000)
100%	0.000 (± 0.0000)	1.000 (± 0.0000)	1.000 (± 0.0000)
with augmentation			
25%	0.000 (± 0.0000)	1.000 (± 0.0000)	1.000 (± 0.0000)
50%	0.000 (± 0.0000)	1.000 (± 0.0000)	1.000 (± 0.0000)
75%	0.000 (± 0.0000)	1.000 (± 0.0000)	1.000 (± 0.0000)
100%	0.000 (± 0.0000)	1.000 (± 0.0000)	1.000 (± 0.0000)

NNs The classification results on NNs are similar to the above SVM results. In local dimension 3, the classifiers barely work, and on $d = 7$, the models are strongly biased.

Table C.7: Classification accuracy of NNs on isotropic states (left) and Werner states (right) with local dimension $d = 3,\ 7$ and $1,000$ test examples per class. The average and std are taken over 5 experiments each.

PPT (%)	SEP (3)	NPPT (3)
0%	0.593 (± 0.015)	1.000 (± 0.000)
	without augmentation	
25%	0.569 (± 0.018)	1.000 (± 0.000)
50%	0.645 (± 0.015)	1.000 (± 0.000)
75%	0.633 (± 0.012)	1.000 (± 0.000)
100%	0.614 (± 0.009)	1.000 (± 0.000)
	with augmentation	
25%	0.735 (± 0.012)	1.000 (± 0.000)
50%	0.483 (± 0.018)	1.000 (± 0.000)
75%	0.572 (± 0.009)	1.000 (± 0.000)
100%	0.475 (± 0.015)	1.000 (± 0.000)
PPT (%)	SEP (7)	NPPT (7)
0%	0.185 (± 0.192)	0.072 (± 0.131)
	without augmentation	
25%	0.272 (± 0.264)	0.014 (± 0.017)
50%	0.230 (± 0.203)	0.056 (± 0.111)
75%	0.130 (± 0.109)	0.203 (± 0.399)
100%	0.104 (± 0.151)	0.047 (± 0.093)
	with augmentation	
25%	0.434 (± 0.344)	0.009 (± 0.017)
50%	0.144 (± 0.277)	0.094 (± 0.176)
75%	0.120 (± 0.136)	0.011 (± 0.022)
100%	0.381 (± 0.207)	0.000 (± 0.000)

PPT (%)	SEP (3)	NPPT (3)
0%	0.699 (± 0.010)	0.926 (± 0.003)
	without augmentation	
25%	0.688 (± 0.012)	0.946 (± 0.005)
50%	0.743 (± 0.015)	0.901 (± 0.011)
75%	0.669 (± 0.021)	0.987 (± 0.004)
100%	0.706 (± 0.015)	0.979 (± 0.003)
	with augmentation	
25%	0.664 (± 0.012)	0.912 (± 0.002)
50%	0.607 (± 0.015)	1.000 (± 0.000)
75%	0.658 (± 0.012)	0.984 (± 0.004)
100%	0.663 (± 0.013)	1.000 (± 0.000)
PPT (%)	SEP (7)	NPPT (7)
0%	0.124 (± 0.114)	0.765 (± 0.302)
	without augmentation	
25%	0.123 (± 0.123)	0.604 (± 0.236)
50%	0.060 (± 0.110)	0.595 (± 0.233)
75%	0.000 (± 0.000)	0.328 (± 0.174)
100%	0.000 (± 0.000)	0.493 (± 0.295)
	with augmentation	
25%	0.288 (± 0.369)	0.430 (± 0.314)
50%	0.175 (± 0.167)	0.597 (± 0.332)
75%	0.201 (± 0.185)	0.756 (± 0.229)
100%	0.285 (± 0.195)	0.608 (± 0.237)

C.4.1 SVMs trained on combined training datasets

Table C.8: Classification accuracy of NNs on Horodecki states with 1,000 test examples per class. The average and std are taken over 5 experiments each.

PPT (%)	SEP (3)	PPT-ENT (3)	NPPT (3)
0%	0.000 (±0.000)	1.000 (±0.000)	1.000 (±0.000)
without augmentation			
25%	0.000 (±0.000)	1.000 (±0.000)	1.000 (±0.000)
50%	0.000 (±0.000)	1.000 (±0.000)	1.000 (±0.000)
75%	0.000 (±0.000)	1.000 (±0.000)	1.000 (±0.000)
100%	0.000 (±0.000)	1.000 (±0.000)	1.000 (±0.000)
with augmentation			
25%	0.000 (±0.000)	1.000 (±0.000)	1.000 (±0.000)
50%	0.000 (±0.000)	1.000 (±0.000)	1.000 (±0.000)
75%	0.000 (±0.000)	1.000 (±0.000)	1.000 (±0.000)
100%	0.000 (±0.000)	1.000 (±0.000)	1.000 (±0.000)

Table C.9: Classification accuracy of SVMs trained on the combined dataset (400/200/200/200) on 3-dimensional Horodecki states with 1,000 test examples per class. The average and std are taken over 5 experiments each.

PPT (%)	SEP (3)	PPT-ENT (3)	NPPT (3)
0%	1.000 (±0.000)	0.000 (±0.000)	1.000 (±0.000)
without augmentation			
25%	1.000 (±0.000)	0.000 (±0.000)	1.000 (±0.000)
50%	1.000 (±0.000)	0.000 (±0.000)	1.000 (±0.000)
75%	1.000 (±0.000)	0.000 (±0.000)	1.000 (±0.000)
100%	0.279 (±0.392)	0.721 (±0.392)	0.600 (±0.490)
with augmentation			
25%	1.000 (±0.000)	0.000 (±0.000)	1.000 (±0.000)
50%	1.000 (±0.000)	0.000 (±0.000)	1.000 (±0.000)
75%	1.000 (±0.000)	0.000 (±0.000)	1.000 (±0.000)
100%	0.000 (±0.000)	1.000 (±0.000)	0.600 (±0.490)

Table C.10: Classification accuracy of SVMs trained on 3- and 7-dimensional combined training data (600/200/200) with 1,000 original test examples per class.

PPT (%)	SEP (3)	PPT-ENT (3)	NPPT (3)	Total (3)
0%	0.944 (±0.003)	0.749 (±0.030)	0.911 (±0.015)	0.868 (±0.014)
		without augmentation		
25%	0.940 (±0.003)	0.831 (±0.013)	0.949 (±0.007)	0.907 (±0.006)
50%	0.938 (±0.005)	0.893 (±0.014)	0.975 (±0.007)	0.935 (±0.008)
75%	0.939 (±0.005)	0.918 (±0.010)	0.982 (±0.006)	0.946 (±0.004)
100%	0.952 (±0.022)	0.917 (±0.018)	0.977 (±0.011)	0.949 (±0.007)
		with augmentation		
25%	0.938 (±0.002)	0.876 (±0.014)	0.963 (±0.007)	0.925 (±0.006)
50%	0.931 (±0.006)	0.951 (±0.017)	0.989 (±0.004)	0.957 (±0.008)
75%	0.932 (±0.007)	0.970 (±0.009)	0.992 (±0.005)	0.965 (±0.005)
100%	**0.932 (±0.005)**	**0.985 (±0.009)**	**0.998 (±0.002)**	**0.972 (±0.005)**

PPT (%)	SEP (7)	PPT-ENT (7)	NPPT (7)	Total (7)
0%	0.814 (±0.086)	0.224 (±0.107)	0.244 (±0.113)	0.427 (±0.045)
		without augmentation		
25%	0.621 (±0.331)	0.438 (±0.321)	0.451 (±0.319)	0.504 (±0.106)
50%	0.653 (±0.030)	0.446 (±0.031)	0.461 (±0.034)	0.520 (±0.012)
75%	0.598 (±0.016)	0.524 (±0.018)	0.527 (±0.016)	0.550 (±0.005)
100%	0.619 (±0.187)	0.482 (±0.241)	0.482 (±0.238)	0.528 (±0.097)
		with augmentation		
25%	**0.530 (±0.013)**	**0.601 (±0.020)**	**0.588 (±0.018)**	**0.573 (±0.010)**
50%	0.877 (±0.247)	0.159 (±0.319)	0.156 (±0.312)	0.397 (±0.128)
75%	1.000 (±0.000)	0.000 (±0.000)	0.000 (±0.000)	0.333 (±0.000)
100%	1.000 (±0.000)	0.000 (±0.000)	0.000 (±0.000)	0.333 (±0.000)